

LAMBDA

Setting up an AWS Lambda Function to Trigger from S3 and Update DynamoDB

This document provides a step-by-step procedure for creating an AWS Lambda function that automatically triggers upon an object upload to an S3 bucket and records the event details in an Amazon DynamoDB table.-----1. **DynamoDB Setup**

1. **Open DynamoDB** in the AWS Management Console.
2. **Create Table:**
 - **Table Name:** `table1`
 - **Partition Key:** `unique` (Type: String)
3. **Create Sample Items** (for initial testing/exploration):
 - **Item 1:**
 - `unique` (Primary Key): `EMP1` (String)
 - `Name`: `abcd` (String)
 - `Phone`: `1234` (Number)
 - `Employer`: (Boolean - set as needed)
 - `Desc`: (String - set as needed)
 - **Item 2:**
 - Create another item with `unique`: `EMP2`.
 - **Optional Operations:**
 - Search items using **Scan** (e.g., search all phone numbers).
 - Search items using **Query** (more efficient, requires key usage).
 - **Edit Item EMP1**: Update the JSON, e.g., set `"Employer": true`.
 - **Edit Datatype** of an attribute (e.g., change `Phone` from Number to String).

-----2. Lambda Function Creation

1. **Open Lambda** in the AWS Management Console.
2. **Create function:**
 - **Author From Scratch.**
 - **Function Name:** (Provide a descriptive name).
 - **Runtime:** Python 3.14 (or later compatible version).
 - **Change Default Execution Role:**
 - Select **Use an existing role:** `LABROLE` (This role must have permissions for S3 read, DynamoDB `PutItem`, and CloudWatch logging).
3. **Initial Code Testing (Optional: Simple Addition Function):**
 - Scroll down to the Code Source.
 - Paste the following test code, click **Deploy**, and **Test** with a sample JSON event like `{"num1": 5, "num2": 10}`.
4. `import json`
- 5.
6. `def lambda_handler(event, context):`

```

7.     a = event.get('num1', 0)
8.     b = event.get('num2', 0)
9.     result = a + b
10.    return {
11.        'statusCode': 200,
12.        'body': json.dumps({
13.            'sum': result
14.        })
15.    }

```

-----3. Integrating Lambda with S3 and DynamoDB

1. **Update Lambda Code (S3-to-DynamoDB Logic):**
 - Replace the simple addition code with the following Python code.
 - **IMPORTANT:** Replace -----tablename----- with your actual DynamoDB table name (e.g., `table1`).

```

2. import boto3
3. from uuid import uuid4
4. import json
5.
6. def lambda_handler(event, context):
7.     s3 = boto3.client("s3")
8.     dynamodb = boto3.resource("dynamodb")
9.     # Replace with your actual table name
10.    dynamoTable = dynamodb.Table("-----tablename-----")
11.
12.    # Check for S3 event records
13.    if "Records" in event:
14.        for record in event["Records"]:
15.            # Extract S3 event details
16.            bucket_name = record["s3"]["bucket"]["name"]
17.            object_key = record["s3"]["object"]["key"]
18.            size = record["s3"]["object"].get("size", -1)
19.            event_name = record.get("eventName", "Unknown")
20.            event_time = record.get("eventTime", "Unknown")
21.
22.            # Write item to DynamoDB
23.            dynamoTable.put_item(
24.                Item={
25.                    "unique": str(uuid4()), # Primary key
26.                    "Bucket": bucket_name,
27.                    "Object": object_key,
28.                    "Size": size,
29.                    "Event": event_name,
30.                    "EventTime": event_time,
31.                }
32.            )
33.    else:

```

- ```
34. print("No 'Records' key found in the event (not an S3 trigger event).")
35.
36. # Return a standard response (optional but good practice)
37. return {
38. 'statusCode': 200,
39. 'body': json.dumps({'message': 'Processed S3 event'})
40. }
```
41. **Deploy Code:** Click **Deploy** to save the changes.
42. **Add S3 Trigger:**
- In the Designer section of the Lambda console, click **Add trigger**.
  - Select **S3** as the trigger type.
  - **Bucket:** Select the S3 bucket you want to monitor.
  - **Event type:** Select the appropriate event, typically **All object create events** (**s3:ObjectCreated:\***).
  - **Versioning:** Ensure the selected **Bucket has versioning enabled** (often a prerequisite for S3 event notifications).
  - Acknowledge any required permissions and click **Add**.
43. **Test and Monitor:**
- **Test:** Manually upload an object to the configured S3 bucket.
  - **Verify:** Check your DynamoDB table (**table1**) to ensure a new item with the S3 object details has been created.
  - **Monitor:** Use the **Monitoring** tab in the Lambda console to view logs (CloudWatch Logs) and troubleshoot any issues.

#### -----Additional Configuration

- **Environment Variables (Optional):** If required, configure environment variables (e.g., DynamoDB table name) in the **Configuration** tab under **Environment variables**.

You have now successfully set up a Lambda function to trigger when an object is uploaded to S3 and update a DynamoDB table with the object's metadata.

SNS SQS

## WEEK-7---SNS,SQS& S3-SNS-SQS-LAMBDA SERVICES

### 1) Simple Notification Service (SNS)

#### A) Create SNS Topic and Send Message to Email

We will:

1. Create an SNS Topic
  2. Create an Email Subscription
  3. Confirm the subscription
  4. Publish a test message
- 

#### Step 1: Create SNS Topic

1. Login to **AWS Console**
2. In search bar → Type **SNS**
3. Open **Amazon SNS (Simple Notification Service)**
4. In left panel → Click **Topics**
5. Click **Create topic**

**Configure:**

- **Type** → Select **Standard**
  - **Name** → MyEmailTopic
  - Leave other settings default (unless required)
  - 6. Click **Create topic**
- 

#### Step 2: Create Email Subscription

1. Open your created topic
2. Go to **Subscriptions** tab
3. Click **Create subscription**

**Configure:**

- **Protocol** → Select **Email**
  - **Endpoint** → Enter your email address
  - 4. Click **Create subscription**
- 

#### Step 3: Confirm Subscription

- Go to your email inbox
- You will receive a mail from **Amazon SNS**
- Click **Confirm subscription**

Now your email is successfully subscribed 🎉

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#### Step 4: Publish Test Message

1. Open your SNS topic
  2. Click **Publish message**
  3. Add:
    - **Subject** → Test Mail
    - **Message body** → Hello this is SNS test
  4. Click **Publish message**
- You will receive email notification.

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## **B) Trigger Email When S3 Object is Uploaded (S3 → SNS → Email)**

Now we extend this setup:

Flow:

S3 Bucket → Event Notification → SNS Topic → Email

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### **Step 1: Create S3 Bucket**

1. Search → **S3**
2. Click **Create bucket**

**Configure:**

- **Bucket name** → my-upload-bucket-2026
  - Choose Region (Important: Same region as SNS topic)
  - Keep default settings (unless specific requirement)
3. Click **Create bucket**

---

### **Step 2: Create SNS Topic (If not already created)**

You can reuse the topic from Part A

OR create new topic like:

- Topic name → S3UploadNotification

Make sure email subscription is confirmed.

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### **Step 3: Allow S3 to Publish to SNS (Access Policy)**

1. Open SNS Topic
2. Go to **Access policy**
3. Click **Edit**

Add permission for S3 to publish.

If using Basic Editor:

Access Policy that needs to be configured on SNS:

a) replace SNS topic ARN

b) Replace s3 bucket ARN

c) Replace account

```
{
 "Version": "2012-10-17",
 "Id": "example-ID",
 "Statement": [
 {
 "Sid": "Example SNS topic policy",
 "Effect": "Allow",
 "Principal": {
 "Service": "s3.amazonaws.com"
 },
 "Action": [
 "SNS:Publish"
],
 "Resource": "SNS topic ARN",
 "Condition": {
 "ArnLike": {
 "aws:SourceArn": "s3 bucket ARN"
 },
 "StringEquals": {
 "aws:SourceAccount": "account id"
 }
 }
 }
]
}
```

(Modern AWS console often auto-generates this when configuring from S3)

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#### Step 4: Configure S3 Event Notification

1. Open your S3 bucket
2. Go to **Properties**
3. Scroll to **Event notifications**
4. Click **Create event notification**

##### Configure:

- **Event name** → UploadNotification
- **Event types** → Select:
  - All object create events
- **Destination**:
  - Select **SNS topic**
  - Choose your topic (S3UploadNotification)
- 5. Click **Save changes**

---

#### Step 5: Test It

1. Upload any file into the S3 bucket
2. After upload completes:
  - S3 triggers event
  - SNS receives publish request
  - SNS sends email

You will receive mail about object upload.

---

### Simple Queue Service (SQS)

#### 2) Create SQS Queue and Send Message

We will:

1. Create an SQS Queue
2. Send a message to the Queue
3. Receive the message from the Queue

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#### Step 1: Create SQS Queue

1. Login to **AWS Console**
2. In the **search bar** → **Type SQS**
3. Open **Amazon Simple Queue Service**
4. In the dashboard → Click **Create queue**

Configure:

- **Type** → **Select Standard**
- **Name** → **MyQueue**
- Leave other settings as **default**
- 5. Click **Create queue**

Your **SQS queue is now created successfully.**

---

### **Step 2: Send Message to Queue**

1. Open the **MyQueue** queue
2. Click **Send and receive messages**
3. In the **Message body**, type:

Hello this is SQS test message

4. Click **Send message**

Your message is now stored in the queue.

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### **Step 3: Receive Message from Queue**

1. In the same page (**Send and receive messages**)
2. Click **Poll for messages**

You will see the message displayed.

Example output:

Hello this is SQS test message

3. After processing, click **Delete message** (optional)
- 

### **Result:**

- SQS queue created
  - Message sent successfully
  - Message received from queue
- 

## **3) S3 Bucket Event → SNS → SQS → Lambda Processing**

S3 bucket event (like file upload) triggers SNS, which sends the notification to SQS. Then **Lambda acts as the consumer and processes the message.**

Services used:

- Amazon S3
  - Amazon Simple Notification Service
  - Amazon Simple Queue Service
  - AWS Lambda
- 

## **S3 → SNS → SQS → Lambda Architecture**

S3 Bucket (File Upload)

↓

SNS Topic

↓

SQS Queue



### Lambda Function (Consumer)

When a file is uploaded, S3 sends an event notification to SNS, which forwards the message to SQS. The Lambda function then reads and processes the message from SQS.

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#### Step 1: Create S3 Bucket

1. Login to AWS Console
2. Search **S3**
3. Open **Amazon S3**
4. Click **Create bucket**

Configure:

- Bucket name → **mys3eventbucket123** (must be unique)
- Region → choose your region

Leave other settings default.

5. Click **Create bucket**

---

#### Step 2: Create SNS Topic

1. Search **SNS** in AWS Console
2. Open **Amazon SNS**
3. Click **Topics** → **Create topic**

Configure:

- Type → **Standard**
- Name → **MyS3SNSTopic**

4. Click **Create topic**

Copy the **SNS Topic ARN**.

---

#### Step 3: Create SQS Queue

1. Search **SQS** in AWS Console
2. Open **Amazon SQS**
3. Click **Create queue**

Configure:

- Type → **Standard**
- Name → **MyS3Queue**

4. Click **Create queue**

Copy the **Queue ARN**.

---

#### Step 4: Subscribe SQS to SNS

1. Open **MyS3SNSTopic**
2. Click **Create subscription**

Configure:

- Protocol → **Amazon SQS**
  - Endpoint → Select **MyS3Queue**
3. Click **Create subscription**
- 

#### Step 5: Add Access Policy to SQS

Open **MyS3Queue** → **Access Policy** → **Edit**

Replace:

- SQS ARN
- SNS ARN

```
{
 "Statement": [
 {
 "Effect": "Allow",
 "Principal": {
 "Service": "sns.amazonaws.com"
 },
 "Action": "sqs:SendMessage",
 "Resource": "SQS ARN",
 "Condition": {
 "ArnEquals": {
 "aws:SourceArn": "SNS ARN"
 }
 }
 }
]
}
```

Click **Save**.

---

## Step 6: Add Access Policy to SNS

Open **SNS Topic** → **Edit access policy**

Replace:

- SNS topic ARN
- S3 bucket ARN
- Account ID

```
{
 "Version": "2012-10-17",
 "Id": "example-ID",
 "Statement": [
 {
 "Sid": "Example SNS topic policy",
 "Effect": "Allow",
 "Principal": {
 "Service": "s3.amazonaws.com"
 },
 "Action": [
 "SNS:Publish"
],
 "Resource": "SNS topic ARN",
 "Condition": {
 "ArnLike": {
 "aws:SourceArn": "S3 bucket ARN"
 },
 "StringEquals": {
 "aws:SourceAccount": "Account ID"
 }
 }
 }
]
}
```

```
}

]

}
```

Save the policy.

---

### Step 7: Configure S3 Event Notification

1. Open **S3 Bucket (mys3eventbucket123)**
2. Go to **Properties**
3. Scroll to **Event notifications**

Click **Create event notification**

Configure:

- Event name → **S3UploadEvent**
- Event type → **All object create events**

Destination:

- **SNS Topic** → **MyS3SNSTopic**

Click **Save changes**.

---

### Step 8: Test the Architecture

1. Open the **S3 bucket**
2. Click **Upload file**
3. Upload any file (example: **test.txt**)

Now:

File Upload → S3

↓

Notification → SNS

↓

Message → SQS

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### Step 9: Verify Message in SQS

1. Open **MyS3Queue**
2. Click **Send and receive messages**
3. Click **Poll for messages**

You will see the **S3 event notification message**.

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## Step 10: Configure Lambda as Consumer

Now configure Lambda to **automatically process messages from SQS**.

### Create Lambda Function

1. Search **Lambda** in AWS Console
2. Open **AWS Lambda**
3. Click **Create Function**

Configure:

- Author from scratch
- Function name → **SQSConsumerFunction**
- Runtime → **Python 3.x**

Click **Create function**

---

### Add SQS Trigger

1. Open **SQSConsumerFunction**
2. Click **Add Trigger**
3. Select **SQS**
4. Choose **MyS3Queue**

Click **Add**

Now Lambda will automatically read messages from SQS.

---

### Lambda Code

Replace code with:

```
def lambda_handler(event, context):

 for record in event['Records']:

print("Message received from SQS:")
 print(record['body'])

 return {
 'statusCode': 200,
 'body': 'Message processed successfully'
 }
```

Click **Deploy**.

---

### Final Architecture

User uploads file



Amazon S3



SNS Topic



SQS Queue



Lambda Function (Consumer)



Processing / Logs / Database

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### Result

When a file is uploaded to S3:

1. S3 sends a notification to SNS
2. SNS forwards the message to SQS
3. SQS stores the message
4. Lambda automatically reads and processes the message

This creates a **serverless event-driven architecture commonly used in real cloud applications**.

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### Scenario Based Questions (SNS, SQS, S3-SNS-SQS-Lambda)

1. A company wants to send notifications to multiple users whenever an important system event occurs. The same message must be delivered to email, SMS, and application subscribers simultaneously.  
Which AWS service can be used to implement this publish-subscribe messaging system?
2. A developer created an SNS topic and added an email subscription, but the subscriber is not receiving any notifications.  
What could be the possible reason for this issue?

3. An organization wants to automatically notify administrators via email whenever a new file is uploaded to an S3 bucket.  
How can this be implemented using AWS services?
4. A developer configured an S3 bucket to trigger an SNS notification on file upload, but the message is not reaching the SNS topic.  
What configuration or permission issue might cause this problem?
5. A company wants to store incoming messages temporarily so that applications can process them later without losing data even if the system is busy.  
Which AWS messaging service should be used for this requirement?
6. A developer sends messages to an SQS queue, but when checking the queue no messages appear.  
What possible reasons could cause this issue?
7. A company wants to process thousands of messages coming from different applications without overloading the processing system.  
How does Amazon SQS help in handling this situation?
8. A developer wants to ensure that messages are processed automatically from an SQS queue without manually polling the queue.  
Which AWS service can be configured to automatically consume messages from SQS?
9. An application uploads files to S3, and the company wants the upload event to trigger a workflow that sends notifications and processes the message later.  
How can this be designed using S3, SNS, SQS, and Lambda?
10. A developer configured an SNS topic to send messages to an SQS queue, but the queue is not receiving any messages.  
What policy configuration might be missing?
11. A company wants to decouple its application components so that failures in one component do not affect others.  
Which AWS service combination can be used to implement loosely coupled communication?
12. A Lambda function is configured to process messages from an SQS queue, but the function is not being triggered.  
What possible configuration issue might cause this problem?
13. An organization wants to ensure that uploaded files in S3 trigger notifications, store messages reliably, and process them asynchronously.  
Which serverless architecture using AWS services can achieve this?
14. A developer wants to monitor the processing of messages handled by a Lambda function triggered by SQS.  
Which AWS service can be used to view logs and troubleshoot the function execution?
15. A company wants to build a fully serverless event-driven system where file uploads automatically trigger notifications and background processing without managing servers.  
Which AWS services can be combined to build this architecture?

16. · A company wants to send a notification to multiple applications whenever a new order is placed in their system. Each application should receive the same message and process it independently.  
Which AWS service can be used to implement this publish-subscribe communication model?
17. · A developer configured an SNS topic and connected it to an SQS queue, but the messages published to SNS are not appearing in the queue.  
What configuration or permission might be missing in this setup?
18. · An application generates a large number of events, and the development team wants to ensure that these events are stored safely until they are processed by another service.  
Which AWS service can be used to reliably store and deliver these messages?
19. · A company uploads files to an S3 bucket and wants to process these files using Lambda, but only after the messages are safely stored to avoid data loss during high traffic.  
Which AWS architecture using SNS and SQS can help achieve this requirement?
20. · A developer notices that messages remain in an SQS queue even after being processed by a Lambda function.  
What configuration or step might be missing to ensure messages are removed after processing?

ELB ASG

## WEEK-9- Implementation of Elastic Load Balancer with Auto Scaling using AWS EC2 Instances.

### Elastic Load Balancer (ALB)

#### Step 1: Launch Two EC2 Instances

##### 1.1 Launch EC2 Instance #1

1. Go to **AWS EC2 Console** → **Launch Instance**.
2. **Name**: webserver-1
3. Choose an **AMI** (Amazon Linux 2).
4. Select **Instance Type** (e.g., t2.micro).
5. Add **Storage** (default is fine).
6. Configure **Security Group**:
  - Allow **SSH (22)** from your IP
  - Allow **HTTP (80)** from anywhere (0.0.0.0/0)
7. **Launch** → Select/Create Key Pair → Launch.

**Run these commands after connecting VM**

```
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd

echo "This is Server 1" > /var/www/html/index.html
```

---

✓ Now you create second **EC2 instances** running Apache, with message **This is Servier**  
**2**each serving a simple web page.

Repeat it for second instance

```
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd

echo "This is Server 2" > /var/www/html/index.html
```

---

#### Step 2: Create a Load Balancer

AWS provides **Elastic Load Balancer (ELB)**. We will create an **Application Load Balancer (ALB)**.

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##### 2.1 Configure Security Group for Load Balancer

- Go to **EC2** → **Security Groups** → **Create Security Group**
    - Name: lb-sg
    - Allow **HTTP (80)** from anywhere
    - Optional: HTTPS (443) if using SSL
- 

## 2.2 Create the Load Balancer

1. Go to **EC2** → **Load Balancers** → **Create Load Balancer** → **Application Load Balancer**
  2. **Basic Configuration:**
    - Name: my-alb
    - Scheme: Internet-facing
    - IP address type: IPv4
  3. **Listeners:**
    - Default: HTTP (80)
- 

## 2.3 Configure Availability Zones

- Select the default **VPC** where your EC2 instances exist
  - Select **two /all subnets** (one for each instance if possible)
- 

## 2.4 Configure Security Groups

- Choose the **lb-sg** security group created earlier
- 

## 2.5 Configure Target Groups

1. **Create a new target group:**
    - Name: web-servers-tg
    - Target type: Instance
    - Protocol: HTTP
    - Port: 80
    - Health checks: / (default)
  2. **Register Targets:**
    - Select EC2 INSTANCES HERE ---**webserver-1** and **webserver-2**
    - Click **Include as pending** → **Register**
- 

## 2.6 Review and Create

- Review all settings → Click **Create Load Balancer**
  - AWS will take a few minutes to provision the ALB
- 

## Step 3: Verify Load Balancer

1. Go to **Load Balancers** → **SelectALB** → **Description** → **DNS name**
2. Copy the **DNS name** (something like my-alb-123456.us-east-1.elb.amazonaws.com)
3. Paste it in a browser → You should see either **WebServer-1** or **WebServer-2** page.
  - Refresh multiple times → Load balancer distributes traffic between the two instances

---

#### Step 4: Optional – Test Health Checks

- Go to **Target Groups** → **web-servers-tg** → **Targets**
- Ensure **Status** of both instances is healthy

If a server is unhealthy, check the bootstrapping script or security group rules.

---

### Step-by-Step: Auto Scaling Group with Load Balancer

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#### Step 1: Create an AMI from Existing EC2 Instance

1. Go to **EC2 Console** → **Instances**
2. Select the EC2 instance you already configured with bootstrapping
3. Click **Actions** → **Image and Templates** → **Create Image**
4. **Provide Image Name:** e.g., my-webserver-AMI
5. Optional: Add description
6. Click **Create Image**
7. Wait for the AMI to become **available** (status: available in **AMIs** section)

✓ This AMI will be used for launching new instances automatically in ASG.

---

#### Step 2: Create a Launch Template

1. Go to **EC2** → **instances->Launch Templates** → **Create Launch Template**
2. **Launch Template Name:** my-launch-template
3. **AMI:** Select the AMI created in Step 1 (my-webserver-AMI)
4. **Instance Type:** t2.micro (or your choice)
5. **Key Pair:** Select your existing key pair
6. **Security Groups:** Select the **Load Balancer Security Group** (allow HTTP 80)
7. **User Data:** Optional (already baked into AMI)
8. Click **Create Launch Template**

✓ Launch Template is ready for Auto Scaling.

---

#### Step 3: Create Auto Scaling Group (ASG)

1. Go to **EC2** → **Auto Scaling Groups** → **Create Auto Scaling Group**
  2. **Name:** webserver-asg
  3. **Launch Template:** Select my-launch-template
  4. **Template Version:** Select **Version 1**
  5. Click **Next**
- 

##### Step 3.1: Network Settings

1. **VPC:** Default VPC (or your existing VPC)
2. **Availability Zones (AZs):** Select **all** subnets (for high availability)
3. **Load Balancer:** Enable **Attach to an existing load balancer**
  - Select your **existing Load Balancer**

4. **Health Check:** HTTP
    - Set **Health Check Grace Period:** 300 seconds
  5. Click **Next**
- 

### Step 3.2: Configure Group Size

1. **Desired Capacity:** 2 (initial instances)
  2. **Minimum Capacity:** 1
  3. **Maximum Capacity:** 4
  4. Click **Next**
- 

### Step 3.3: Review and Create ASG

1. Review all settings
  2. Click **Create Auto Scaling Group**
- ✔ Auto Scaling Group is now ready and attached to your Load Balancer.
- 

### Step 4: Verify Load Balancer and Auto Scaling

1. Go to **Load Balancers** → **Select your Load Balancer**
  2. Click **Instances** → **Target Group** → **Registered Targets**
    - You will see **2 healthy instances** initially
  3. Check **Auto Scaling Group** → **Instances**
    - You may see up to **4 instances** if scaling triggers
  4. Test **auto recovery**:
    - **Terminate one of the instances**
    - After a few seconds, **ASG will launch a new instance automatically**
    - Traffic will automatically be **distributed by the Load Balancer**
- 

### Step 5: Test the Load Balancer

1. Copy the **Load Balancer DNS Name** from ALB
  2. Open in a browser → You should see your web page from one of the instances
  3. Refresh multiple times → traffic alternates between instances
- ✔ This confirms **Auto Scaling + Load Balancer** is working correctly.
- 

### Summary of Flow

1. **Create AMI** → snapshot of working EC2
  2. **Create Launch Template** → using AMI, security group
  3. **Create Auto Scaling Group** → attach Launch Template, set min/desired/max, attach Load Balancer
  4. **Verify instances** → ASG launches/replaces instances automatically
  5. **Test Load Balancer** → distributes traffic between all active instances
- 

### Scenario Based Questions (ELB & Auto Scaling)

- 1) A company hosts a web application on two EC2 instances, but sometimes one server becomes overloaded while the other remains idle.  
Which AWS service can distribute incoming traffic evenly across both servers?
- 2) A developer created multiple EC2 instances for a web application but wants users to access the application using a single entry point instead of multiple server IP addresses.  
Which AWS service can solve this problem?
- 3) An organization notices that during peak hours their application receives heavy traffic and servers become slow. During non-peak hours, many servers remain unused.  
Which AWS feature can automatically increase or decrease the number of servers based on demand?
- 4) A company deployed a load balancer but users are still getting errors because traffic is being sent to an unhealthy server.  
Which load balancer feature helps detect and stop sending traffic to failed servers?
- 5) A developer wants to automatically launch additional EC2 instances when CPU utilization exceeds 70%.  
Which AWS service configuration should be used to implement this requirement?
- 6) An application receives traffic from users worldwide, and the company wants to ensure that requests are distributed across multiple EC2 instances for high availability.  
How can Elastic Load Balancer help in this architecture?
- 7) A developer configured an Auto Scaling Group, but new instances are not launching even when traffic increases.  
What configuration might be missing or incorrect?
- 8) A company wants to maintain a minimum of two running servers at all times, but automatically add more servers when traffic increases.  
Which Auto Scaling configuration parameters should be defined?
- 9) A web application running on EC2 instances behind a load balancer suddenly crashes when traffic spikes.  
How can combining Elastic Load Balancer with Auto Scaling improve system reliability?
- 10) A developer created an Auto Scaling Group but forgot to attach it to a load balancer.  
What potential problem might occur in handling user requests?
- 11) A company wants to replace unhealthy EC2 instances automatically to maintain application availability.  
Which AWS feature can automatically detect and replace failed instances?
- 12) A developer wants to monitor the performance of EC2 instances and trigger scaling actions based on metrics such as CPU usage or network traffic.  
Which AWS service can provide these monitoring metrics?
- 13) A startup expects sudden spikes in website traffic during a promotional event and wants the infrastructure to automatically handle the load.  
How can Auto Scaling help manage this situation?
- 14) A developer wants to ensure that incoming user requests are routed to servers based on HTTP or HTTPS protocols.  
Which type of AWS load balancer should be used?

15) A company wants to build a highly available architecture where traffic is automatically balanced across multiple servers and new servers are launched when demand increases.

Which AWS services should be combined to achieve this solution?

elastic bean stack

## WEEK-10--Elasticbeanstalk

### Prerequisites

- A web application ready (e.g., .war, .zip, or code)

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h1>My first PHP page</h1>
```

```
<?php
```

```
echo "Hello World!";
```

```
?>
```

```
</body>
```

```
</html>
```

---

### Steps to Create Elastic Beanstalk Application

#### Step 1: Open Elastic Beanstalk

1. Login to AWS Console
2. Under Compute section ->Search for **Elastic Beanstalk**
3. Click **Create Application**

---

#### Step 2: Create Application

1. Enter **Application Name** (e.g., MyApp)
2. Add description (optional)
3. Click **Create**

---

#### Step 3: Create Environment

1. Click **Create Environment**
2. Choose:
  - **Web Server Environment** (for web apps)
3. Click **Select**

---

#### Step 4: Configure Environment

1. Enter **Environment Name**
2. Choose **Platform**:
  - Java (for .war)
  - Node.js / Python / PHP / etc.

3. Under **Application Code**:
    - Select **Upload your code**
    - Upload your .war or .zip file
- 

### Step 5: Configure Service Access

1. Service Role → Select existing -à**LabRole**
  2. EC2 Key Pair → Optional (for SSH)
  3. EC2 Instance Profile → select (e.g., LabInstanceProfile)
- 

### Step 6: Configure Network

1. Select **VPC** (default is fine)
  2. Select **Subnets**
  3. Enable **Public IP**
- 

### Step 7: Instance & Scaling Setup

1. Choose **Instance type** (e.g., t2.micro – free tier)
  2. Set:
    - Min instances: 1
    - Max instances: 2 (or more)
  3. Load Balancer → Enabled (optional but recommended)
- 

### Step 8: Review & Create

1. Click **Review**
  2. Click **Create**
- 

### Deployment Process

- AWS automatically creates:
    - EC2 instances
    - Load Balancer
    - Auto Scaling Group
    - Security Groups
- 

### Access Your Application

- After deployment:
    - You will get a **URL (domain link)**
    - Open it in browser → Your app will run
- 

### Updating Application

1. Go to your environment
  2. Click **Upload and Deploy**
  3. Upload new version
-



amazon LEX

### Step 1: Open Amazon Lex

1. Login to AWS Console
  2. Search **Amazon Lex**
  3. Click **Create bot**
- 

### Step 2: Create Bot

1. Choose:
    - **Create a blank bot**
  2. Enter:
    - Bot name: HotelBookingBot
  3. IAM role → Create new role
  4. Remaining default next
  5. Language: English
  6. Voice (optional)
  7. Click **Done**
- 

### Step 3: Create Intent

1. Go to **Intents**
  2. Name: BookHotel
  3. Click **Create**
- 

### Step 4: Add Sample Utterances

Add user sentences (how user talks):

I want to book a hotel

Book a room

Reserve hotel

I need a room

These are **Utterances**

---

### Step 5: Create Slots (User Inputs)

Slots collect user information.

## Add Slot

1. Click **Add slot**
2. Fill:
  - Slot name: age
  - Slot type: AMAZON.Number
  - Prompt: *Which city do you want?*

Mark as **Required**

## Add IF Conditions (Conditional Logic)

Used to control flow based on slot values

### Example Condition:

If {age} < 18

### Steps:

1. Go to **Intent** → **Slots** -> under **Advance options**
2. Scroll to **Slot capture: success response** -> **Conditional branching**
3. Click **Add condition**

### Example:

#### Condition:

If {age}< 18

#### Response:

Your not eligible for hoot booking?

---

## Step 6: Create Slots (User Inputs)

Slots collect user information.

## Add Slot

3. Click **Add slot**
4. Fill:
  - Slot name: location
  - Slot type: AMAZON.City
  - Prompt: *Which city do you want?*
5. Mark as **Required**

Repeat for other slots.

1. Click **Add slot**
  2. Fill details:
    - **Slot name:**checkin
    - **Slot type:**AMAZON.Date
    - **Prompt:**  
*"What is your check-in date?"*
  3. Mark as **Required**  
  
(Optional but recommended)
    - Add **Retry prompt:**  
*"Please provide a valid date (e.g., 2026-03-25)"*
  4. Click **Save**
- 

### Slot 7: Number of Nights (nights)

#### Steps:

1. Click **Add slot** again
  2. Fill details:
    - **Slot name:** nights
    - **Slot type:**AMAZON.Number
    - **Prompt:**  
*"How many nights will you stay?"*
  3. Mark as **Required**
  4. Click **Save**
- 

### Step 7: Create Custom Slot Type

#### Save intent

#### Click on Back to intents

If you want custom values:

1. Go to **Slot Types**
2. Click **Add slot type ->add blank slot type**
3. Name: RoomType

Add values:

Single

Double

Suite

Save slot type

Now use this in slot: go back to intent

- Slot name: RoomType
  - Slot type: RoomType
- 

### **Step 9: Add Response Cards (Card Groups)**

Used for buttons (quick replies)

#### **Steps:**

1. Go to **slot prompts** -> **more promptoptions**
2. Click **Add** -> **add card groups**

#### **Example:**

**Card Title:** Select Room Type  
**click on add Buttons:**

- Single → "Single"
  - Double → "Double"
  - Suite → "Suite"
- 

### **Step 10: Configure Responses**

#### **Initial Response:**

Welcome to Hotel Booking! What is your name?

#### **Confirmation:**

Do you want to confirm booking in {location} for {nights} nights?

---

### **Step 11: Build & Test**

1. Click **Build**
2. Use **Test chatbot** panel
3. Try:

Book a hotel

---

### Sample output

User: Book a hotel

Bot: What is your age?

User: 25

Bot: Which city do you want?

User: Hyderabad

Bot: Check-in date?

User: Tomorrow

Bot: Number of nights?

User: 2

Bot: Select room type

User: Double

Bot: Do you want to confirm booking in Hyderabad for 2 nights?

User: Yes

Bot: Booking confirmed.

---

## SCENARIO BASED QUESTIONS

### Scenario 1: Bot Not Responding

You created a chatbot, but when you test it, the bot does not respond to user input.

- What could be the possible reason for no response?
  - Did you build and publish the bot correctly?
  - How would you troubleshoot this issue?
- 

### Scenario 2: Incorrect Intent Matching

A user types a valid query, but the chatbot responds with the wrong intent.

- What could cause incorrect intent recognition?
  - How do utterances affect intent matching?
  - How can you improve accuracy?
- 

### Scenario 3: Missing Slot Values

The chatbot asks for user input, but fails to capture required slot values.

- What is a slot in Amazon Lex?
- Why might slot values not be captured properly?
- How would you fix slot configuration issues?

---

#### Scenario 4: Fulfillment Not Working

The bot collects all user inputs but does not perform the expected action.

- What is fulfillment in Amazon Lex?
- Could Lambda integration be misconfigured?
- How would you verify the backend processing?

---

#### Scenario 5: Bot Not Accessible

You created the bot successfully, but users are unable to access it.

- Did you deploy or publish the bot?
- Could permissions or channel integration be missing?
- How would you make the bot accessible to users?

---

#### Scenario 6: Wrong Language Understanding

The chatbot fails to understand user queries written in slightly different formats.

- How does Amazon Lex process natural language?
- Why is training with multiple utterances important?
- How can you improve NLP performance?

---

#### Scenario 7: Versioning Issue

You updated the bot, but users still experience the old behavior.

- Did you create a new version of the bot?
- Why is alias configuration important?
- How do you ensure users access the latest version?

---

#### Scenario 8: Lambda Permission Error

The chatbot is connected to a Lambda function, but it fails during execution.

- What type of permission error might occur?
- How are IAM roles used in this integration?
- How would you fix the permission issue?

---

#### Scenario 9: Bot Gives Default Response

The chatbot always gives a fallback/default response instead of proper answers.

- What is a fallback intent?
- Why does the bot fail to match user input?
- How can you improve intent recognition?

---

#### Scenario 10: Multi-turn Conversation Issue

The chatbot fails to continue conversation properly after asking follow-up questions.

- What is dialog management in Amazon Lex?
  - How are slots used in multi-turn conversations?
  - How can conversation flow be improved?
- 

#### Scenario 11: Integration with Web Application

A company wants to integrate the chatbot into their website.

- How can Amazon Lex be integrated with web applications?
  - What services or APIs are required?
  - What challenges might occur during integration?
- 

#### Scenario 12: Performance Issue

The chatbot responds slowly when many users interact simultaneously.

- What could cause performance delays?
- How can scalability be handled in Amazon Lex?
- What AWS services can support high traffic handling?

IAM- users

## WEEK-12

### IAM

#### 1. Administrative Setup (Root User)

Step 1: Log in to the AWS Management Console as the Root User.

Step 2: Search for IAM under the "Security, Identity, & Compliance" category.

Step 3: Navigate to Users and click Create user.

Step 4: User Details: Enter a name (e.g., S3\_Specialist). Check the box for "Provide user access to the AWS Management Console." – custom password.

Step 5: Set Permissions: Select "Attach policies directly." Search for and select AmazonS3FullAccess.

Step 6: Review & Create: Complete the wizard. On the final page, click Download .csv file.

#### Now you see CC-LAB user created

Step 7: Copy your 12-digit Account ID (found in the top-right dropdown or the .csv) and Sign Out.

Step 8: Use the Sign-in URL from the .csv (or go to the AWS sign-in page and select IAM User).

Step 9: Enter the Account ID, then the IAM Username and Password from the .csv.

**Test 1 (Failure):** Navigate to **EC2**. Try to view instances. You will see "API Error" or "Access Denied." This proves the user is restricted.

**Test 2 (Success):** Navigate to **S3**. Create a new bucket. It will work perfectly because of the attached policy.

#### Part B: CLI Access (Programmatic Access)

Objective: Access AWS resources from a standalone system (CMD/Terminal) using security keys.

##### 1. Credential Generation

Step 1: While logged in as an Admin/Root, go to IAM > Users and click on your user.

Step 2: Go to the Security credentials tab. Scroll to Access keys and click Create access key.

Step 3: Select Command Line Interface (CLI). Acknowledge the warning and click Next.

Step 4: Download the .csv file. > Critical: This contains the Access Key ID and Secret Access Key. If lost, you must delete and recreate them.

## 2. Standalone System Configuration

Step 5: Install the AWS CLI Tool from google on your computer

Download <https://awscli.amazonaws.com/AWSCLIV2.msi>

Step 6: Open your Command Prompt (CMD) or Terminal.

Step 7: Type aws configure and press Enter.

Step 8: Provide the following from your .csv: Access Key ID  
Secret Access Key

Default region: (e.g., ap-south-1)

Default output: json

Run aws ec2 describe-nstances.

Result: You will get an (AccessDenied) error.

This confirms that the CLI is respecting the same IAM policy as the GUI.

## 3. Verification via Commands

Step 9: Run aws s3 ls. This should list your buckets

Step 10: Run aws s3 mb s3://your-unique-bucket-name. (Success)

Check in console you see the bucket created from CLI

IAM - ec2

# Roles

Procedure: Attaching a Role to an EC2 Instance

1. **Create the Role:** Go to IAM > Roles > Create role
2. Select Trusted Entity: Choose AWS Service.
3. Under the "Service or use case" dropdown, select EC2. (This tells AWS: "I trust the EC2 service to use this role").
4. Add Permissions: Search for and check AmazonS3FullAccess.
5. Name & Review: Name it (e.g., EC2-S3-ReadOnly-Role) and click Create role.
6. Role has been created
7. Create the ec2 instance and and connect to it and check for aws ec2 describe-instances
8. Attach to Instance:
9. Go to the EC2 Dashboard. Select your running instance.
10. Click Actions > Security > Modify IAM role.
11. .Select the role you just created and click Update IAM role
12. Verify via CLI:
13. SSH/Connect into your EC2 instance. Type `aws s3 ls`.
14. The Result: The command works immediately.
15. We can here role has only s3 full access only here--  
`aws ec2 describe-instances`
16. You did not have to provide an Access Key or Secret Key. AWS automatically provided.

# SCREENSHOTS



LAMBDA1

Launch AWS Academy Learner | x Service | Amazon DynamoDB M x +

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#service

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022~vrushankmit@gmail.com

## DynamoDB

- Dashboard
- Tables
- Explore items
- PartiQL editor
- Backups
- Exports to S3
- Imports from S3
- Integrations
- Reserved capacity
- Settings

### ▼ DAX

- Clusters
- Subnet groups

# Amazon DynamoDB

## A fast and flexible NoSQL database service for any scale

DynamoDB is a fully managed, key-value, and document database that delivers single-digit-millisecond performance at any scale.

### Get started

Create a new table to start exploring DynamoDB.

Create table

### Pricing

DynamoDB charges for reading, writing, and storing data in your DynamoDB tables, along with any optional features you choose to turn on. DynamoDB has on-demand capacity

### How it works

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Launch AWS Academy Learner | x List tables | Amazon DynamoDB x +

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#tables

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022~vrushankmit@gmail.com

## DynamoDB

- Dashboard
- Tables
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- Imports from S3
- Integrations
- Reserved capacity
- Settings

### ▼ DAX

- Clusters
- Subnet groups

We are updating the infrastructure to improve the DynamoDB console experience. We have identified that your current setup might cause permission issues when access the Management Console using VPC/PrivateLink. 1) Please update your identity-based or resource-based policy to use both aws:SourceIp and aws:SourceVpc to control your VPC related access 2) Onboard to the AWS Management Console Private Access feature to access the AWS Management Console through a VPC endpoint. 3) Ensure VPN and firewall configuration allows /api-proxy/prod/query access. [Learn more about resolution paths](#). After October 15, 2024, the old console experience will be deprecated.

## Tables

Find tables

Filter by tag Any tag key

Filter by tag value Any tag va...

Loading...

Actions Delete Create table

Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read c
Loading tables								

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Launch AWS Academy Learner | x Create table | Amazon DynamoDB | x +

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#create-table

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankmit@gmail.com

## Create table

**Table details** [info](#)

DynamoDB is a schemaless database that requires only a table name and a primary key when you create the table.

**Table name**  
This will be used to identify your table.

Table1

Between 3 and 255 characters, containing only letters, numbers, underscores (\_), hyphens (-), and periods (.).

**Partition key**  
The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.

unique String

1 to 255 characters and case sensitive.

**Sort key - optional**  
You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.

Enter the sort key name String

1 to 255 characters and case sensitive.

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Type here to search

Launch AWS Academy Learner | x View table | Amazon DynamoDB | x +

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#table?name=Table1

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankmit@gmail.com

DynamoDB <

- Dashboard
- Tables**
- Explore items
- PartiQL editor
- Backups
- Exports to S3
- Imports from S3
- Integrations
- Reserved capacity
- Settings

▼ DAX

- Clusters
- Subnet groups

**Tables (1)** <

Filter by tag Any tag key

Filter by tag value Any tag value

Find tables

< 1 > ⚙️

Table1 ☆

## Table1

Last updated February 17, 2026, 12:13 (UTC+5:30) ⌛ ⭐ Actions Explore table items

< Settings Indexes Monitor Global tables Backups Exports >

**Protect your DynamoDB table from accidental writes and deletes** Edit PITR ✕

When you turn on point-in-time recovery (PITR), DynamoDB backs up your table data automatically so that you can restore to any given second in the preceding 1 to 35 days. Additional charges apply. [Learn more](#)

**General information** [info](#) [Get live item count](#)

**Partition key**  
unique (String)

**Capacity mode**  
[On-demand](#)

**Alarms**

**Sort key**  
-

**Table status**  
Active

**Point-in-time recovery (PITR)** [info](#)

https://us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#item-explorer?maximize=true&table=Table1 © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

Type here to search

Launch AWS Academy Learner | x Items | Amazon DynamoDB M... x +

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#item-explorer?maximize=true&table=Table1

Account ID: 8457-5218-6127  
voclabs/user4578022~vrushankmit@gmail.com

DynamoDB > Explore items > Table1

Run Reset

Completed - Items returned: 0 - Items scanned: 0 - Efficiency: 100% - RCUs consumed: 2

Table: Table1 - Items returned (0)

Scan started on February 17, 2026, 12:14:23

unique (String)

No items

No items to display.

Create item

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Type here to search

Launch AWS Academy Learner | x Create item | Amazon Dynamol... x +

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#edit-item?itemMode=1&route=ROUTE\_ITEM\_EXPLORER&table=Table1

Account ID: 8457-5218-6127  
voclabs/user4578022~vrushankmit@gmail.com

DynamoDB > Explore items: Table1 > Create item

### Create item

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Form JSON view

Attributes

Attribute name	Value	Type
unique - Partition key	Emp1	String

Add new attribute

- String
- Number
- Boolean
- Binary
- Null
- String set
- Number set
- Binary set
- List
- Map

CloudShell Feedback

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Type here to search

Launch AWS Academy Learner

Create item | Amazon Dynamol

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#edit-item?itemMode=1&route=ROUTE\_ITEM\_EXPLORER&table=Table1

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankkmit@gmail.com

DynamoDB

Explore items: Table1

Create item

Attributes

Add new attribute

Attribute name	Value	Type	Remove
unique - Partition key	Emp1	String	
Name	abcd	String	Remove
Phone	12345	Number	Remove
Employeeer	<input checked="" type="radio"/> True <input type="radio"/> False	Boolean	Remove
Description	CEO	String	Remove

Cancel

Create item

CloudShell

Feedback

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Type here to search

12:16 PM  
17-02-2025

Launch AWS Academy Learner | x Items | Amazon DynamoDB M... x +

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#item-explorer?table=Table1

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankkmit@gmail.com

DynamoDB > Explore items > Table1

**DynamoDB**

- Dashboard
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- Imports from S3
- Integrations
- Reserved capacity
- Settings

▼ DAX

- Clusters
- Subnet groups

Table1

Filters - optional

Run Reset

Completed - Items returned: 0 - Items scanned: 0 - Efficiency: 100% - RCUs consumed: 2

Table: Table1 - Items returned (1)

Scan started on February 17, 2026, 12:14:23

Actions Create item

unique (String) Description Employee Name Pho

Emp1 CEO true abcd 123

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Launch AWS Academy Learner | x Create item | Amazon Dynamol... x +

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#edit-item?itemMode=1&route=ROUTE\_ITEM\_EXPLORER&table=Table1

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankkmit@gmail.com

DynamoDB > Explore items: Table1 > Create item

**Create item**

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Form JSON view

Add new attribute

Attributes

Attribute name	Value	Type	Remove
unique - Partition key	Emp2	String	
Name	XYZ	String	Remove
Phone	55555	Number	Remove
Employeeer	<input checked="" type="radio"/> True <input type="radio"/> False	Boolean	Remove
Description	Assistant	String	Remove

CloudShell Feedback

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## Search All phone no. by scan

The screenshot shows the AWS DynamoDB console interface. The left sidebar contains navigation links for Dashboard, Tables, Explore Items, PartiQL editor, Backups, Exports to S3, Imports from S3, Integrations, Reserved capacity, and Settings. The main area is titled 'Table1' and shows a 'Scan' operation. The 'Scan' button is selected, and the 'Query' button is disabled. The 'Select a table or index' dropdown is set to 'Table - Table1'. The 'Select attribute projection' dropdown is set to 'Specific attributes'. The 'Specific attributes to project' field contains 'Phone'. The 'Run' button is highlighted. Below the scan operation, a green status bar indicates 'Completed - Items returned: 2 - Items scanned: 2 - Efficiency: 100% - RCUs consumed: 2'. The results table shows two items with the attribute 'Phone'.

Phone
55555
12345

## Search phone no by query

The screenshot shows the AWS DynamoDB console interface. The left sidebar contains navigation links for Dashboard, Tables, Explore Items, PartiQL editor, Backups, Exports to S3, Imports from S3, Integrations, Reserved capacity, and Settings. The main area is titled 'Table1' and shows a 'Query' operation. The 'Query' button is selected, and the 'Scan' button is disabled. The 'Select a table or index' dropdown is set to 'Table - Table1'. The 'Select attribute projection' dropdown is set to 'Specific attributes'. The 'Specific attributes to project' field contains 'Phone'. The 'Partition key' section shows the attribute 'unique' with the value 'Emp1'. The 'Add filter' button is highlighted.

## Edit datatype of attribute

Launch AWS Academy Learner

Items | Amazon DynamoDB

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#item-explorer?maximize=true&operation=SCAN&pkAttr1Name=unique&pkAttr1Value=...

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankkmit@gmail.com

DynamoDB

Explore items

Table1

DynamoDB

Dashboard

Tables

Explore items

PartiQL editor

Backups

Exports to S3

Imports from S3

Integrations

Reserved capacity

Settings

DAX

Clusters

Subnet groups

Filters

Run

Reset

Completed · Items returned: 2 · Items scanned: 2 · Efficiency: 100% · RCUs consumed: 2

Table: Table1 - Items returned (1/2)

Scan started on February 17, 2026, 12:30:00

Edit item

Duplicate item

Delete items

Download selected items to CSV

Download results to CSV

	unique (String)	De	Name	Pho
<input type="checkbox"/>	<a href="#">Emp2</a>	As	YZ	555
<input checked="" type="checkbox"/>	<a href="#">Emp1</a>	CEO	true	abcd 123

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Type here to search

Launch AWS Academy Learner | Edit item | Amazon DynamoDB

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#edit-item?itemMode=2&pk=Emp1&route=ROUTE\_ITEM\_EXPLORER&table=Table1

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankmit@gmail.com

DynamoDB > Explore items: Table1 > Edit item

### Edit item

You can add, remove, or edit the attributes of an item. You can nest attributes inside other attributes up to 32 levels deep. [Learn more](#)

Form JSON view

Attributes View DynamoDB JSON

```
1 {
2 "unique": {
3 "S": "Emp1"
4 },
5 "Description": {
6 "S": "CEO"
7 },
8 "Employee": {
9 "BOOL": true
10 },
11 "Name": {
12 "S": "abcd"
13 },
14 "Phone": {
15 "N": "12345"
16 }
17 }
```

Launch AWS Academy Learner | Edit item | Amazon DynamoDB

us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#edit-item?itemMode=2&pk=Emp1&route=ROUTE\_ITEM\_EXPLORER&table=Table1

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankmit@gmail.com

DynamoDB > Explore items: Table1 > Edit item

### Edit item

Attributes View DynamoDB JSON

```
1 {
2 "unique": {
3 "S": "Emp1"
4 },
5 "Description": {
6 "S": "CEO"
7 },
8 "Employee": {
9 "S": "Yes"
10 },
11 "Name": {
12 "S": "abcd"
13 },
14 "Phone": {
15 "N": "12345"
16 }
17 }
```

Open Lambda

Launch AWS Academy Learner | Items | Amazon DynamoDB M... | us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#item-explorer?table=Table1

aws | Search: lambda | Ask Amazon Q | United States (N. Virginia) | Account ID: 8457-5218-6127 | voclabs/user4578022=vrushankmit@gmail.com

**DynamoDB**

- Dashboard
- Tables
- Explore items
- PartiQL editor
- Backups
- Exports to S3
- Imports from S3
- Integrations
- Reserved capacity
- Settings

**DAX**

- Clusters
- Subnet groups

**Services**

- Lambda**  
Run code without thinking about servers
- CodeBuild**  
Build and Test Code
- AWS Signer**  
Ensuring trust and integrity of your code

**Features**

- Lambda Insights**  
CloudWatch feature
- Object Lambda Access Points**

Were these results helpful?

VCUs consumed: 2

Name	Photo
XYZ	555
abcd	123

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12:33 PM 17-02-2026

Launch AWS Academy Learner | Functions | Lambda | us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions

aws | Search | [Alt+S] | United States (N. Virginia) | Account ID: 8457-5218-6127 | voclabs/user4578022=vrushankmit@gmail.com

**Lambda**

- Dashboard
- Applications
- Functions

**Additional resources**

- Capacity providers [New](#)
- Code signing configurations
- Event source mappings
- Layers
- Replicas

**Related AWS resources**

- Step Functions state machines

Documentation [L](#) Serverless workshops [New](#)

**Functions (5)** Last fetched 2/17/2026, 12:34:25 PM

Search by attributes or search by keyword

<input type="checkbox"/>	Function name	Description	Package type	Runtime	Type	Last modified
<input type="checkbox"/>	<a href="#">RoleCreationFunction</a>	Create SLR if absent	Zip	Python 3.10	Standard	1 month ago
<input type="checkbox"/>	<a href="#">RedshiftOverwatch</a>	Deletes Redshift Cluster if the count is more than 2.	Zip	Python 3.10	Standard	1 month ago
<input type="checkbox"/>	<a href="#">ModLabRole</a>	updates LabRole to allow it to assume itself	Zip	Python 3.10	Standard	1 month ago
<input type="checkbox"/>	<a href="#">MainMonitoringFunction</a>	-	Zip	Python 3.10	Standard	1 month ago
<input type="checkbox"/>	<a href="#">RedshiftEventSubscription</a>	Create Redshift event subscription to SNS Topic.	Zip	Python 3.10	Standard	1 month ago

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12:35 PM 17-02-2026

Launch AWS Academy Learner | x Create function | Functions | La x +

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/create/function?intent=authorFromScratch

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankmit@gmail.com

Lambda > Functions > Create function

### Create function [Info](#)

Choose one of the following options to create your function.

☒ **Author from scratch**  
Start with a simple Hello World example.

☐ **Use a blueprint**  
Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**  
Select a container image to deploy for your function.

#### Basic information

**Function name**  
Enter a name that describes the purpose of your function.

New Function for Dynamo

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (\_).

**Runtime** [Info](#)  
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.14

**Durable execution** - [new](#) [Info](#)  
Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#)

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Launch AWS Academy Learner | x Create function | Functions | La x +

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/create/function?intent=authorFromScratch

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankmit@gmail.com

Lambda > Functions > Create function

by default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

#### ▼ Change default execution role

**Execution role**  
Choose a role that defines the permissions of your function. To create a custom role, go to the [IAM console](#).

☐ Create a new role with basic Lambda permissions

☒ Use an existing role

☐ Create a new role from AWS policy templates

**Existing role**  
Choose an existing role that you've created to be used with this Lambda function. The role must have permission to upload logs to Amazon CloudWatch Logs.

LabRole

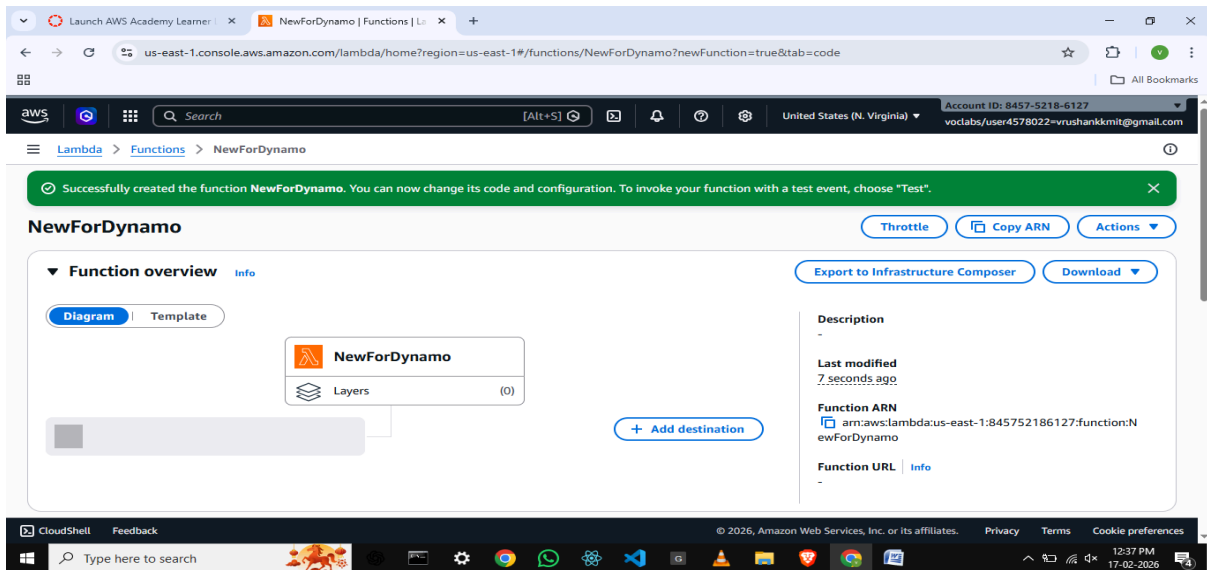
[View the LabRole role](#) on the IAM console.

#### ▶ Additional configurations

Use additional configurations to set up networking, security, and governance for your function. These settings help secure and customize your Lambda function deployment.

[Cancel](#) [Create function](#)

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import json

def lambda\_handler(event, context):

    a = event.get('num1', 0)

    b = event.get('num2', 0)

    result = a + b

    return {

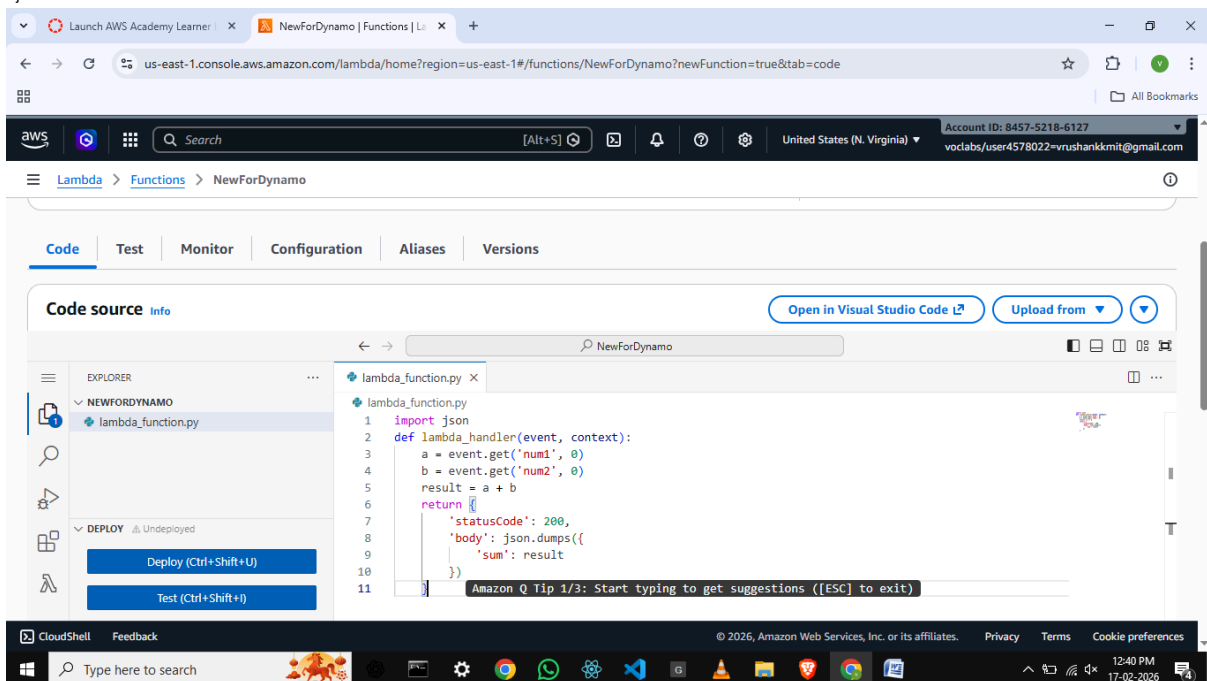
        'statusCode': 200,

        'body': json.dumps({

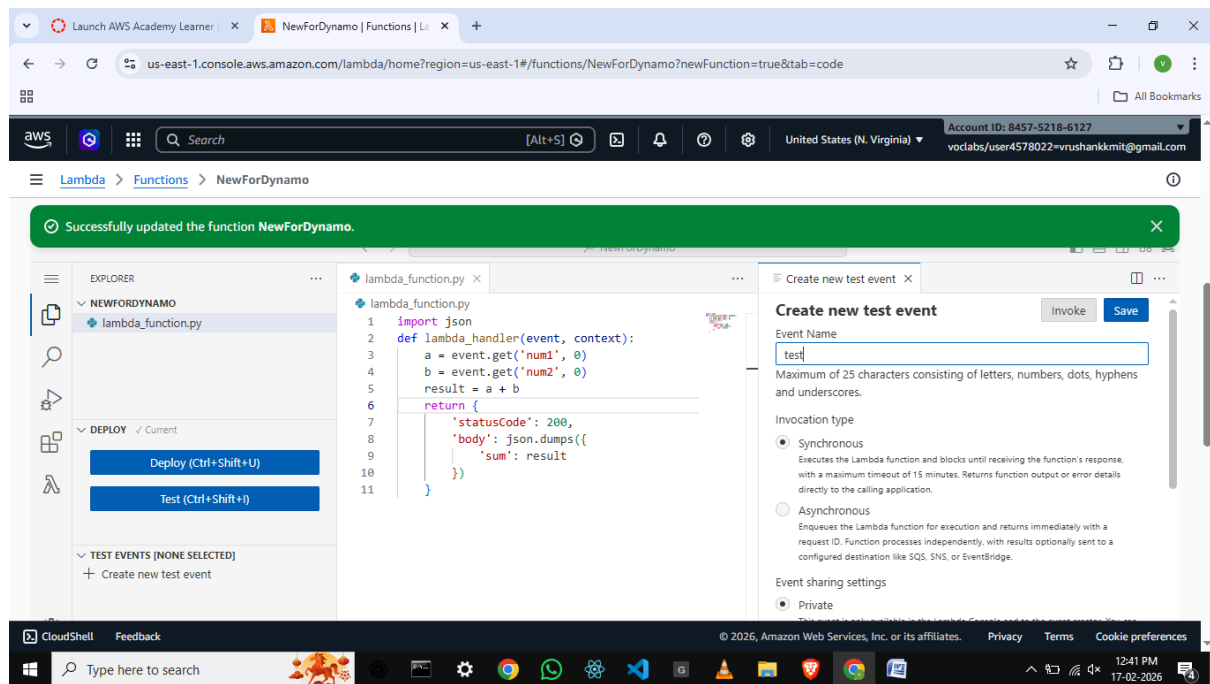
            'sum': result

        })

    }



Click deploy and test



Test again



Link S3 bucket and DynamoDB

import boto3

from uuid import uuid4

def lambda\_handler(event, context):

s3 = boto3.client("s3")

dynamodb = boto3.resource("dynamodb")

dynamoTable = dynamodb.Table("-----tablename-----")

# Check if the 'Records' key is present in the event

if "Records" in event:

for record in event["Records"]:

```

bucket_name = record["s3"]["bucket"]["name"]
object_key = record["s3"]["object"]["key"]
size = record["s3"]["object"].get("size", -1)
event_name = record.get("eventName", "Unknown")
event_time = record.get("eventTime", "Unknown")

```

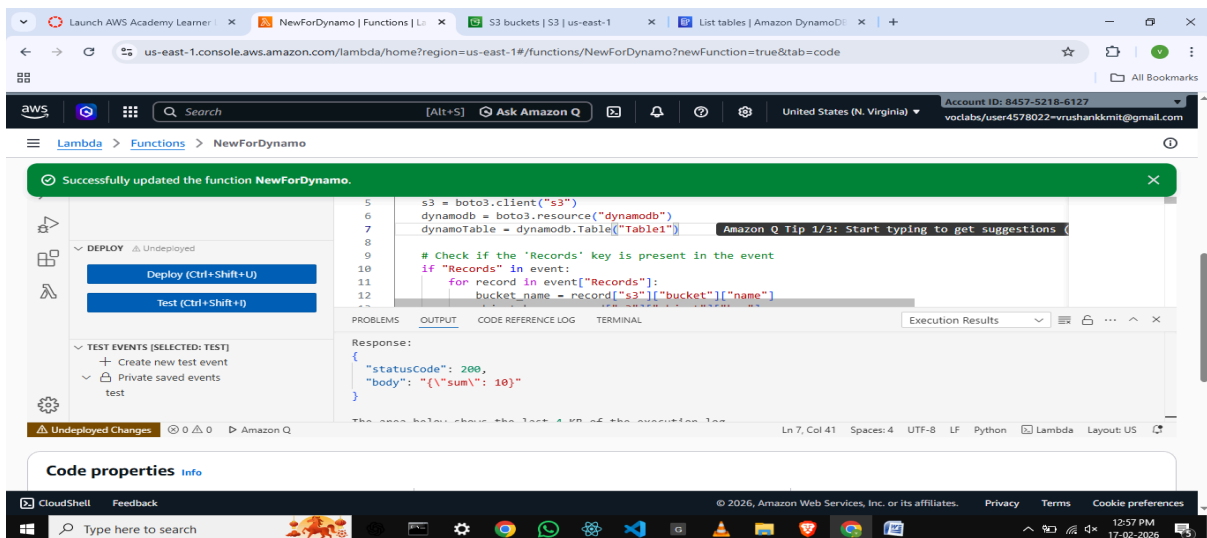
```

dynamoTable.put_item(
 Item={
 "unique": str(uuid4()),
 "Bucket": bucket_name,
 "Object": object_key,
 "Size": size,
 "Event": event_name,
 "EventTime": event_time,
 }
)

```

else:

```
print("No 'Records' key found in the event.")
```



## Add trigger

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/NewForDynamo?newFunction=true&tab=code

Successfully updated the function **NewForDynamo**.

## NewForDynamo

Throttle Copy ARN Actions

### Function overview

Diagram Template

**NewForDynamo**

Layers (0)

+ Add trigger + Add destination

**Description**

**Last modified**  
1 minute ago

**Function ARN**  
arn:aws:lambda:us-east-1:845752186127:function:NewForDynamo

**Function URL** Info

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us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/add...

## Add trigger

### Trigger configuration

S3  
aws asynchronous storage

**Bucket**  
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.

Search Bucket

Bucket must be in region us-east-1

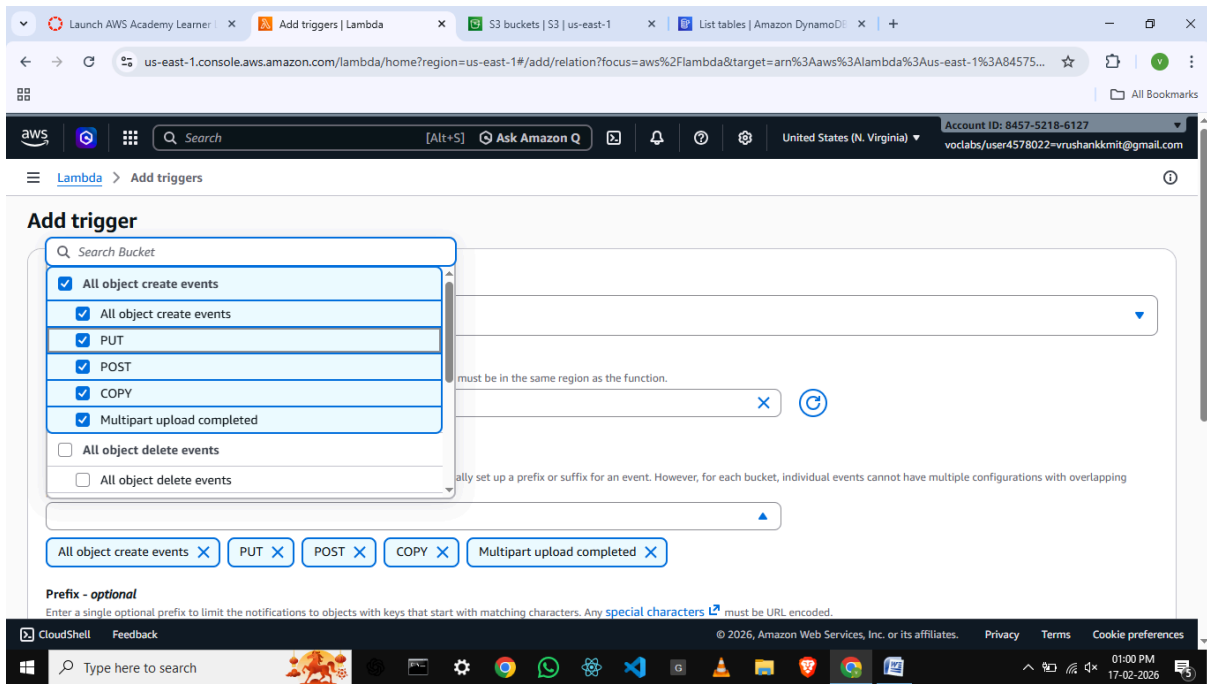
**Event types**  
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.

All object create events

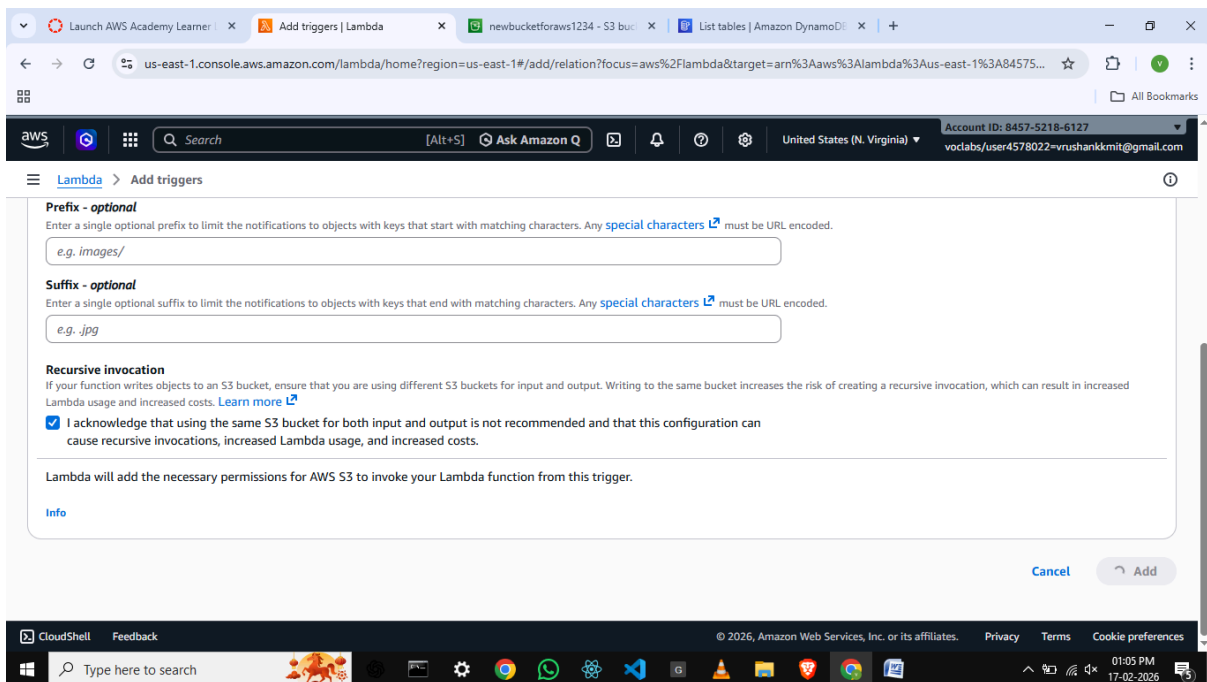
**Prefix - optional**  
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any special characters must be URL encoded.

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## Bucket should have versioning enable



us-east-1.console.aws.amazon.com/s3/upload/newbucketforaws1234?region=us-east-1

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankkmit@gmail.com

Amazon S3 > Buckets > newbucketforaws1234 > Upload

### Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

#### Files and folders (1 total, 19.0 B)

All files and folders in this table will be uploaded.

<input type="checkbox"/>	Name	Folder	Type	Size
<input type="checkbox"/>	share.txt	-	text/plain	19.0 B

#### Destination Info

Destination  
[s3://newbucketforaws1234](#)

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us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#item-explorer?maximize=true&table=Table1

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankkmit@gmail.com

DynamoDB > Explore items > Table1

**DynamoDB**

- Dashboard
- Tables
- Explore items**
- PartiQL editor
- Backups
- Exports to S3
- Imports from S3
- Integrations
- Reserved capacity
- Settings

**DAX**

- Clusters
- Subnet groups

**Run** **Reset**

**Completed** · Items returned: 3 · Items scanned: 3 · Efficiency: 100% · RCUs consumed: 2

#### Table: Table1 - Items returned (3)

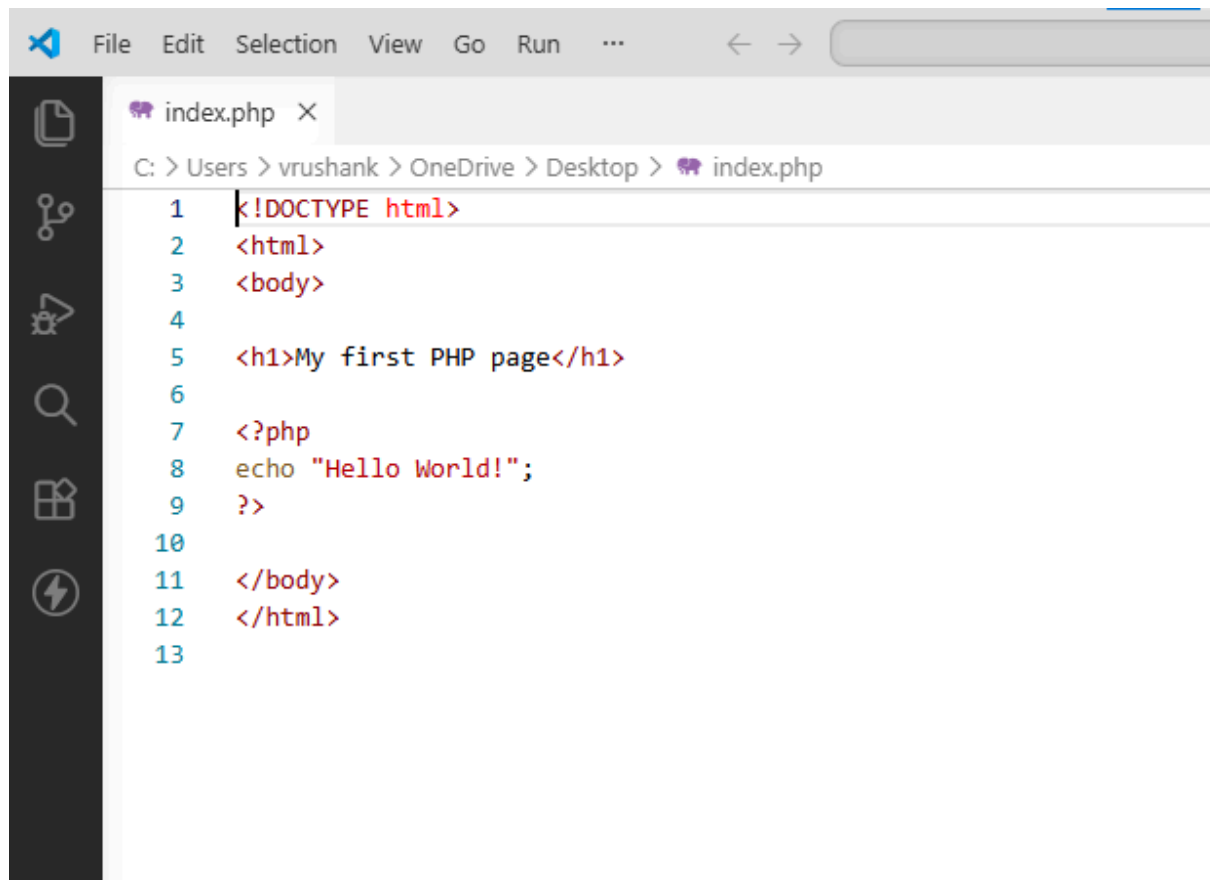
Scan started on February 17, 2026, 13:07:52

<input type="checkbox"/>	unique (String)	Bucket	Description	Employee	Even
<input type="checkbox"/>	<a href="#">Emp2</a>		Assistant	true	
<input type="checkbox"/>	<a href="#">Emp1</a>		CEO	Yes	
<input type="checkbox"/>	<a href="#">68348c86-2ae8-4728...</a>	newbucketf...			Obj

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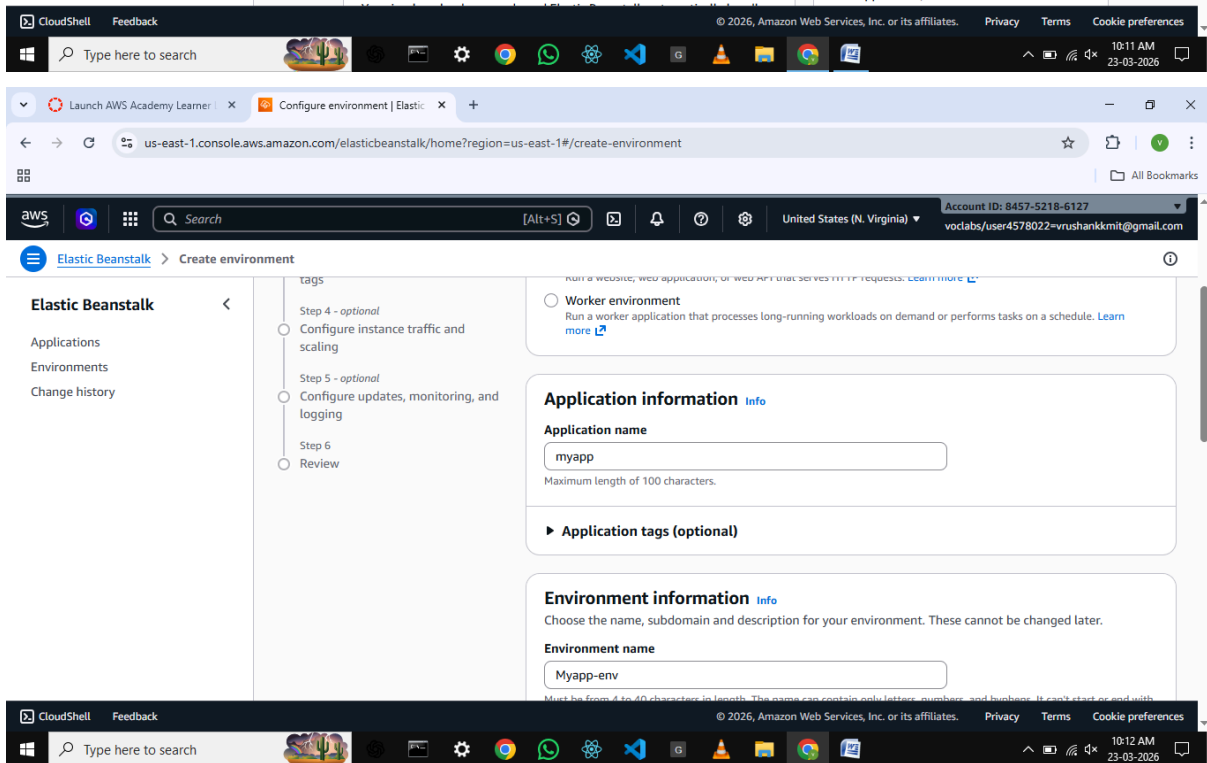
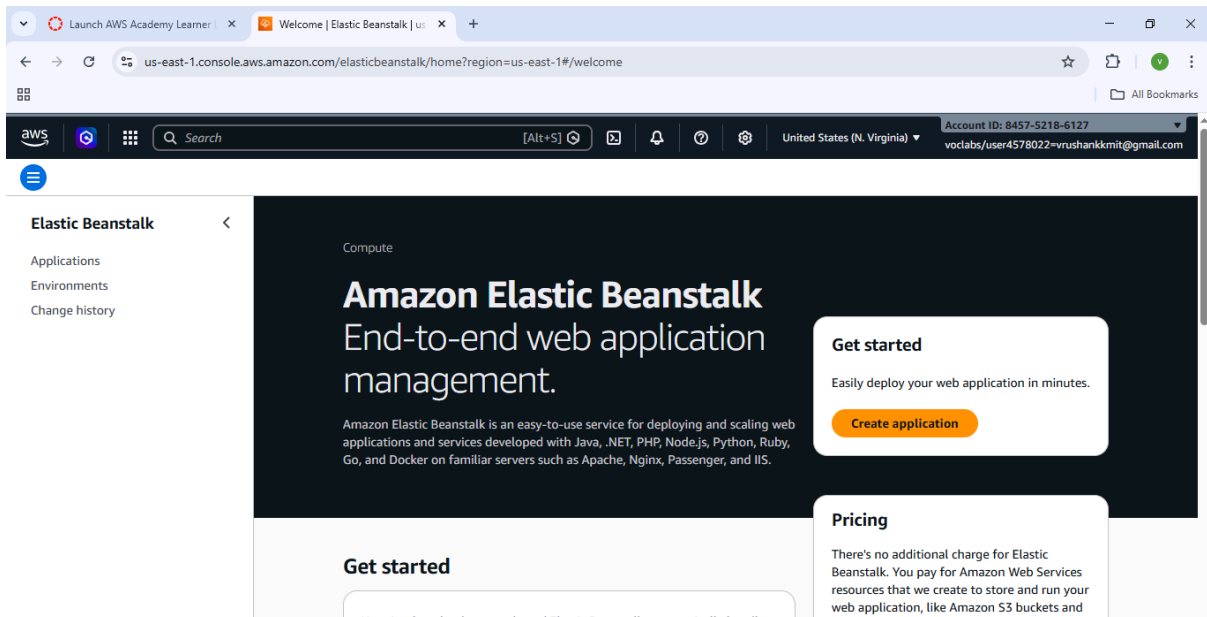
# ELASTIC BEANSTACK

Create index.php and compress the folder

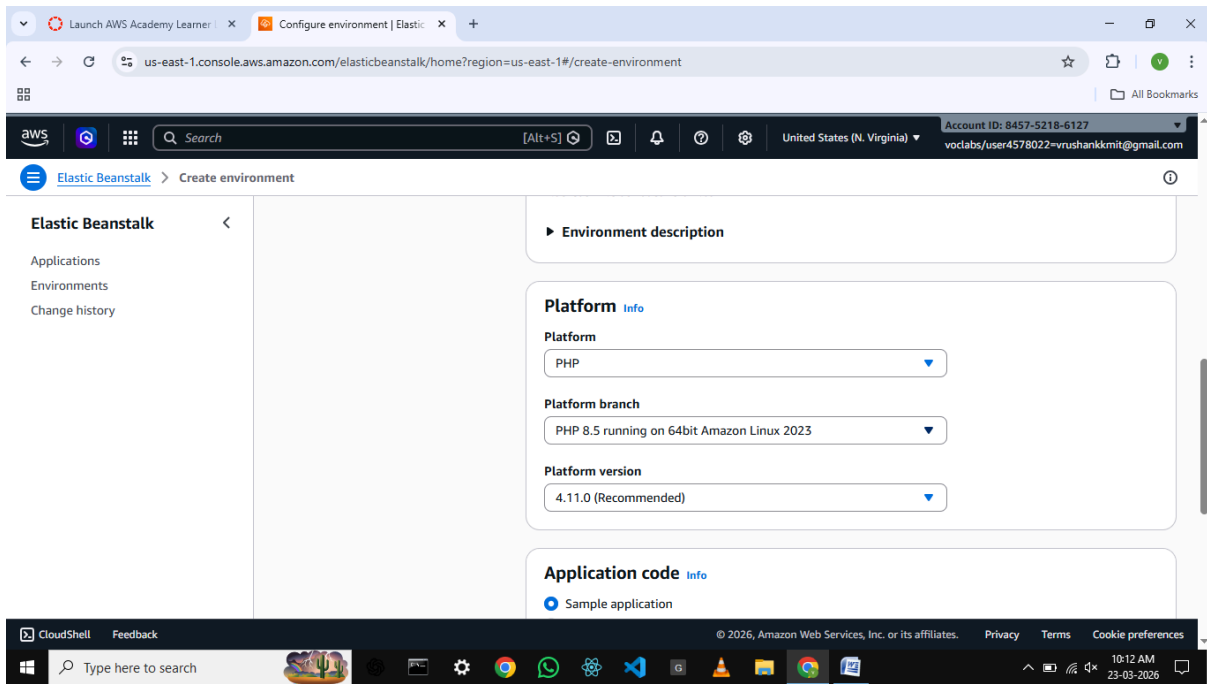


The screenshot shows the Visual Studio Code editor interface. The top menu bar includes 'File', 'Edit', 'Selection', 'View', 'Go', 'Run', and a search icon. The left sidebar contains icons for Explorer, Search, Run and Debug, Source Control, Extensions, and Testing. The main editor area displays a file named 'index.php' with the following code:

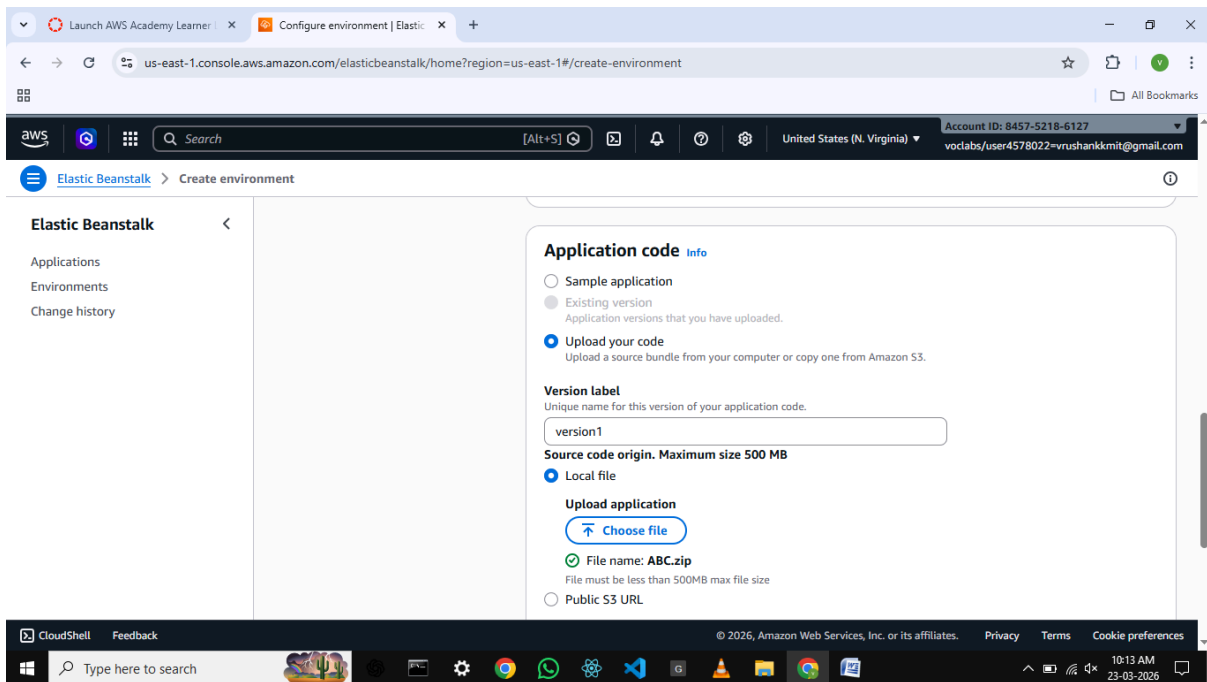
```
1 <!DOCTYPE html>
2 <html>
3 <body>
4
5 <h1>My first PHP page</h1>
6
7 <?php
8 echo "Hello World!";
9 ?>
10
11 </body>
12 </html>
13
```

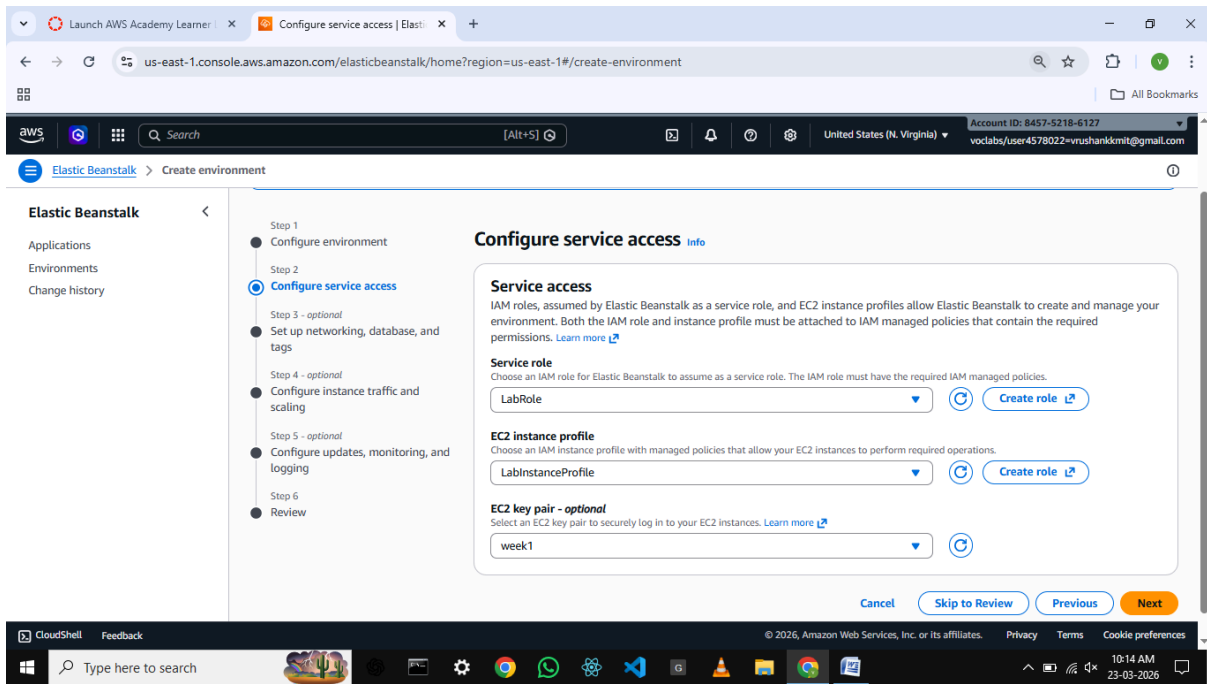


Select platform php

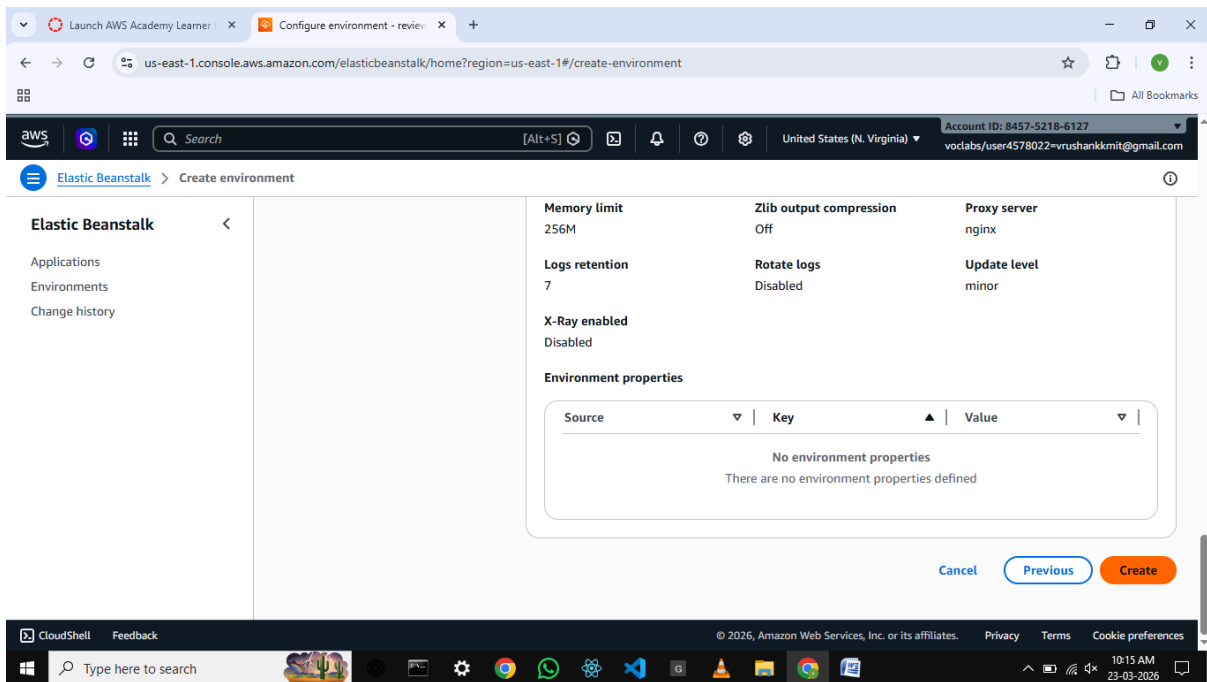


## Upload zip file

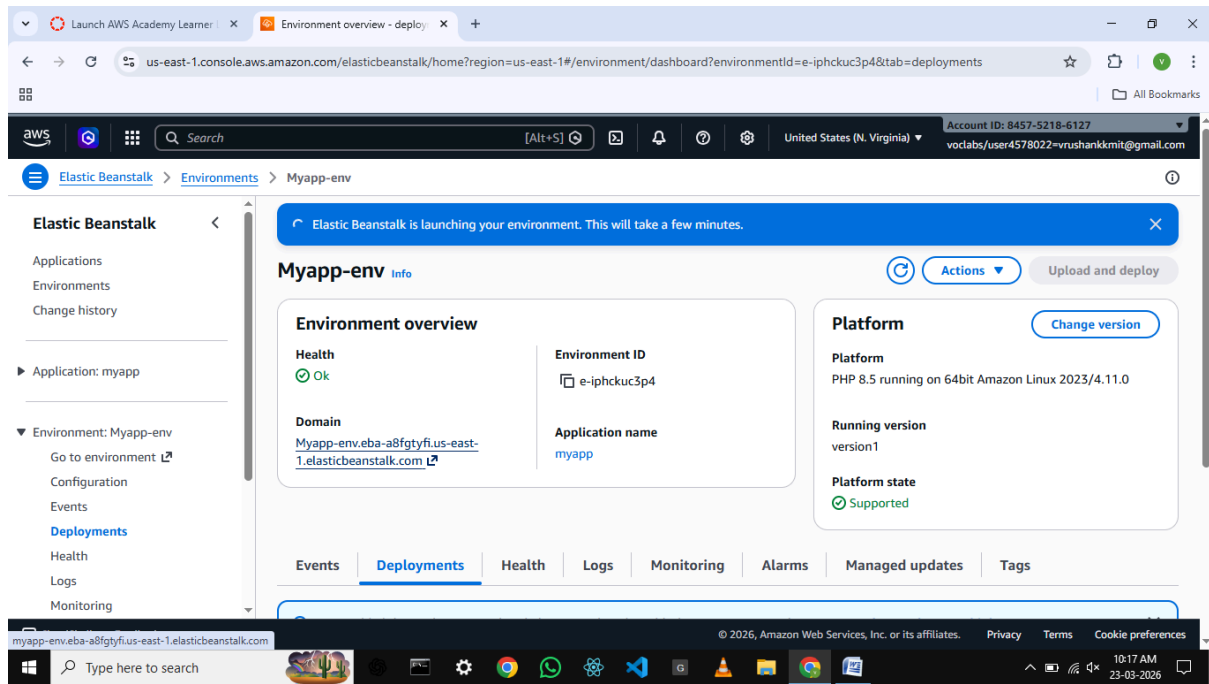




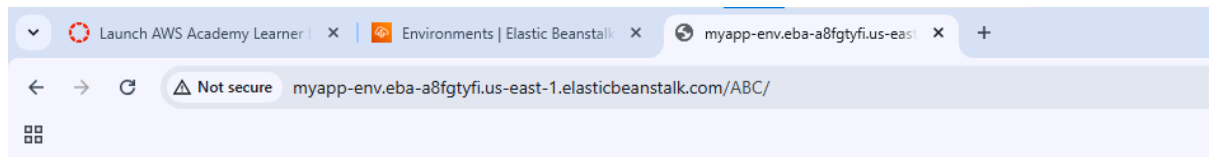
## Create



## Open domain url



Add '/Foldername' to url

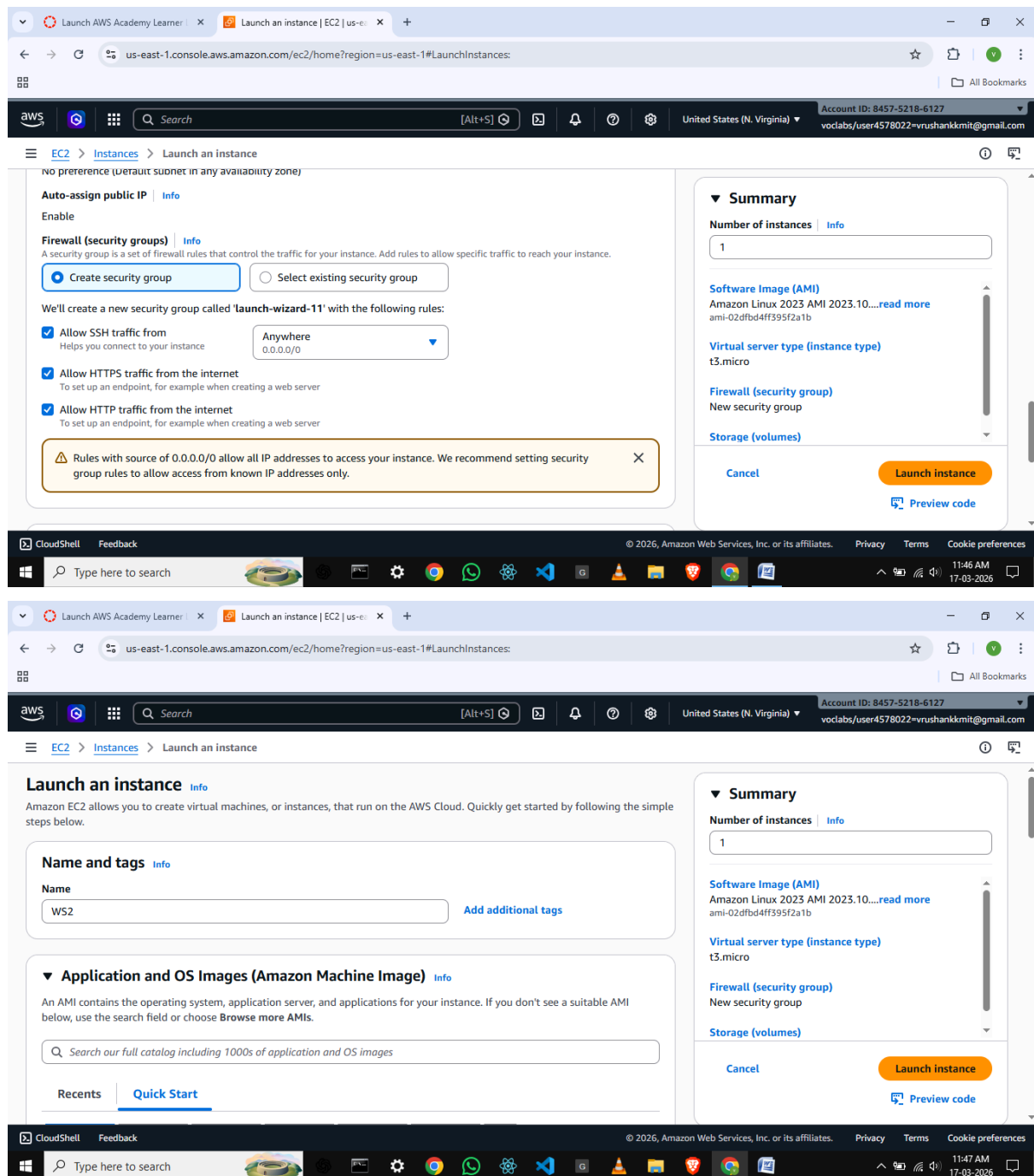


## My first PHP page

Hello World!

# Elastic Load balancer

# WS1 & WS2 instance



Connect both instances

Sudo su

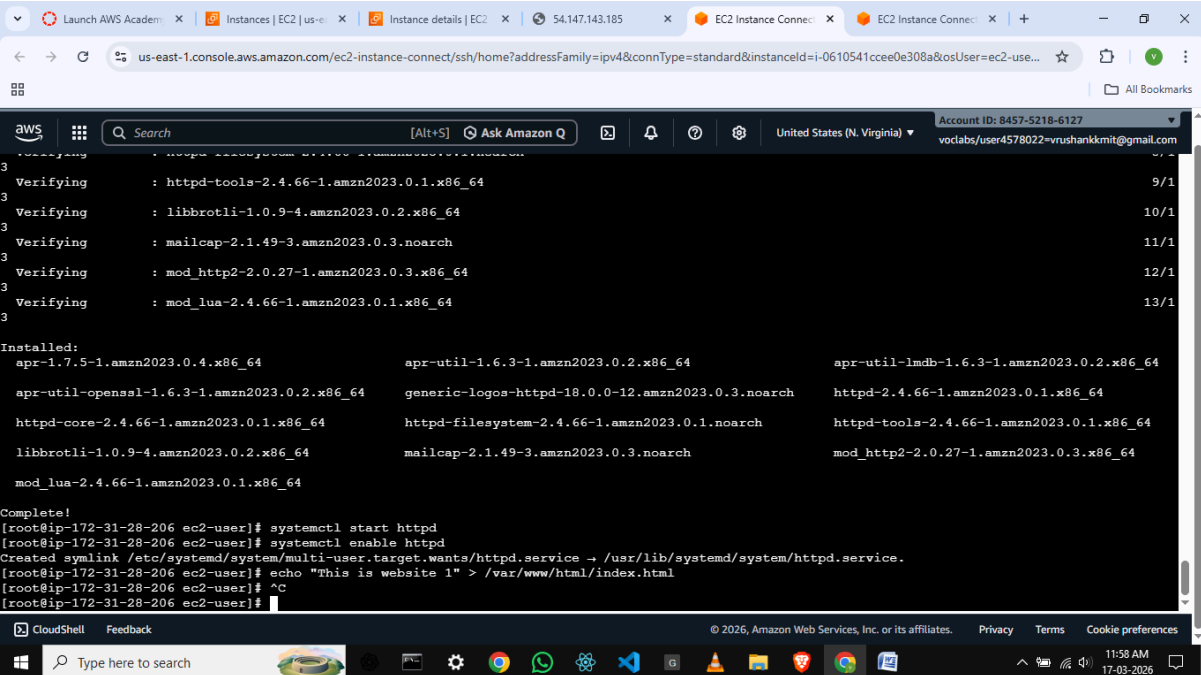
Yum update -y

Yum install httpd -y

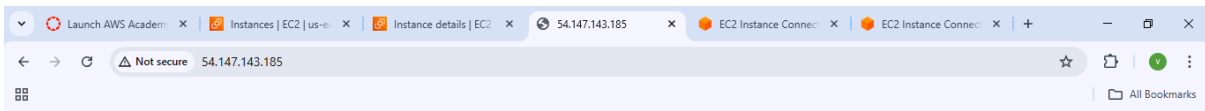
Systemctl start httpd

Systemctl enable httpd

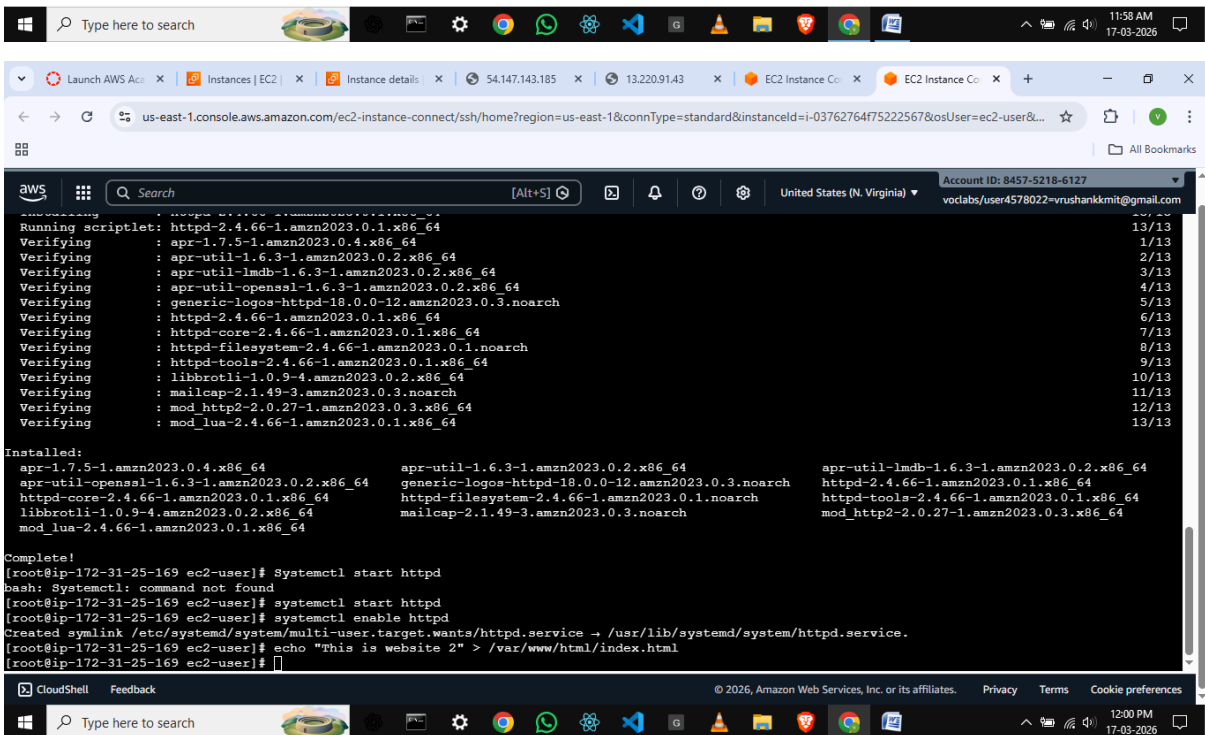
echo "This is website 1" > /var/www/html/index.html

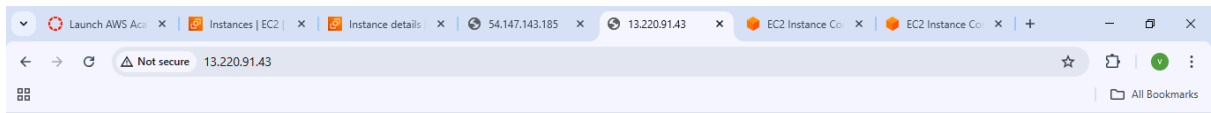


```
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0610541ccee0e308a&osUser=ec2-use...
aws [Search] [Alt+S] Ask Amazon Q United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com
3 Verifying : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 9/1
3 Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/1
3 Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/1
3 Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/1
3 Verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/1
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64 apr-util-1.6.3-1.amzn2023.0.2.x86_64 apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch httpd-2.4.66-1.amzn2023.0.1.x86_64
httpd-core-2.4.66-1.amzn2023.0.1.x86_64 httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch httpd-tools-2.4.66-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64 mailcap-2.1.49-3.amzn2023.0.3.noarch mod_http2-2.0.27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64
Complete!
[root@ip-172-31-28-206 ec2-user]# systemctl start httpd
[root@ip-172-31-28-206 ec2-user]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-28-206 ec2-user]# echo "This is website 1" > /var/www/html/index.html
[root@ip-172-31-28-206 ec2-user]# ^C
[root@ip-172-31-28-206 ec2-user]#
```



This is website 1





Create security group

Launch AWS Academy Learner | x Instances | EC2 | us-east-1 x Instance details | EC2 | us-east-1 x +

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Instances

### Instances (2) Info

Last updated 5 minutes ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#) < 1 > [Settings](#)

<input type="checkbox"/>	Name <a href="#">↗</a>	Instance ID	Instance state <a href="#">↗</a>	Instance type <a href="#">↗</a>	Status check <a href="#">↗</a>	Alarm status <a href="#">↗</a>	Availability <a href="#">↗</a>
<input type="checkbox"/>	WS2	i-03762764f75222567	Running <a href="#">↗</a>	t3.micro	3/3 checks passed <a href="#">↗</a>	<a href="#">View alarms +</a>	us-east-1c
<input type="checkbox"/>	WS1	i-0610541ccee0e308a	Running <a href="#">↗</a>	t3.micro	3/3 checks passed <a href="#">↗</a>	<a href="#">View alarms +</a>	us-east-1c

Select an instance [Settings](#) [↕](#)

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12:01 PM 17-03-2026

Launch AWS Academy Learner | x CreateSecurityGroup | EC2 | us- x Instance details | EC2 | us-east-1 x +

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateSecurityGroup:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Security Groups > Create security group

### Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

#### Basic details

**Security group name** [Info](#)

Name cannot be edited after creation.

**Description** [Info](#)

**VPC** [Info](#)

#### Inbound rules

[Info](#)

This security group has no inbound rules.

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12:02 PM 17-03-2026

Launch AWS Academy Learner | CreateSecurityGroup | EC2 | us-east-1 | Instance details | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateSecurityGroup:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Security Groups > Create security group

vpc-06a49e9d40ba8221f

### Inbound rules

Type	Protocol	Port range	Source	Description - optional
SSH	TCP	22	My IP	
			152.57.179.196/32	
HTTP	TCP	80	Any...	
			0.0.0.0/0	
HTTPS	TCP	443	Any...	
			0.0.0.0/0	

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

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Type here to search 12:03 PM 17-03-2026

## Create Target Groups

Launch AWS Academy Learner | x Step 1 Create target group | EC2 | us-east-1 | x +

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTargetGroup:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Target groups > Create target group

Step 1  
Create target group  
Step 2 - recommended  
Register targets  
Step 3  
Review and create

## Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

**Settings - immutable**  
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

**Target type**  
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**  
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.  
Suitable for: ALB NLB GWLB

☐ **IP addresses**  
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.  
Suitable for: ALB NLB GWLB

☐ **Lambda function**  
Supports load balancing to a single Lambda function. ALB required as traffic source.  
Suitable for: ALB

☐ **Application Load Balancer**  
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

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Type here to search

Launch AWS Academy Learner | x Step 1 Create target group | EC2 | us-east-1 | x +

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTargetGroup:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Target groups > Create target group

☐ **Lambda function**  
Supports load balancing to a single Lambda function. ALB required as traffic source.  
Suitable for: ALB

☐ **Application Load Balancer**  
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.  
Suitable for: NLB

**Target group name**  
Name must be unique per Region per AWS account.  
my target  
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 9/32

**Protocol**  
Protocol for communication between the load balancer and targets.  
HTTP

**Port**  
Port number where targets receive traffic. Can be overridden for individual targets during registration.  
80  
1-65535

**IP address type**  
Only targets with the indicated IP address type can be registered to this target group.

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Type here to search

Launch AWS Academy Learner | x Step 2 Create target group | EC2 | us-east-1 | x +

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTargetGroup:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Target groups > Create target group

Step 1  
Create target group  
Step 2 - recommended  
**Register targets**  
Step 3  
Review and create

### Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Filter instances

<input checked="" type="checkbox"/>	Instance ID	Name	State	Security groups
<input checked="" type="checkbox"/>	i-03762764f75222567	WS2	Running	launch-wizard-12
<input checked="" type="checkbox"/>	i-0610541ccee0e308a	WS1	Running	launch-wizard-11

2 selected

Ports for the selected instances  
Ports for routing traffic to the selected instances.

80

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Type here to search

## Create load balancer(application)

Launch AWS Academy Learner | x Compare and select load balancer | EC2 | us-east-1 | x +

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#SelectCreateELBWizard:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Load balancers > Compare and select load balancer type

### Compare and select load balancer type

A complete feature-by-feature comparison along with detailed highlights is also available. [Learn more](#)

#### Load balancer types

##### Application Load Balancer Info

The diagram shows a user icon connected to an ALB box. Below the ALB box are two boxes labeled 'HTTP' and 'HTTPS'. Dashed lines connect the user to the ALB, and the ALB to the target instances (represented by circles with targets).

##### Network Load Balancer Info

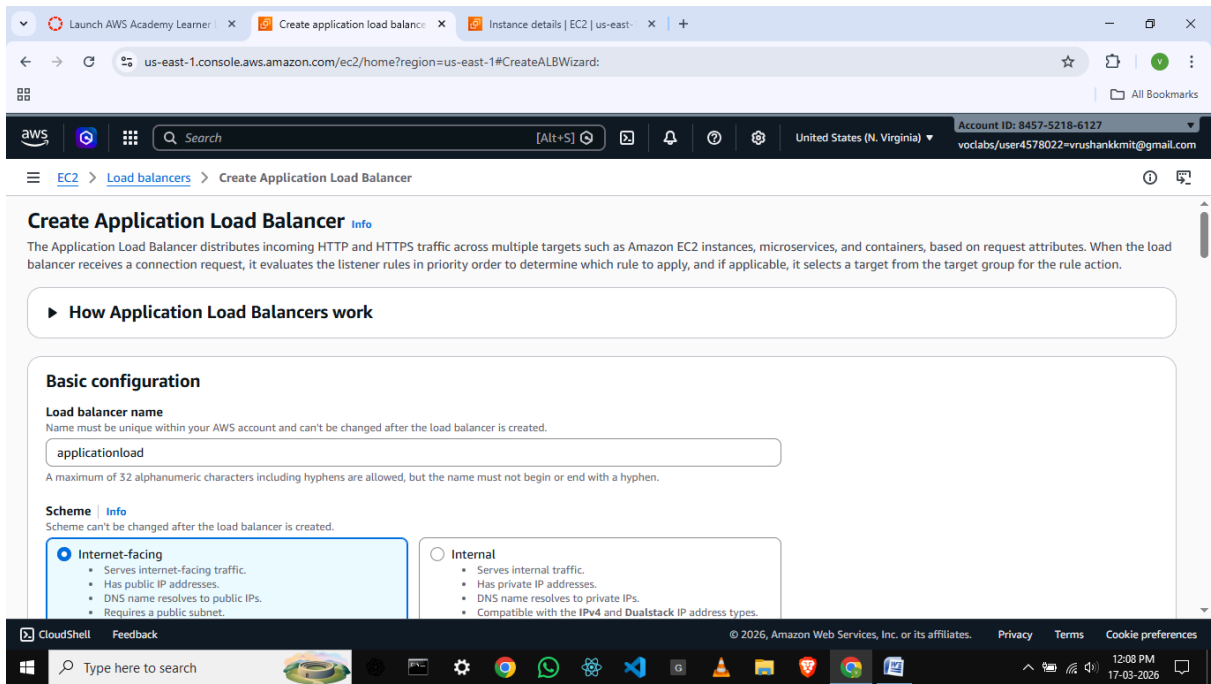
The diagram shows a user icon connected to a VPC/VPCe box. Below this is an NLB box. Below the NLB box are three boxes labeled 'TCP', 'UDP', and 'TLS'. Dashed lines connect the user to the VPC/VPCe box, the VPC/VPCe box to the NLB, and the NLB to the target instances.

##### Gateway Load Balancer Info

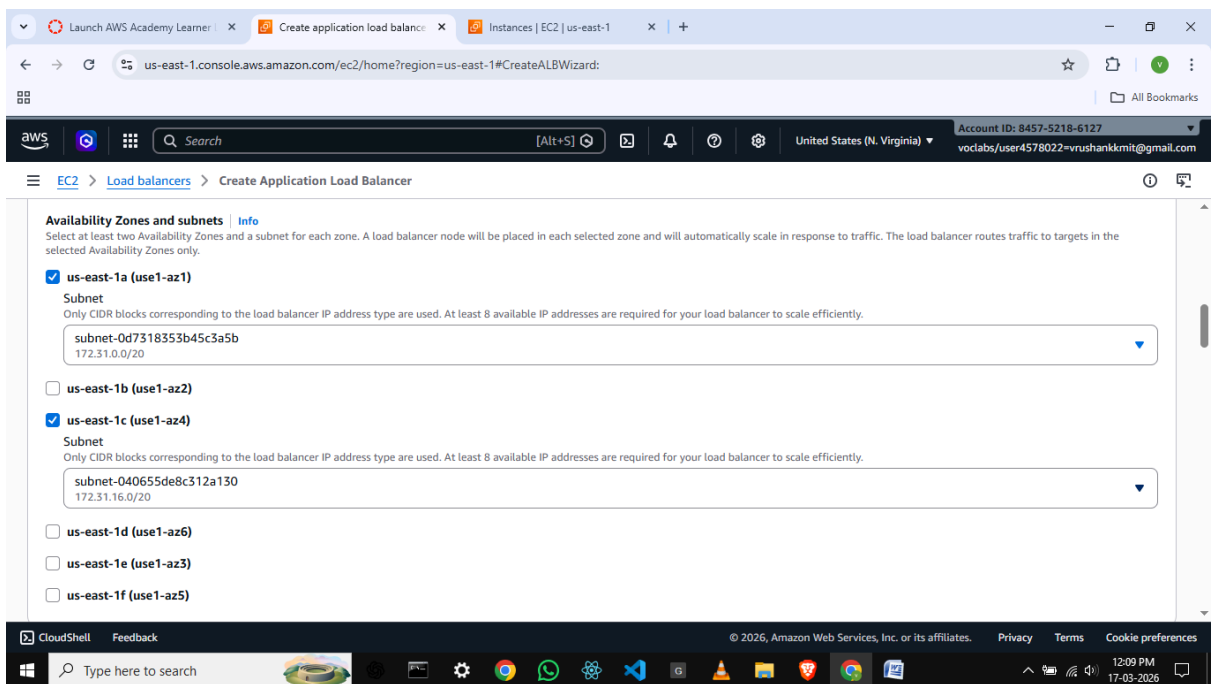
The diagram shows a user icon connected to a GWLB box. A dashed line connects the user to the GWLB, and another dashed line connects the GWLB to the target instances.

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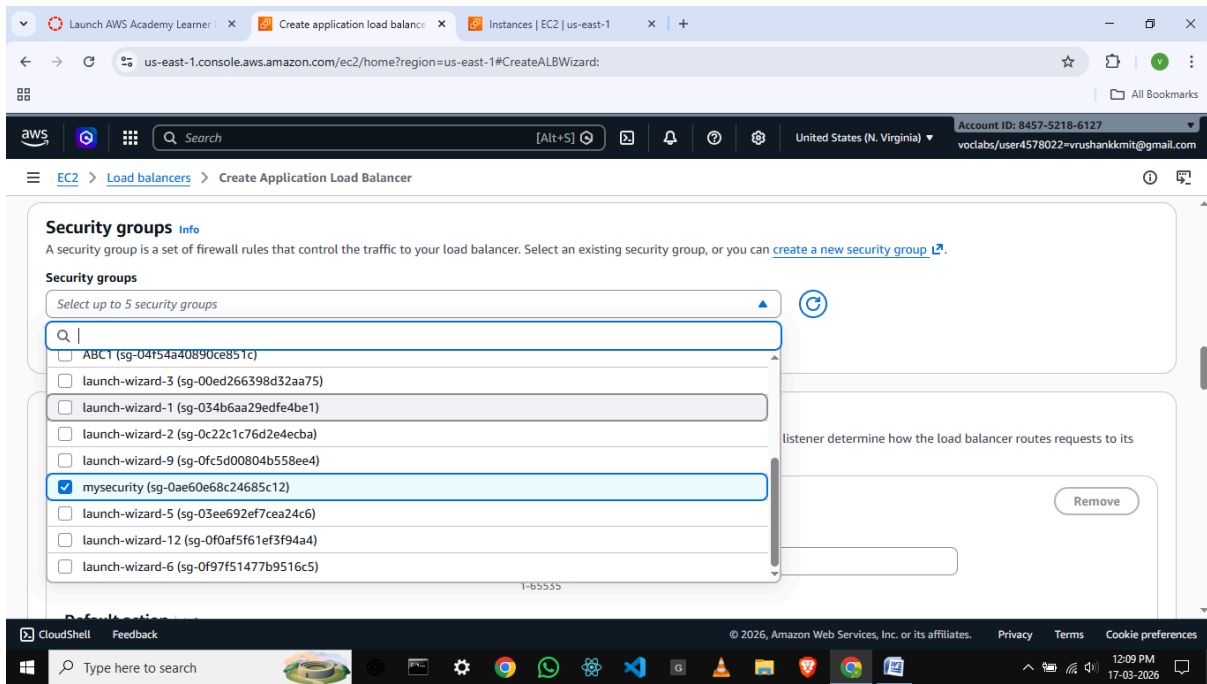
Type here to search



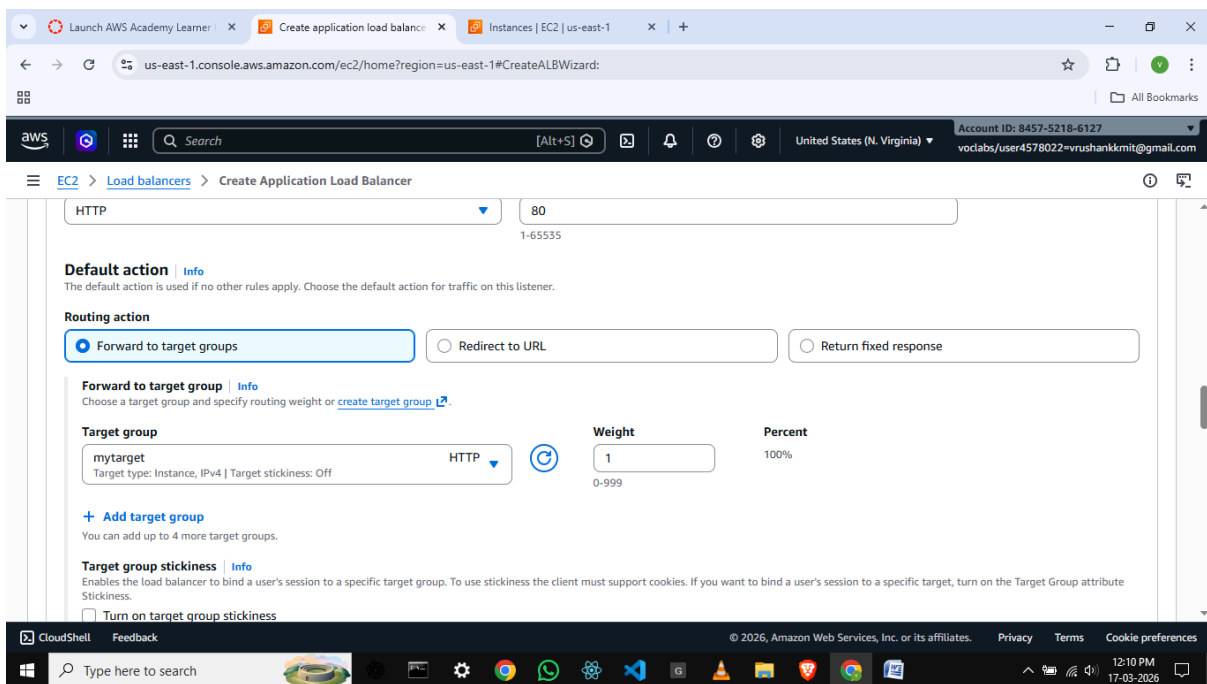
## Select availability zone of instances and 1 more



## Add security group



## Add target group



- Click **Include as pending** → **Register**

## Open dns

Launch AWS Academy Learner | x Load balancers | EC2 | us-east-1 x Load balancers | EC2 | us-east-1 x 503 Service Temporarily Unavailable x +

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancers:

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankmtr@gmail.com

EC2 > Load balancers

**Load balancers (1)** [What's new?](#) [Actions](#) [Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

<input type="checkbox"/>	Name	Availability Zones	Security groups	ARN	Date created
<input type="checkbox"/>	applicationload	2 Availability Zones	2 Security groups	applicationload-168741101...	March 17, 2026

**0 load balancers selected**

Select a load balancer above.

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Type here to search

Launch AWS Academy | x Load balancers | EC2 | x Target group details | x applicationload-1687 x Instance details | EC2 | x 54.147.143.185 x +

Not secure applicationload-1687411011.us-east-1.elb.amazonaws.com

This is website 1

## Auto Scaling Group with Load Balancer

Launch AWS Academy x Instances | EC2 | us-east-1 x Target group details x applicationload-1687 x Instance details | EC2 x 54.147.143.185 x + - x

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Instances

EC2

- Dashboard
- AWS Global View
- Events
- ▼ Instances
  - Instances
  - Instance Types
  - Launch Templates
  - Spot Requests
  - Savings Plans
  - Reserved Instances
  - Dedicated Hosts
  - Capacity Reservations
  - Capacity Manager
- ▼ Images

Instances (1/2) Info Last updated 28 minutes ago

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status
WS2	i-03762764f75222567	Running	t3.micro	3/3
WS1	i-0610541ccee0e308a	Running		

WS1 i-0610541ccee0e308a (WS1)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0610541ccee0e308a	54.147.143.185   open address	172.31.28.206

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Launch template

Launch AWS Academy x Launch templates | E x Target group details x applicationload-1687 x Instance details | EC2 x 54.147.143.185 x + -

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchTemplates:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankmtr@gmail.com

EC2

- Dashboard
- AWS Global View
- Events
- ▼ Instances
  - Instances
  - Instance Types
  - Launch Templates
  - Spot Requests
  - Savings Plans
  - Reserved Instances
  - Dedicated Hosts
  - Capacity Reservations
  - Capacity Manager New
- ▼ Images

Compute

# EC2 launch templates

Streamline, simplify and standardize instance launches

Use launch templates to automate instance launches, simplify permission policies, and enforce best practices across your organization. Save launch parameters in a template that can be used for on-demand launches and with managed services, including EC2 Auto Scaling and EC2 Fleet. Easily update your launch parameters by creating a new launch template version.

New launch template

Create launch template

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Type here to search

Launch AWS Academy x Create launch template x Target group details x applicationload-1687 x Instance details | EC2 x 54.147.143.185 x + -

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankmtr@gmail.com

EC2 > Launch templates > Create launch template

Recents My AMIs Quick Start

☐ Don't include in launch template ☒ Owned by me

☐ Shared with me

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

myimage  
ami-0a239278d5943d33b  
2026-03-17T06:55:54.000Z Virtualization: hvm ENA enabled: true  
Root device type: ebs Boot mode: uefi-preferred

Description  
this is desc

Architecture AMI ID  
x86\_64 ami-0a239278d5943d33b

▼ Summary

Software Image (AMI)  
this is desc  
ami-0a239278d5943d33b

Virtual server type (instance type)  
-

Firewall (security group)  
-

Storage (volumes)  
1 volume(s) - 8 GiB

Cancel Create launch template

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Type here to search

Launch AWS Academ... Create launch templa... Target group details... applicationload-1687... Instance details | EC2... 54.147.143.185... us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Launch templates > Create launch template

**Instance type** Info | Get advice Advanced

**Instance type**

t2.micro  
Family: t2 1 vCPU 1 GiB Memory Current generation: true  
On-Demand Linux base pricing: 0.0116 USD per Hour  
On-Demand Windows base pricing: 0.0162 USD per Hour  
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour  
On-Demand SUSE base pricing: 0.0116 USD per Hour  
On-Demand RHEL base pricing: 0.026 USD per Hour

☐ All generations [Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

**Key pair (login)** Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

**Key pair name**  
week1 [Create new key pair](#)

**Summary**

**Software Image (AMI)**  
this is desc  
ami-0a239278d5943d33b

**Virtual server type (instance type)**  
t2.micro

**Firewall (security group)**  
-

**Storage (volumes)**  
1 volume(s) - 8 GiB

[Cancel](#) [Create launch template](#)

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Type here to search

Launch AWS Academ... Create launch templa... Target group details... applicationload-1687... Instance details | EC2... 54.147.143.185... us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Launch templates > Create launch template

**Network Settings** Info

**Subnet** Info  
Don't include in launch template [Create new subnet](#)  
When you specify a subnet, a network interface is automatically added to your template.

**Availability Zone** Info  
Don't include in launch template [Enable additional zones](#)

**Firewall (security groups)** Info  
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.  
☒ Select existing security group ☐ Create security group

**Security groups** Info  
Select security groups [Compare security group rules](#)  
mysecurity sg-0ae60e68c24685c12 [X](#)  
VPC: vpc-06a49e9d40ba8221f

[Advanced network configuration](#)

**Summary**

**Software Image (AMI)**  
this is desc  
ami-0a239278d5943d33b

**Virtual server type (instance type)**  
t2.micro

**Firewall (security group)**  
mysecurity

**Storage (volumes)**  
1 volume(s) - 8 GiB

[Cancel](#) [Create launch template](#)

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Type here to search

Auto scaling groups



Launch AWS Academy x Create Auto Scaling x Target group details x applicationload-1687 x Instance details | EC2 x 54.147.143.185 x + -

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Auto Scaling groups > Create Auto Scaling group

Integrate with other services

Step 4 - optional  
Configure group size and scaling

Step 5 - optional  
Add notifications

Step 6 - optional  
Add tags

Step 7  
Review

**Auto scaling group name**  
Enter a name to identify the group.  
myautoscale  
Must be unique to this account in the current Region and no more than 255 characters.

**Launch template info** [Switch to launch configuration](#)

**Launch template**  
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.  
mytemplate [Create a launch template](#)

**Version**  
Default (1) [Create a launch template version](#)

**Description**  
-

**Launch template**  
mytemplate  
lt-0676fec89413015d2

**Instance type**  
t2.micro

**AMI ID**

**Security groups**

**Request Spot Instances**

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Type here to search

Launch AWS Academy x Create Auto Scaling x Target group details x applicationload-1687 x Instance details | EC2 x 54.147.143.185 x + -

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup:

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Auto Scaling groups > Create Auto Scaling group

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-06a49e9d40ba8221f  
172.31.0.0/16 Default [Create a VPC](#)

**Availability Zones and subnets**  
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.  
Select Availability Zones and subnets

use1-az4 (us-east-1c) | subnet-040655de8c312a130  
172.31.16.0/20 Default

use1-az6 (us-east-1d) | subnet-09e8fb247527a5569  
172.31.32.0/20 Default [Create a subnet](#)

**Availability Zone distribution - new**  
Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**  
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**  
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

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Type here to search

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup:

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 2: Choose instance launch options  
Step 3 - optional: **Integrate with other services**  
Step 4 - optional: Configure group size and scaling  
Step 5 - optional: Add notifications  
Step 6 - optional: Add tags  
Step 7: Review

Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

### Load balancing

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

**Select Load balancing options**

☐ No load balancer  
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ **Attach to an existing load balancer**  
Choose from your existing load balancers.

☐ Attach to a new load balancer  
Quickly create a basic load balancer to attach to your Auto Scaling group.

**Attach to an existing load balancer**

**Select the load balancers to attach**

☒ **Choose from your load balancer target groups**  
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

**Existing load balancer target groups**  
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

mytarget | HTTP  
Application Load Balancer: applicationload

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup:

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 4 - optional: **Configure group size and scaling**  
Step 5 - optional: Add notifications  
Step 6 - optional: Add tags  
Step 7: Review

### Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

### Desired capacity

Specify your group size.

2

### Scaling

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

**Scaling limits**  
Set limits on how much your desired capacity can be increased or decreased.

**Min desired capacity**  
1  
Equal or less than desired capacity

**Max desired capacity**  
4  
Equal or greater than desired capacity

**Automatic scaling - optional**  
Choose whether to use a target tracking policy | Info

Stop instance 1

Launch AWS Academy x Auto Scaling groups x Instances | EC2 | us-east-1 x applicationload-1687 x Instance details | EC2 x 54.147.143.185 x + -

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Instances

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Alarm status	Availability
WS2	i-03762764f75222567	Running	t3.m	View alarms +	us-east-1c
WS1	i-0610541ccee0e308a	Running	t3.m	View alarms +	us-east-1c

i-0610541ccee0e308a (WS1)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0610541ccee0e308a

Public IPv4 address 54.147.143.185 | open address

Private IPv4 addresses 172.31.28.206

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Type here to search

Launch AWS Academy x Auto Scaling groups x Instances | EC2 | us-east-1 x applicationload-1687 x Instance details | EC2 x 54.147.143.185 x + -

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankkmit@gmail.com

EC2 > Instances

Successfully initiated stopping of i-0610541ccee0e308a

Instances (1/4) Info

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
WS2	i-03762764f75222567	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1c
WS1	i-0610541ccee0e308a	Stopped	t3.micro	-	View alarms +	us-east-1c
i-075094a4377cee077	i-075094a4377cee077	Running	t2.micro	Initializing	View alarms +	us-east-1c

i-075094a4377cee077

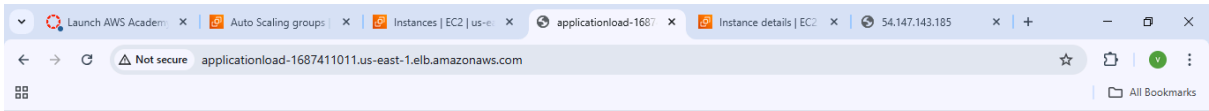
Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

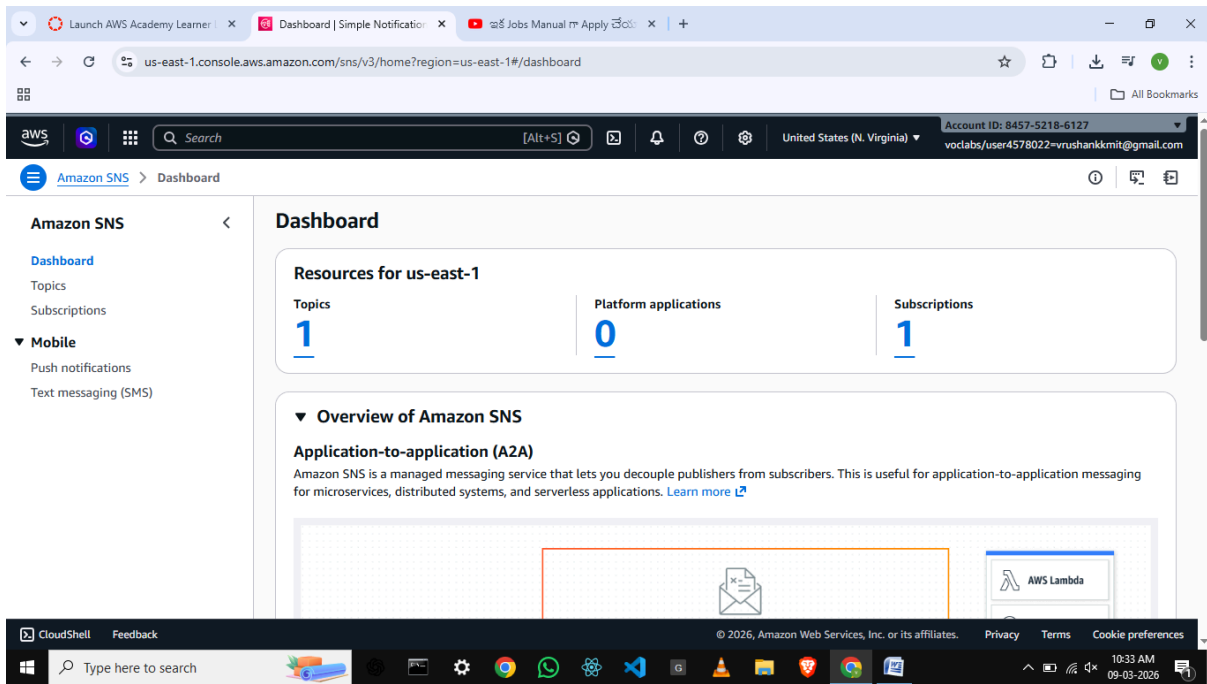
CloudShell Feedback

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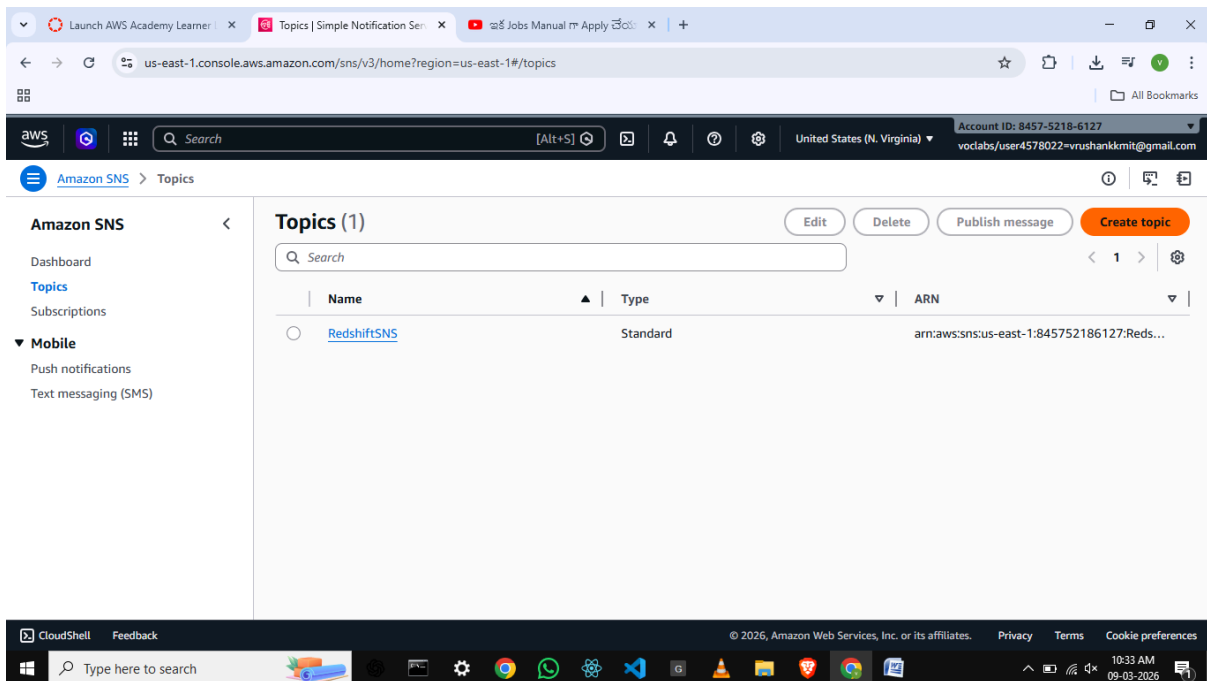
Type here to search

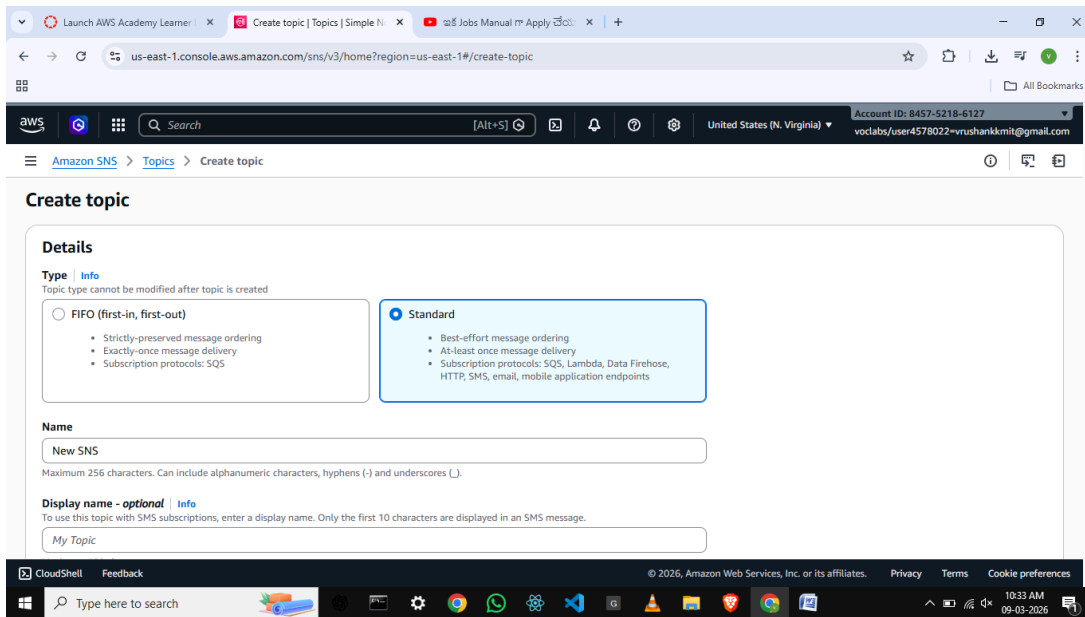


SNS-SQS

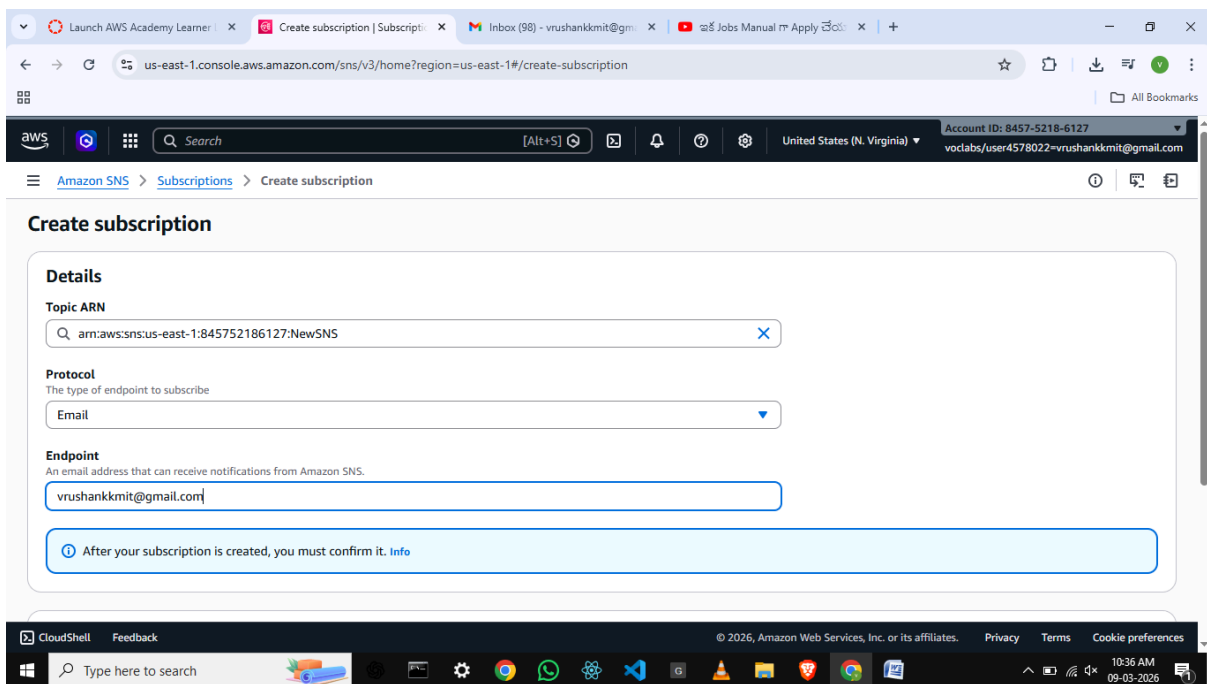


## Create topic

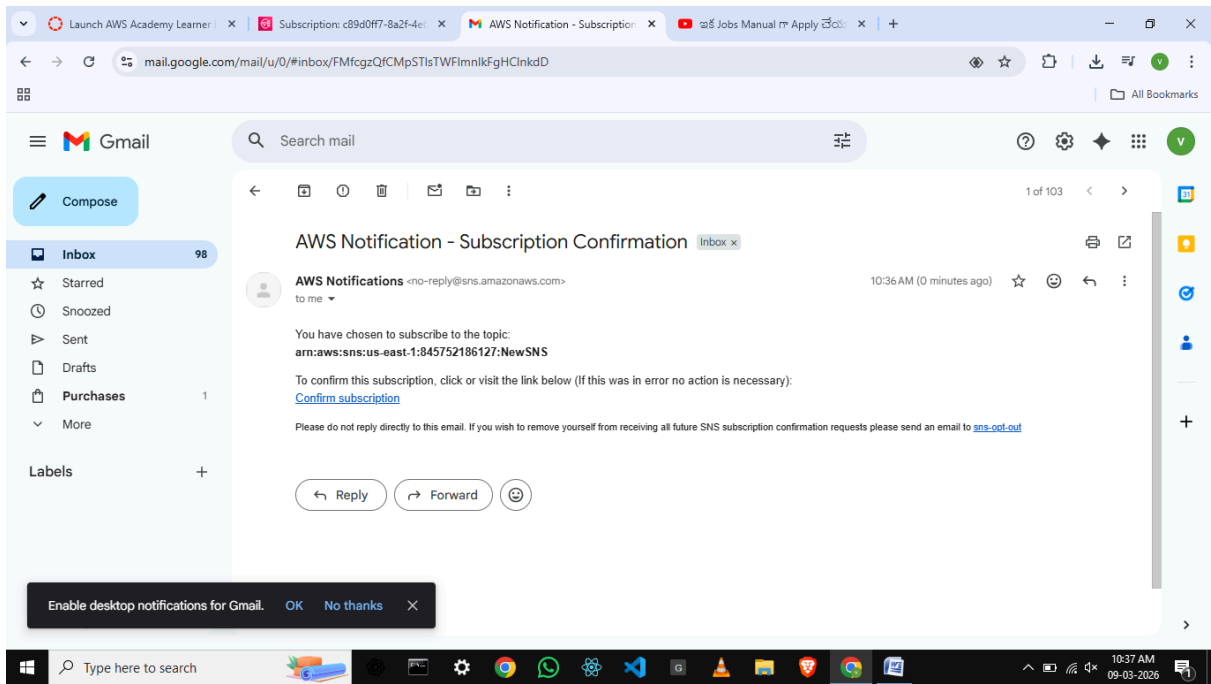




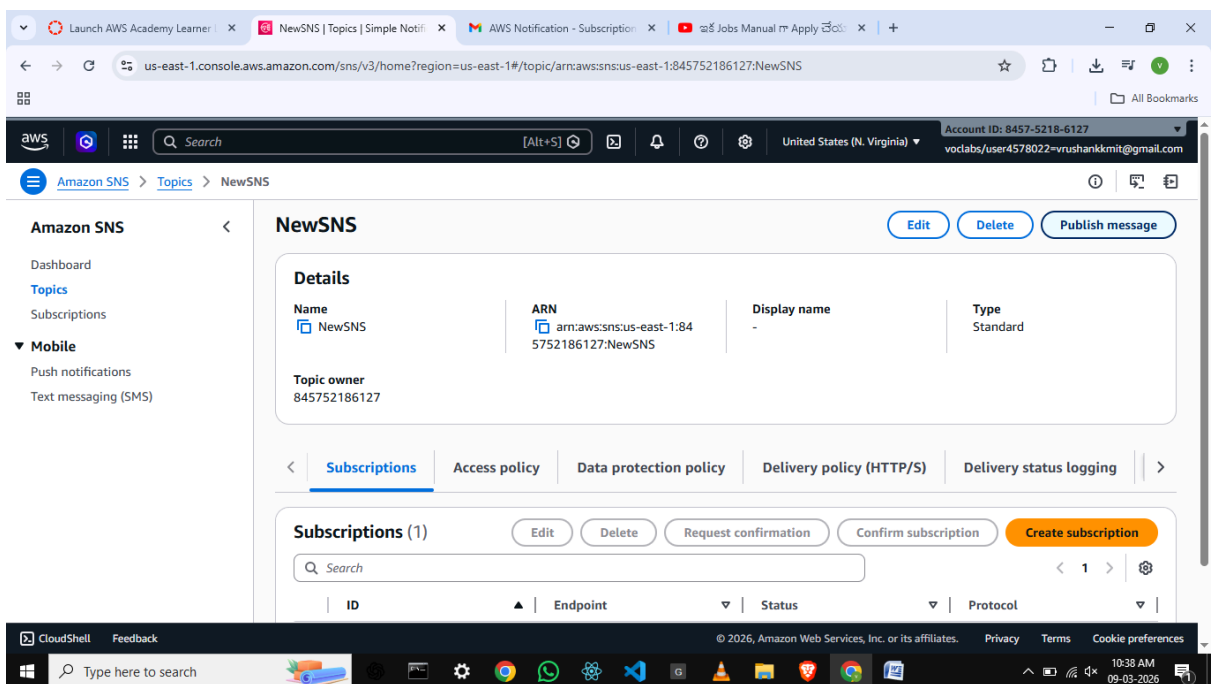
## Create Subscription

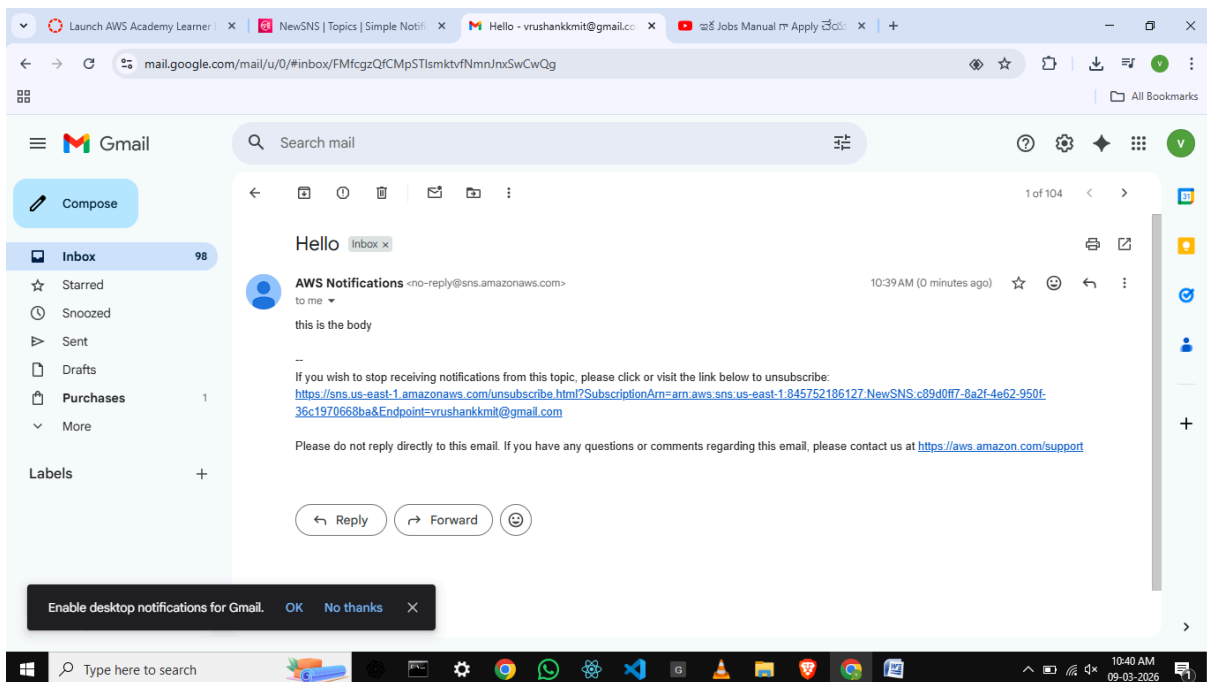
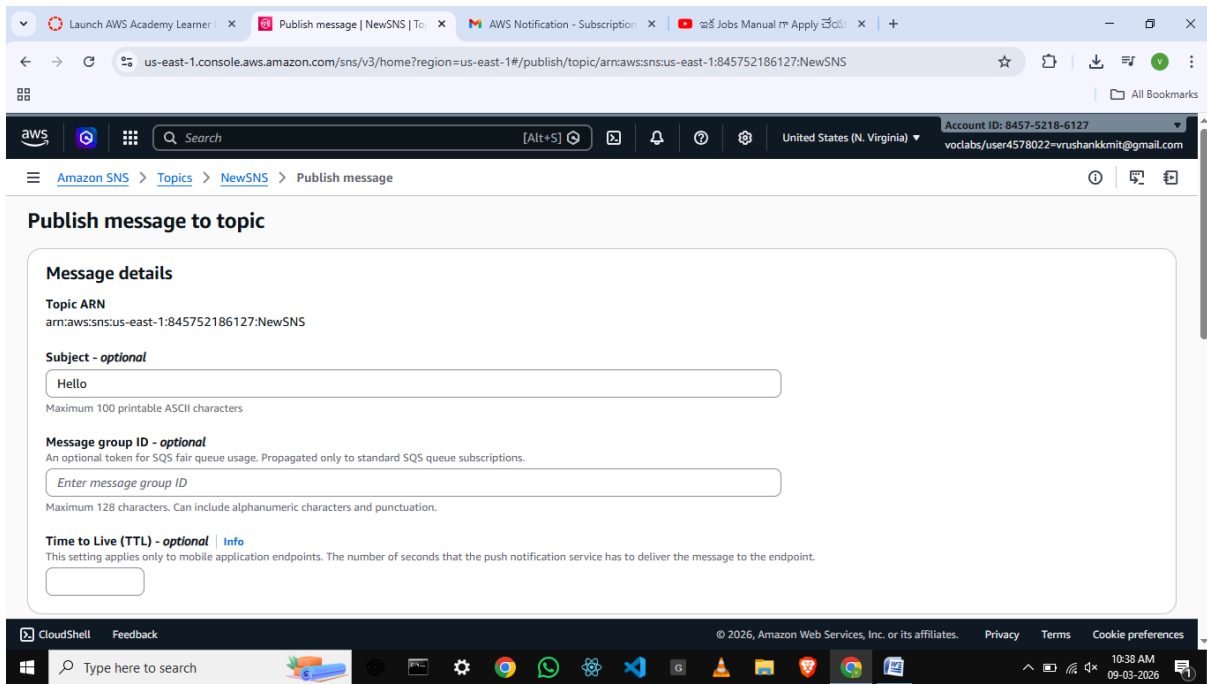


## Confirm the subscription



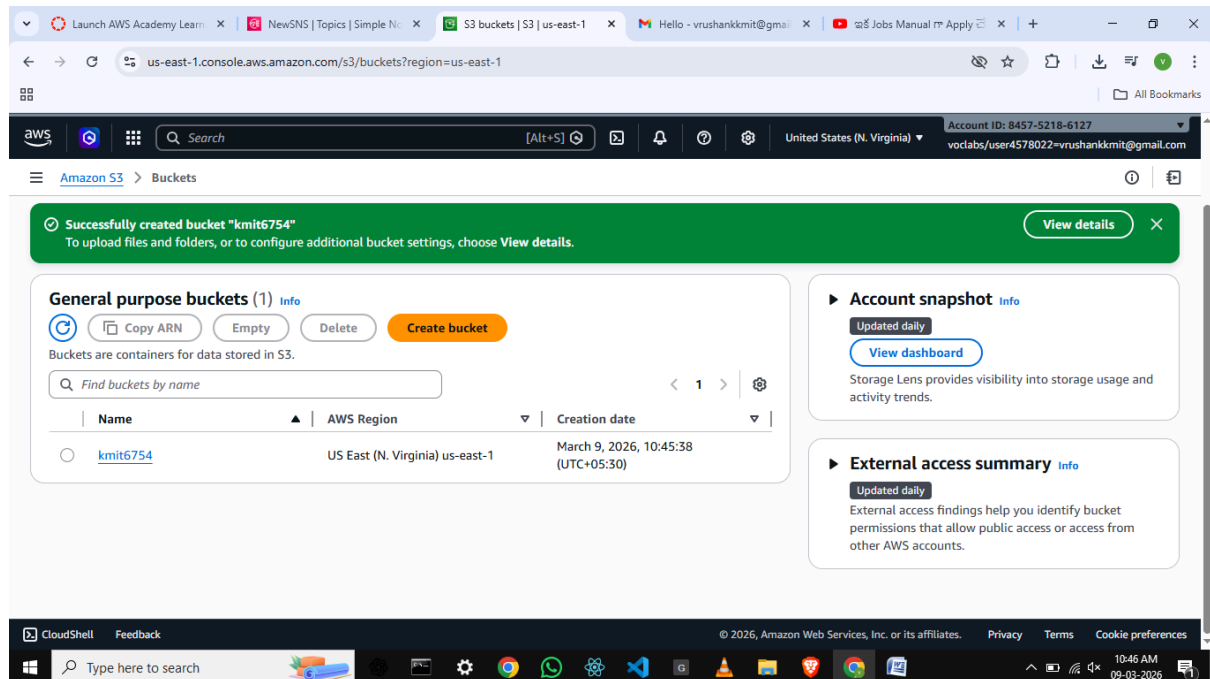
## Publish message





## Get notifications for S3 buckets

# 1)create bucket



# 2)Create Lambda function

Launch AWS Academy x NewSNS | Topics | S3 buckets | S3 | us-east-1 x Create function | Functions | Hello - vrushankmit x Jobs Manual m A x +

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/create/function?intent=authorFromScratch

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankmit@gmail.com

Lambda > Functions > Create function

### Create function [Info](#)

Choose one of the following options to create your function.

☒ **Author from scratch**  
Start with a simple Hello World example.

☐ **Use a blueprint**  
Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**  
Select a container image to deploy for your function.

#### Basic information

**Function name**  
Enter a name that describes the purpose of your function.  
  
Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (\_).

**Runtime** [Info](#)  
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

**Durable execution** - [new](#) [Info](#)  
Enable durable execution to simplify building resilient multi-step applications that checkpoint progress and resume after interruptions. Supports Python and Node.js runtimes. [View pricing](#)

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Launch AWS Academy x NewSNS | Topics | S3 buckets | S3 | us-east-1 x Create function | Functions | Hello - vrushankmit x Jobs Manual m A x +

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/create/function?intent=authorFromScratch

aws Search [Alt+S] United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankmit@gmail.com

Lambda > Functions > Create function

By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

#### ▼ Change default execution role

**Execution role**  
To access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

☐ Create default role ☒ Use another role

[Create new role](#)

[View role details in IAM](#)

#### ► Additional configurations

Use additional configurations to set up networking, security, and governance for your function. These settings help secure and customize your Lambda function deployment.

[Cancel](#) [Create function](#)

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# add Trigger

The screenshot shows the AWS Lambda console for a function named 'forSNS'. At the top, a green banner indicates that the function was successfully created. Below this, the 'Function overview' section is visible, showing the function's name, icon, and a 'Layers' section with '(0)' layers. To the right, the 'Configuration' tab is active, displaying the function's description, last modified time (2 minutes ago), function ARN, and function URL. The bottom of the screen shows the Windows taskbar with various application icons and the system clock.

Successfully created the function **forSNS**. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

**forSNS** [Throttle](#) [Copy ARN](#) [Actions](#)

**Function overview** [Info](#)

[Diagram](#) | [Template](#)

[+ Add trigger](#) [+ Add destination](#)

**Description**

**Last modified**  
2 minutes ago

**Function ARN**  
[arn:aws:lambda:us-east-1:845752186127:function:forSNS](#)

**Function URL** [Info](#)

10:49 AM  
09-03-2026

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/add/relation?focus=aws%3Alambda&target=arn%3Aaws%3Alambda%3Aus-east-1...

**Add trigger**

**Trigger configuration** [Info](#)

**S3**  
aws asynchronous storage

**Bucket**  
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.  
s3/kmit6754  
Bucket region: us-east-1

**Event types**  
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.  
All object create events

**Prefix - optional**  
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any [special characters](#) must be URL encoded.

us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/forSNS?tab=configure

**Function overview** [Info](#)

**Diagram** | **Template**

**forSNS**  
Layers (0)

**S3**  
+ Add trigger

**+ Add destination**

**Description**  
-

**Last modified**  
3 minutes ago

**Function ARN**  
arn:aws:lambda:us-east-1:845752186127:function:fo  
rSNS

**Function URL** | [Info](#)  
-

**Code** | **Test** | **Monitor** | **Configuration** | **Aliases** | **Versions**

## add destination

The screenshot shows the 'Add destination' page in the AWS Lambda console. The breadcrumb navigation is 'Lambda > Functions > forSNS > Add destination'. The page has several sections:

- Event source mapping invocation**: A radio button is selected for 'On success'.
- Condition**: A note says 'Choose whether to send invocation records for event processing failures or for successful invocations.' The 'On success' radio button is selected.
- Destination type**: A dropdown menu is set to 'SNS topic'. A warning box below states 'SNS topic has a 256KB payload limit'.
- Destination**: A text field contains the ARN 'arn:aws:sns:us-east-1:845752186127:NewSNS'.
- Permissions**: A note states 'If your execution role doesn't have the required permissions for the selected destination, then Lambda will attempt to add the permissions to the role.'

The bottom of the screenshot shows the Windows taskbar with various application icons and the system clock showing 10:52 AM on 09-03-2026.

The screenshot shows the 'forSNS' function overview page in the AWS Lambda console. A green banner at the top says 'Your changes have been saved.' The page includes:

- Function overview**: A diagram showing the function 'forSNS' connected to an 'S3' trigger and an 'SNS' destination. Buttons for 'Add trigger' and 'Add destination' are present.
- Function details**: A sidebar on the right showing 'Description', 'Last modified' (4 minutes ago), 'Function ARN' (arn:aws:lambda:us-east-1:845752186127:function:forSNS), and 'Function URL'.
- Actions**: Buttons for 'Throttle', 'Copy ARN', 'Export to Infrastructure Composer', and 'Download' are visible.

The bottom of the screenshot shows the Windows taskbar with various application icons and the system clock showing 10:52 AM on 09-03-2026.

Upload any file to bucket to get notification

Launch AWS Academy x NewSNS | Topics | S... x kmit6754 - S3 bucket x forSNS | Functions | L x Hello - vrushankkmit x Jobs Manual m A x + -

us-east-1.console.aws.amazon.com/s3/buckets/kmit6754?region=us-east-1&tab=objects

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankkmit@gmail.com

Amazon S3 > Buckets > kmit6754

### kmit6754 Info

Objects Metadata Properties Permissions Metrics Management Access Points

**Objects (1)** [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open L](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

<input type="checkbox"/>	Name	Type	Last modified	Size	Storage class
<input type="checkbox"/>	<a href="#">WhatsApp Image 2026-02-21 at 10.22.05 AM.jpeg</a>	jpeg	March 9, 2026, 10:53:38 (UTC+05:30)	177.8 KB	Standard

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Type here to search

Launch AWS Academy x NewSNS | Topics | S... x kmit6754 - S3 bucket x forSNS | Functions | L x AWS Notification Me x Jobs Manual m A x + -

mail.google.com/mail/u/0/#inbox/FMfcgzQfCmpSdrhWfWcNBnqKxDDZRQR

Gmail

Search mail

Compose

Inbox 98

Starred

Snoozed

Sent

Drafts

Purchases 1

More

Labels +

**AWS Notification Message** [Inbox x](#)

**AWS Notifications** <no-reply@sns.amazonaws.com> to me 10:53 AM (0 minutes ago)

`{ "version": "1.0", "timestamp": "2026-03-09T05:23:39.652Z", "requestContext": { "requestId": "621869bc-95e3-42ce-aa56-808ecd01ab2", "functionArn": "arn:aws:lambda:us-east-1:845752186127:function:forSNS:SLATEST", "condition": "Success", "approximateInvokeCount": 1 }, "requestPayload": { "Records": [ { "eventVersion": "2.1", "eventSource": "aws:s3", "awsRegion": "us-east-1", "eventTime": "2026-03-09T05:23:37.921Z", "eventName": "ObjectCreated:Put", "userIdentity": { "principalId": "AWS:AROAAJ2WCGUHVST3Q75YTYuser4578022=vrushankkmit@gmail.com", "requestParameters": { "sourceIPAddress": "152.59.205.96", "responseElements": { "x-amz-request-id": "RWZBVM0H0A3QQ71B", "x-amz-id-2": "GyeLU5HtHyfPO4Lb3UNJUUsXGc0GJo51L7nL3wzNTToHhAIBus2sAb1hJDyqfI4qD24q0swZillmctSoVxDROTomMSRE0", "s3": { "s3SchemaVersion": "1.0", "configurationId": "b430e47c-a069-4559-8180-248038622525", "bucket": { "name": "kmit6754", "ownerIdentity": { "principalId": "A54VX8HWOYEKN", "arn": "arn:aws:s3::kmit6754", "object": { "key": "WhatsApp+Image+2026-02-21+at+10.22.05+AM.jpeg", "size": 182063, "eTag": "56009706fc2fa29309a2c72770807887", "sequencer": "0069AE5959DAD81617"} } } } } } ] } } }`

If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:  
<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:845752186127:NewSNS:c89d0f7-8a2f-4e62-950f-36c1970668ba&Endpoint=vrushankkmit@gmail.com>

Enable desktop notifications for Gmail. OK No thanks forward

Now SQS

us-east-1.console.aws.amazon.com/sqs/v3/home?region=us-east-1#/homepage

aws Search [Alt+S]

United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankmit@gmail.com

# Amazon SQS

## A message queuing service

Amazon SQS provides queues for high-throughput, system-to-system messaging. You can use queues to decouple heavyweight processes and to buffer and batch work. Amazon SQS stores messages until microservices and serverless applications process them.

**Get started**

Learn how to use Amazon SQS by creating a queue, sending a message to the queue, and receiving and processing the message.

[Create queue](#)

**Pricing**

You can get started with Amazon SQS for free. All customers can make 1 million Amazon SQS requests for free each month. Some applications might be able to

**How it works**

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Type here to search

us-east-1.console.aws.amazon.com/sqs/v3/home?region=us-east-1#/create-queue

aws Search [Alt+S]

United States (N. Virginia) Account ID: 8457-5218-6127 voclabs/user4578022=vrushankmit@gmail.com

Amazon SQS > Queues > Create queue

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (\_).

### Configuration [Info](#)

Set the maximum message size, visibility to other consumers, and message retention.

**Visibility timeout** [Info](#)

15 Seconds

Should be between 0 seconds and 12 hours.

**Message retention period** [Info](#)

4 Days

Should be between 1 minute and 14 days.

**Delivery delay** [Info](#)

0 Seconds

Should be between 0 seconds and 15 minutes.

**Maximum message size** [Info](#)

1024 KiB

Should be between 1 KiB and 1024 KiB.

**Receive message wait time** [Info](#)

0 Seconds

Should be between 0 and 20 seconds.

### Er...

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Type here to search

Launch AWS Acc...NewSNS | Topic...Queue details | kmit6754 - S3 b...forSNS | Functio...AWS Notificatio...Jobs Manual

us-east-1.console.aws.amazon.com/sqs/v3/home?region=us-east-1#/queues/https%3A%2F%2Fsqs.us-east-1.amazonaws.com%2F845752186127%2FMyQueue

us-east-1.console.aws.amazon.com/sqs/v3/home?region=us-east-1#/queues/https%3A%2F%2Fsqs.us-east-1.amazonaws.com%2F845752186127%2FMyQueue

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankkmit@gmail.com

Amazon SQS > Queues > MyQueue

Queue MyQueue created successfully  
You can now send and receive messages.

MyQueue

EditDeletePurgeSend and receive messagesStart DLQ redrive

Details info

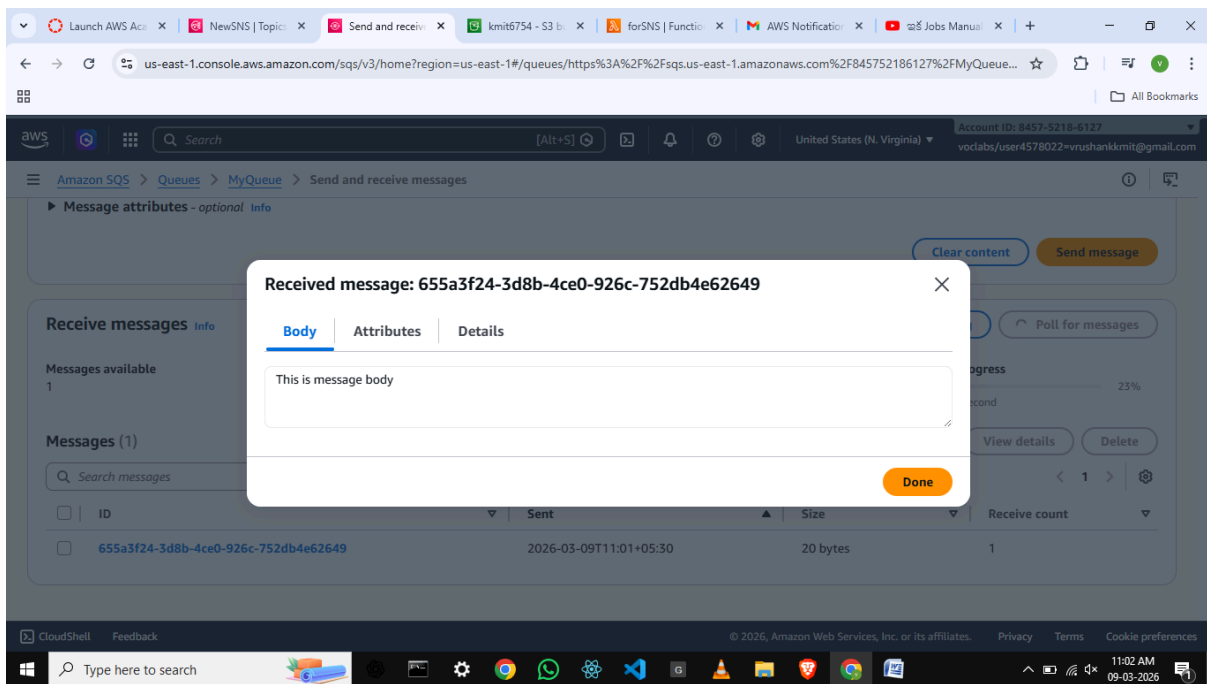
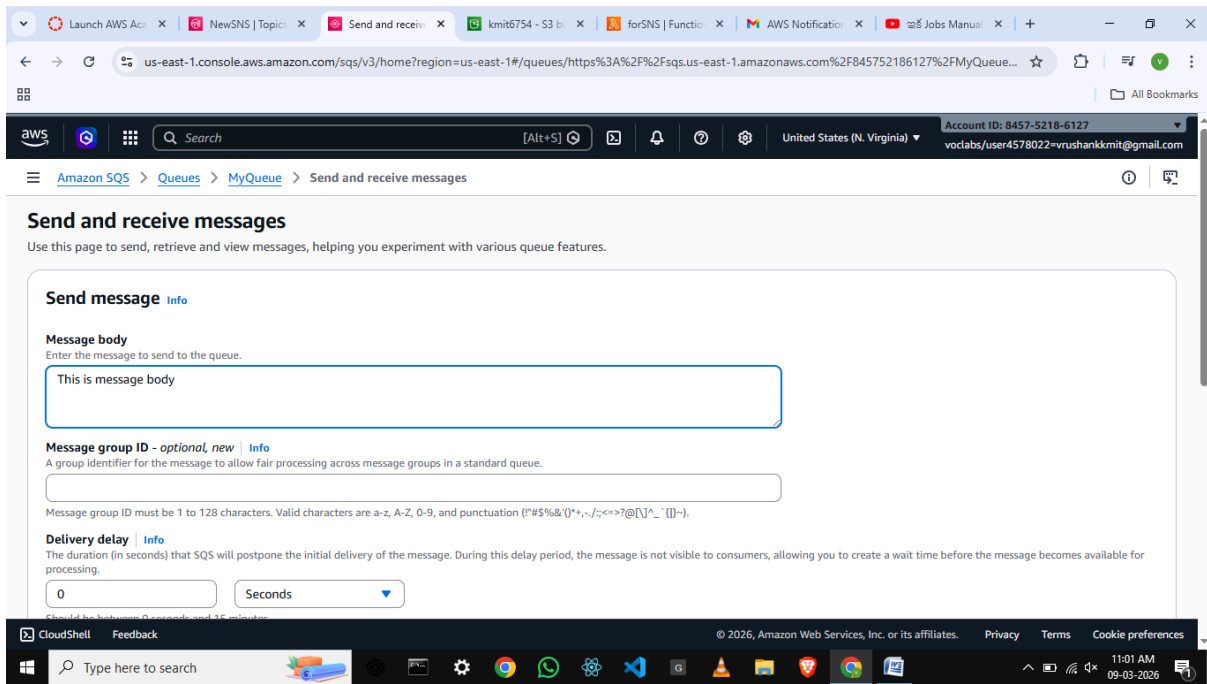
Name MyQueue	Type Standard	ARN arn:aws:sqs:us-east-1:845752186127:MyQueue
Encryption Amazon SQS key (SSE-SQS)	URL https://sqs.us-east-1.amazonaws.com/845752186127/MyQueue	Dead-letter queue -

More

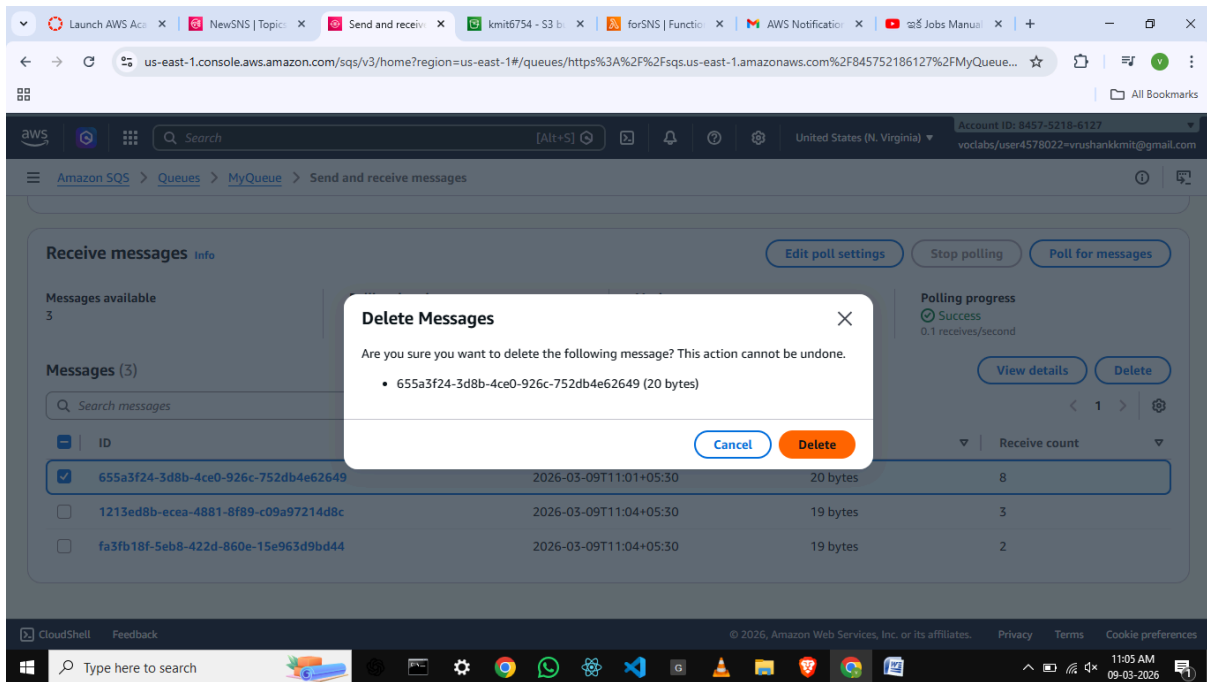
Queue policiesMonitoringSNS subscriptionsLambda triggersEventBridge PipesDead-letter queueTaggingEncryption

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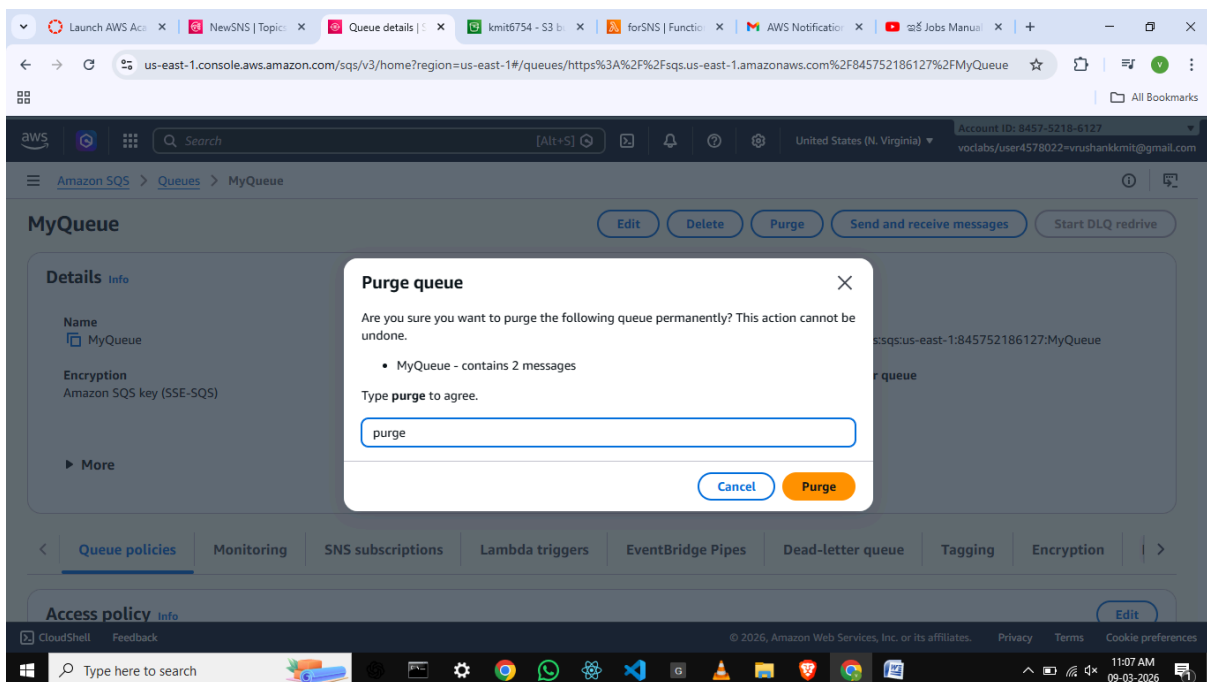
Type here to search11:01 AM09-03-2026



## Delete specific message



## Click purge to delete all messages



## Trigger SQS for S3

## Create topic in SNS

# Create Queue in SQS

## Create new Lambda function

## Create subscription

The screenshot shows the 'Create topic' page in the AWS Management Console. At the top, there's a blue banner for a 'New Feature' about High Throughput FIFO topics. Below this, the 'Create topic' section has a 'Details' tab. Under 'Type', the 'Standard' option is selected, showing its characteristics: Best-effort message ordering, At-least once message delivery, and Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints. The 'Name' field is filled with 'NewSQS'. A taskbar at the bottom shows various application icons and the system clock at 10:58 AM on 10-03-2026.

us-east-1.console.aws.amazon.com/sns/v3/home?region=us-east-1#/create-topic

Amazon SNS > Topics > Create topic

**New Feature**  
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

### Create topic

**Details**

Type | Info  
Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

**Name**

NewSQS

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (\_).

The screenshot shows the 'Create subscription' page in the AWS Management Console. It features the same 'New Feature' banner at the top. The 'Create subscription' section has a 'Details' tab. The 'Topic ARN' field is populated with 'arn:aws:sns:us-east-1:845752186127:NewSQS'. The 'Protocol' dropdown is set to 'Amazon SQS'. The 'Endpoint' field contains 'arn:aws:sqs:us-east-1:845752186127:MyQueue'. There is an unchecked checkbox for 'Enable raw message delivery'. The taskbar at the bottom is identical to the previous screenshot.

us-east-1.console.aws.amazon.com/sns/v3/home?region=us-east-1#/create-subscription

Amazon SNS > Subscriptions > Create subscription

**New Feature**  
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

### Create subscription

**Details**

**Topic ARN**

arn:aws:sns:us-east-1:845752186127:NewSQS

**Protocol**  
The type of endpoint to subscribe

Amazon SQS

**Endpoint**  
Only Amazon SQS standard queues will be listed and can receive notifications from an Amazon SNS standard topic.

arn:aws:sqs:us-east-1:845752186127:MyQueue

☐ Enable raw message delivery

## Edit access policy for topic SNS,SQS:

us-east-1.console.aws.amazon.com/sns/v3/home?region=us-east-1#/edit-topic/arn:aws:sns:us-east-1:845752186127:NewSQS

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankmit@gmail.com

Amazon SNS > Topics > NewSQS > Edit topic

**New Feature**  
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

This policy defines who can access your topic. By default, only the topic owner can publish or subscribe to the topic.

**JSON editor**

```
11 {
12 "Action": [
13 "SNS:Publish"
14],
15 "Resource": "arn:aws:sns:us-east-1:845752186127:NewSQS",
16 "Condition": {
17 "ArnLike": {
18 "aws:SourceArn": "arn:aws:s3:::newkmit6754"
19 },
20 "StringEquals": {
21 "aws:SourceAccount": "845752186127"
22 }
23 }
24 }
25 }
```

20:45 JSON

CloudShell Feedback

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Type here to search

us-east-1.console.aws.amazon.com/sqs/v3/home?region=us-east-1#/queues/https%3A%2F%2Fsqs.us-east-1.amazonaws.com%2F845752186127%2FMyQueue/edit

Account ID: 8457-5218-6127  
voclabs/user4578022=vrushankmit@gmail.com

Amazon SQS > Queues > MyQueue > Edit

☐ AWS Key Management Service key (SSE-KMS)  
An encryption key protected by AWS Key Management Service (AWS KMS).

**Access policy** [Info](#)  
Define who can access your queue.

**JSON editor**

```
1 {
2 "Version": "2008-10-17",
3 "Statement": [
4 {
5 "Effect": "Allow",
6 "Principal": {
7 "Service": "sns.amazonaws.com"
8 },
9 "Action": "sqs:SendMessage",
10 "Resource": "arn:aws:sqs:us-east-1:845752186127:MyQueue",
11 "Condition": {
12 "ArnEquals": {
13 "aws:SourceArn": "arn:aws:sns:us-east-1:845752186127:NewSQS"
14 }
15 }
16 }
17]
18 }
```

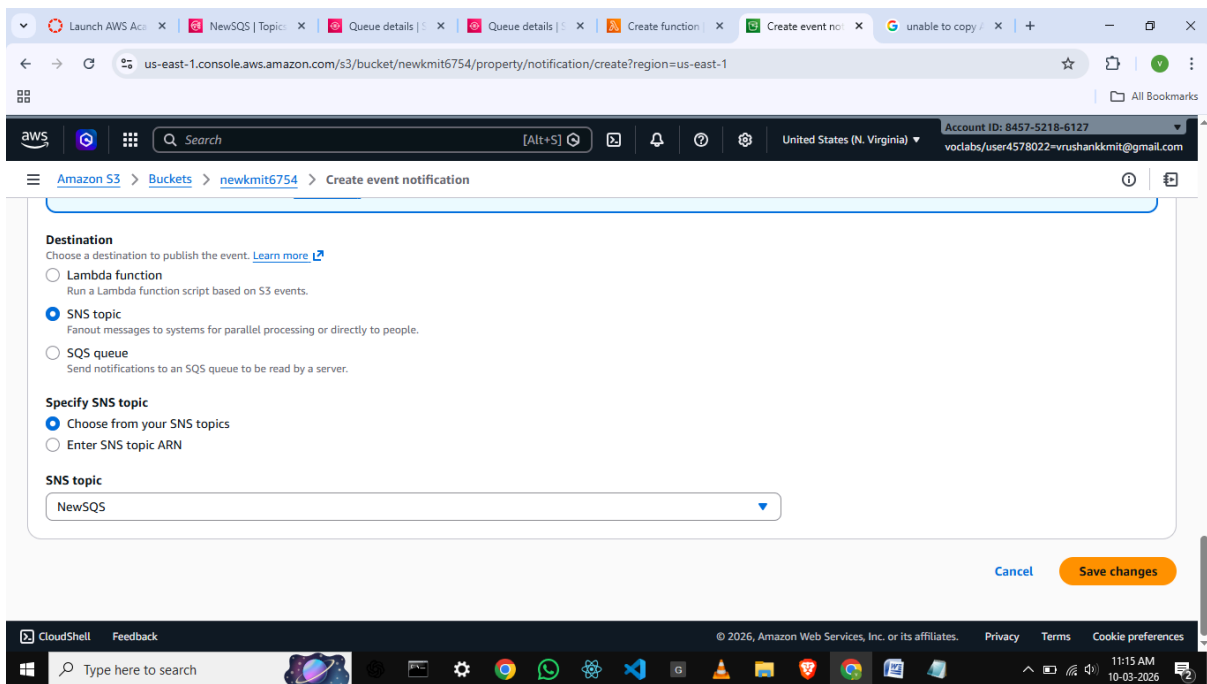
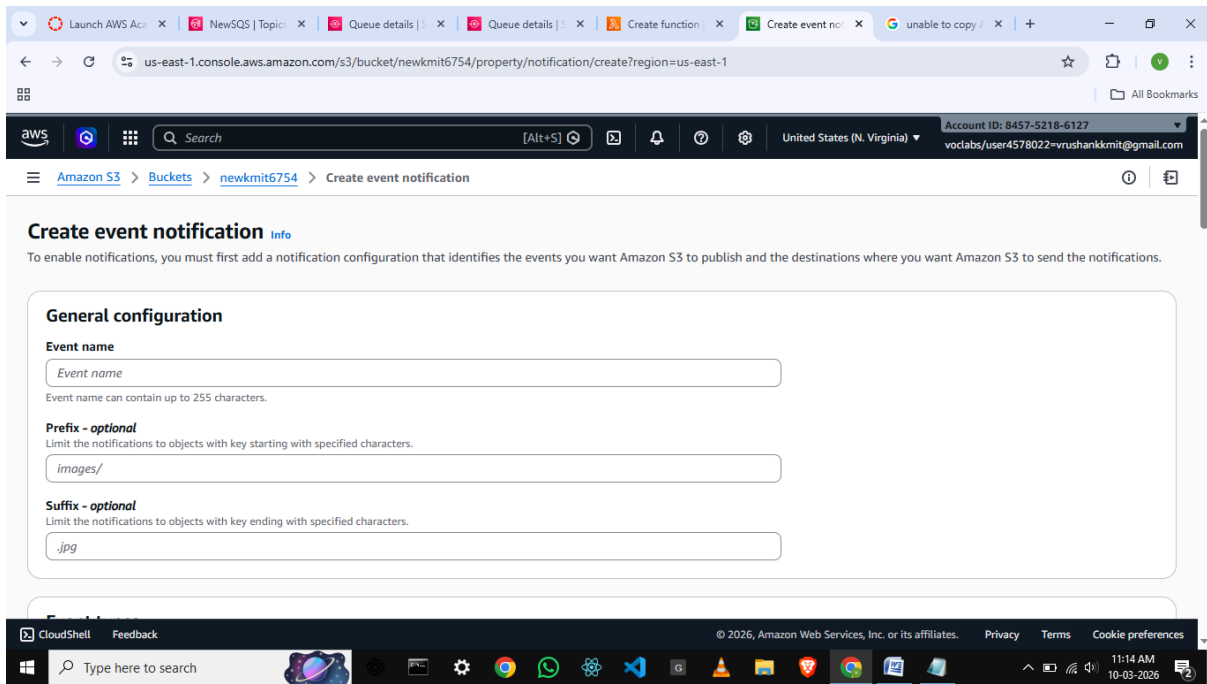
1:1 JSON

CloudShell Feedback

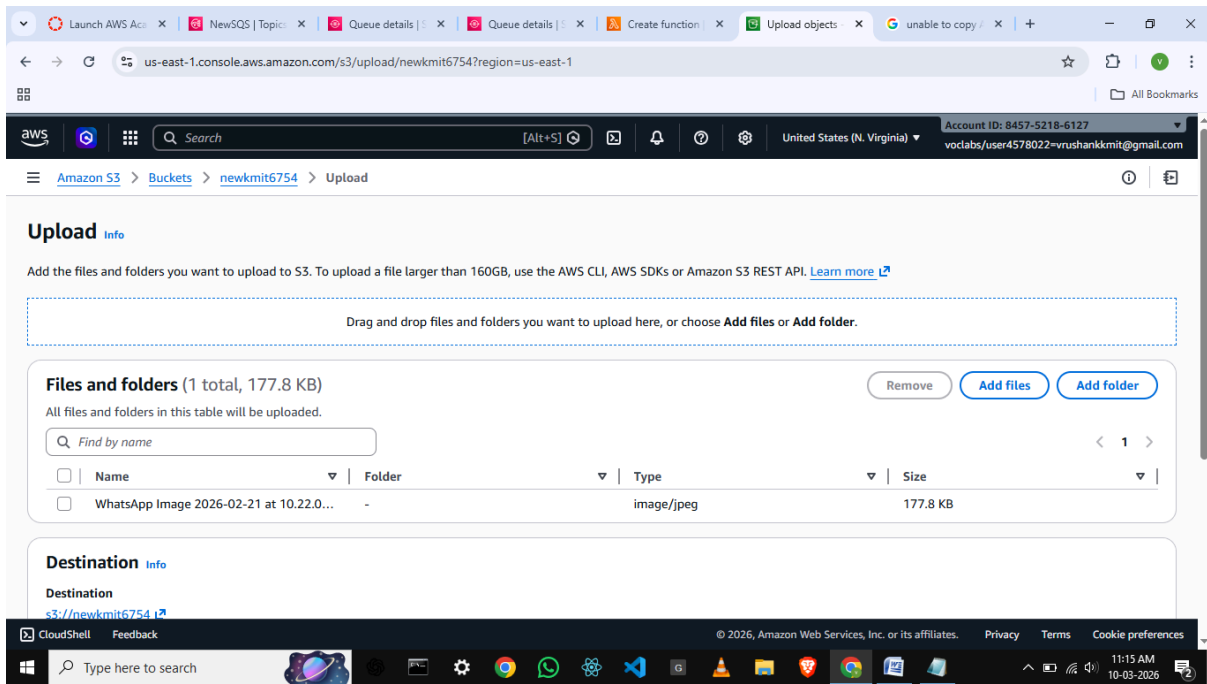
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Type here to search

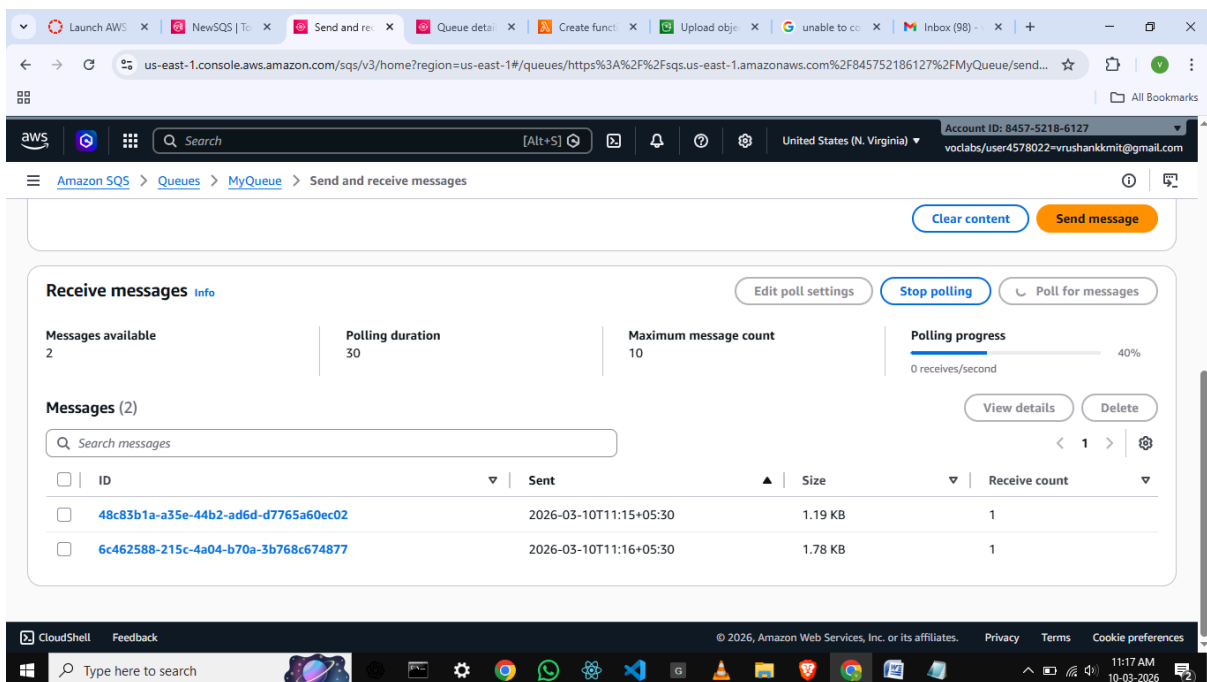
## Create Event Notification



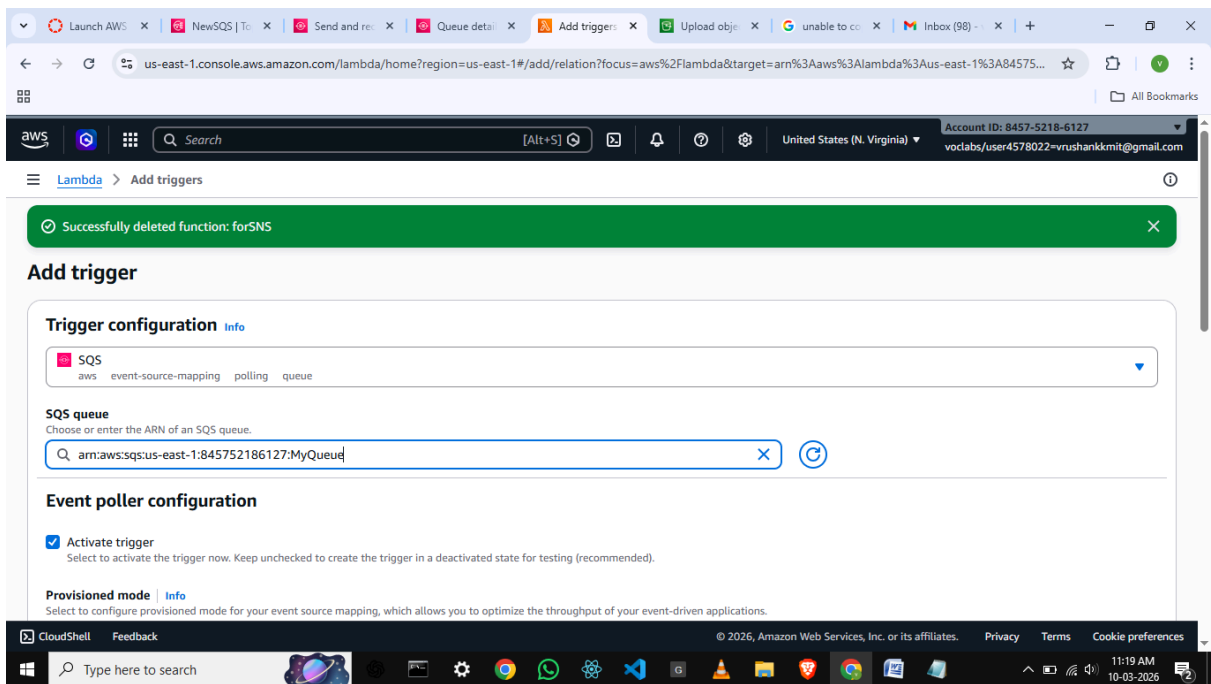
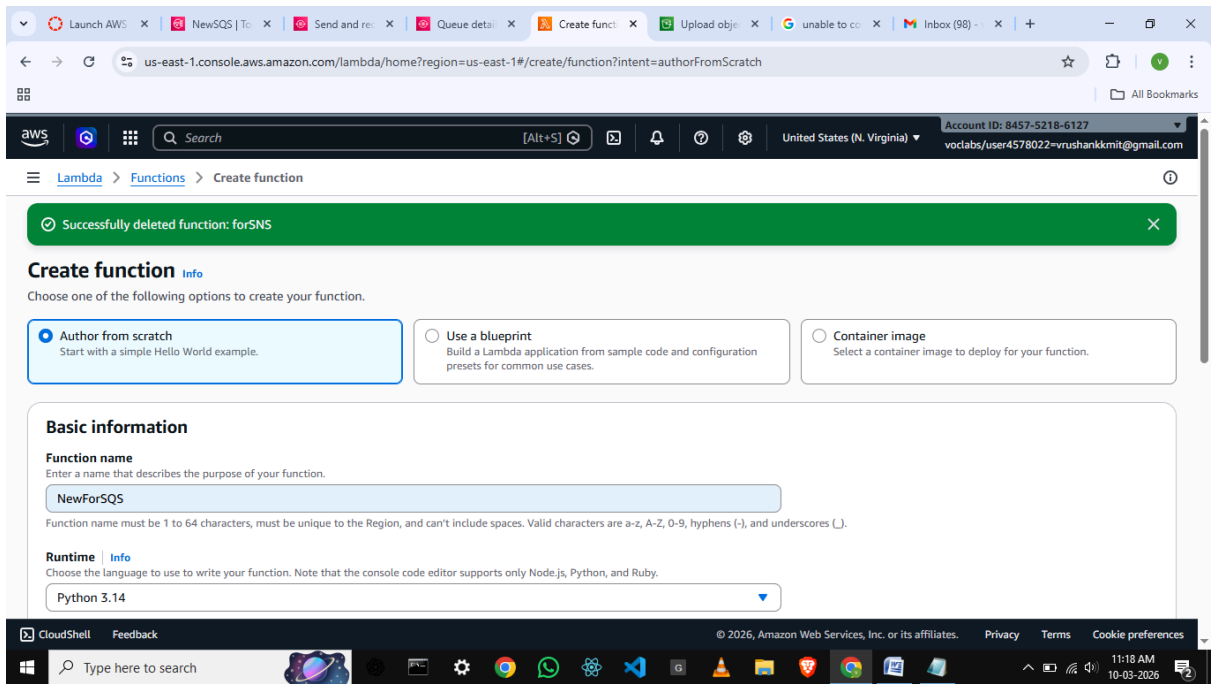
# Upload file in bucket

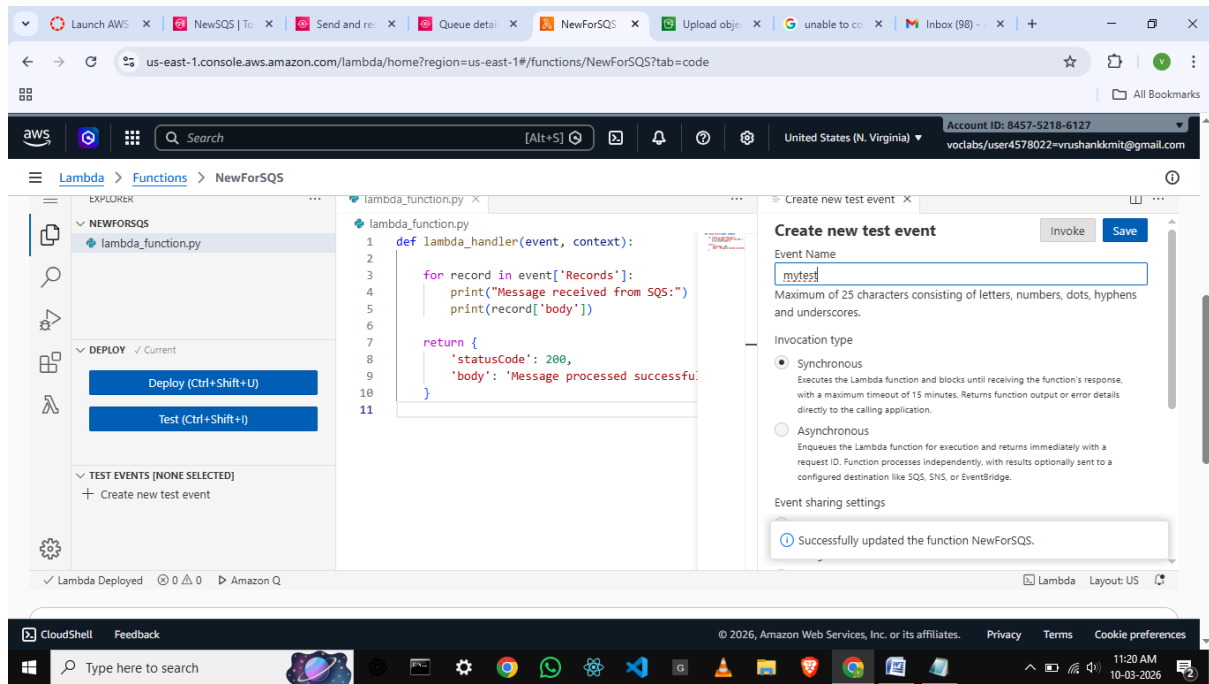


## We will get message in QUEUE poll



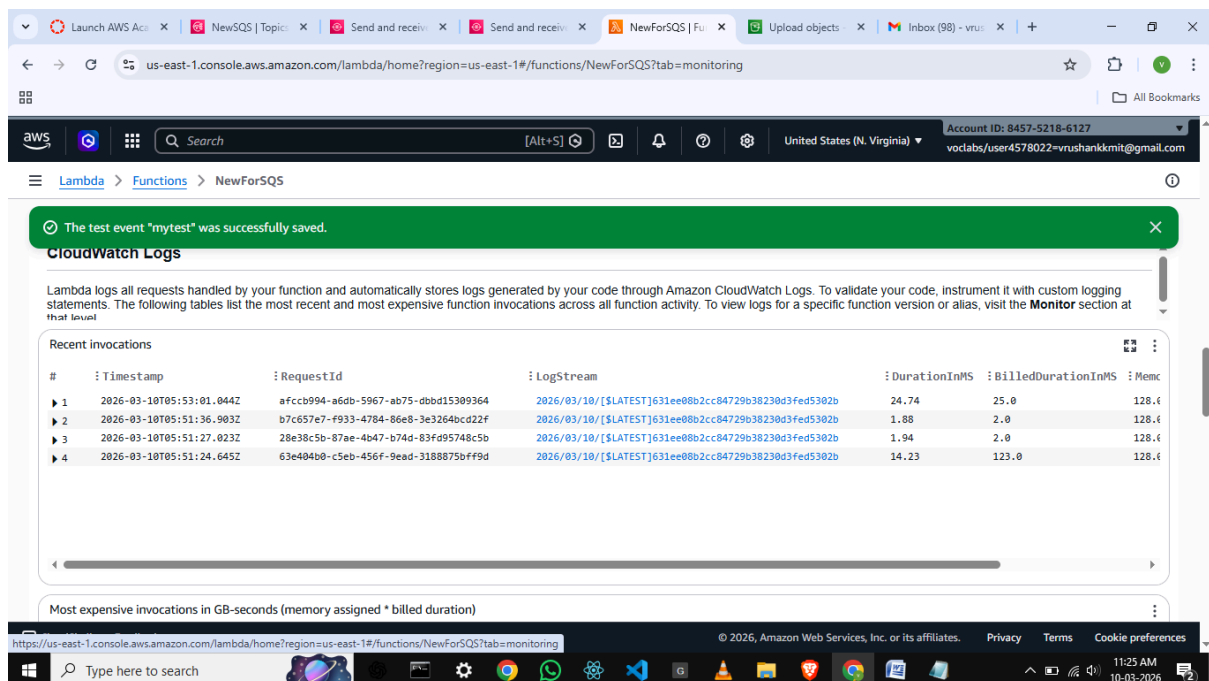
## Now lambda function





# Upload file in S3 again

# Check monitor



## ACCESSPOLICY FOR SQS:

```
{
```

```
"Statement": [
 {
 "Effect": "Allow",
 "Principal": {
 "Service": "sns.amazonaws.com"
 },
 "Action": "sqs:SendMessage",
 "Resource": "SQS ARN",
 "Condition": {
 "ArnEquals": {
 "aws:SourceArn": "SNS ARN"
 }
 }
 }
]
}
```

## **ACCESS POLICY FOR SNS:**

```
{
 "Version": "2012-10-17",
 "Id": "example-ID",
 "Statement": [
 {
 "Sid": "Example SNS topic policy",
 "Effect": "Allow",
 "Principal": {
 "Service": "s3.amazonaws.com"
 },
 "Action": [
 "SNS:Publish"
],
 "Resource": "SNS topic ARN",
 }
]
}
```

```
"Condition": {
 "ArnLike": {
 "aws:SourceArn": "S3 bucket ARN"
 },
 "StringEquals": {
 "aws:SourceAccount": "Account ID"
 }
}
}
}
]
}
```

## LAMBDA CODE:

```
def lambda_handler(event, context):
 for record in event['Records']:
 print("Message received from SQS:")
 print(record['body'])
 return {
 'statusCode': 200,
 'body': 'Message processed successfully'
 }
```

AmazonLEXchatbot

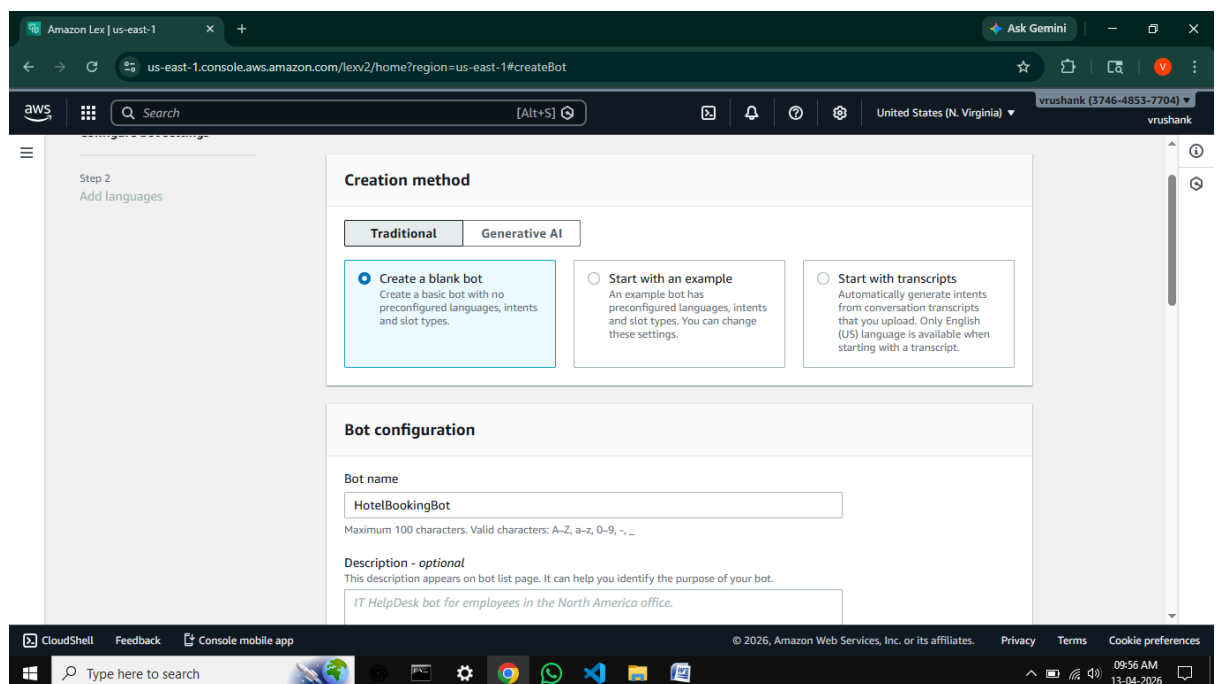
## WEEK-11—AmazonLEX

### Step 1: Open Amazon Lex

1. Login to AWS Console
  2. Search **Amazon Lex**
  3. Click **Create bot**
- 

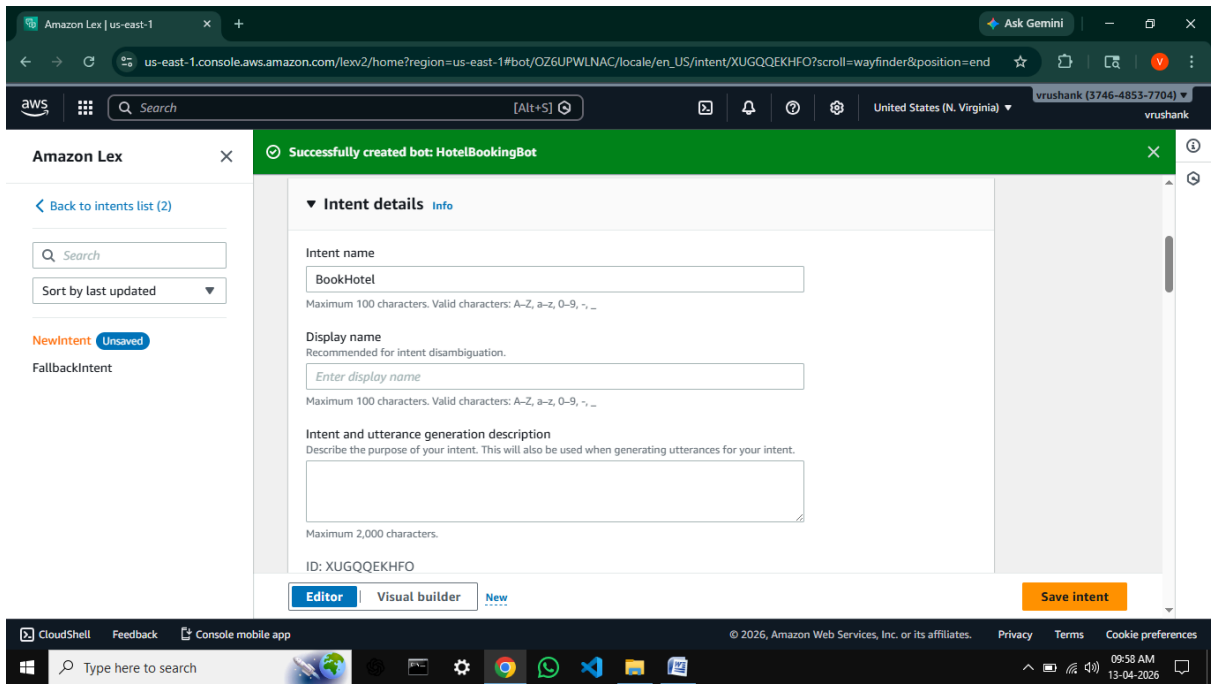
### Step 2: Create Bot

1. Choose:
  - **Create a blank bot**
2. Enter:
  - Bot name: HotelBookingBot
3. IAM role → Create new role
4. Remaining default → next
5. Language: English
6. Voice (optional)
7. Click **Done**



### Step 3: Create Intent

1. Go to **Intents**
2. Name: BookHotel
3. Click **Create**



#### Step 4: Add Sample Utterances

Add user sentences (how user talks):

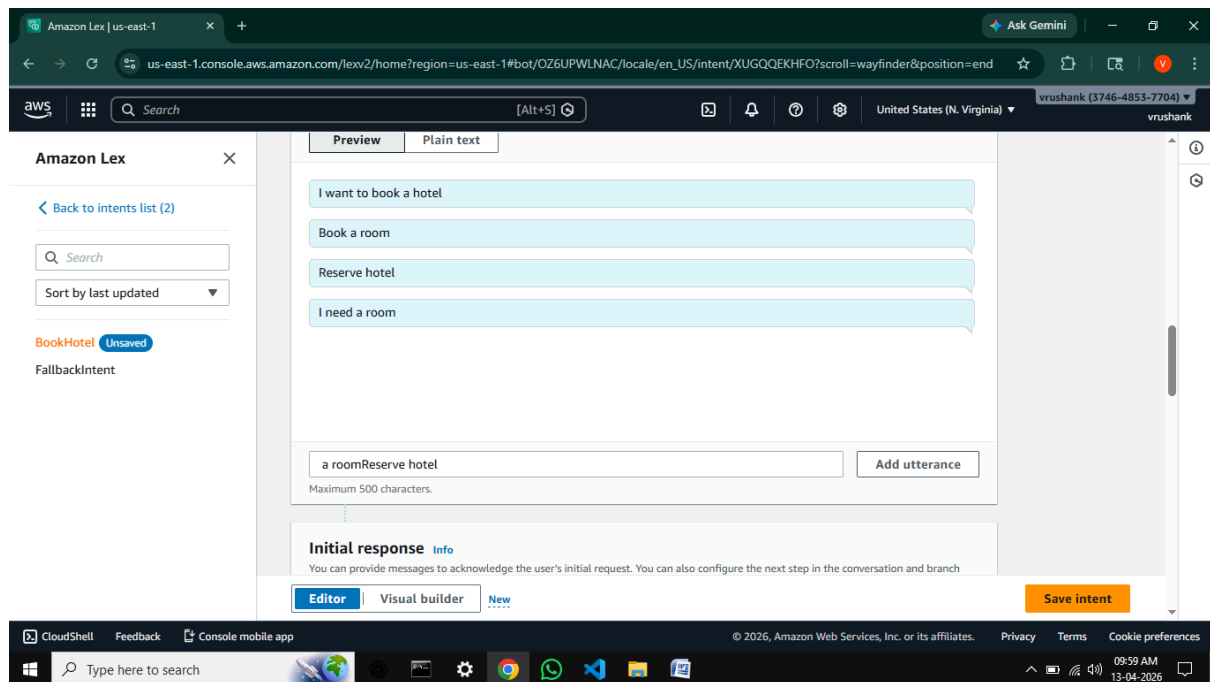
I want to book a hotel

Book a room

Reserve hotel

I need a room

These are **Utterances**

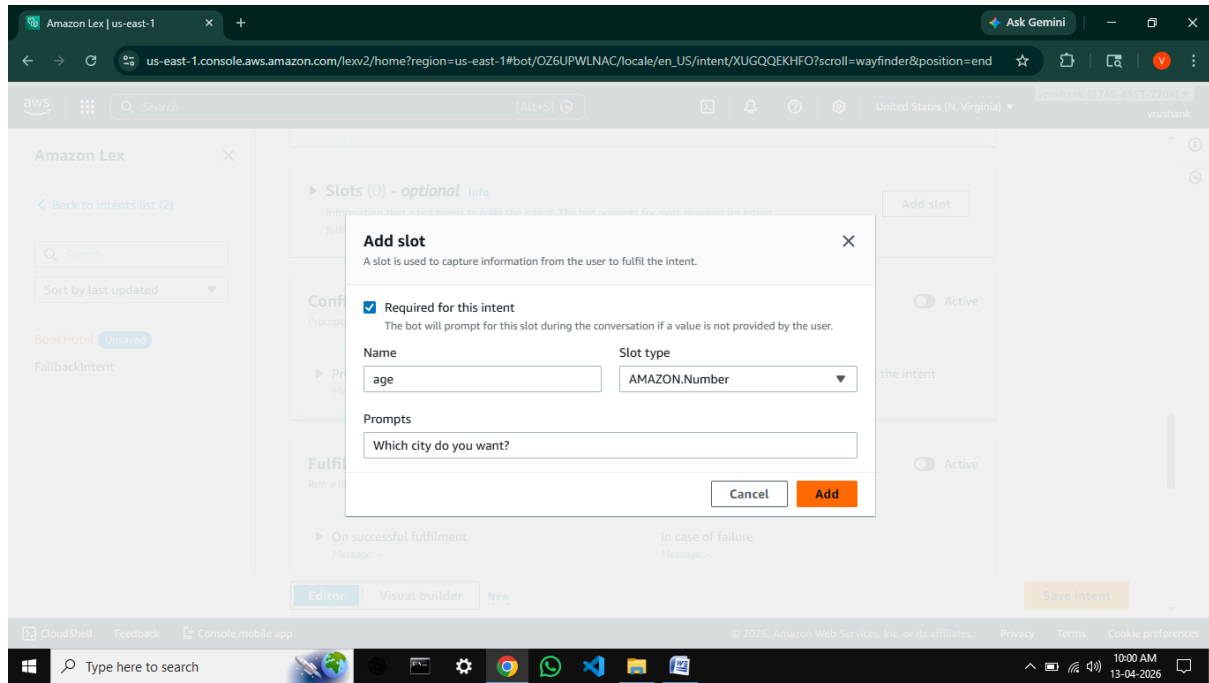


#### Step 5: Create Slots (User Inputs)

Slots collect user information.

## Add Slot

1. Click **Add slot**
2. Fill:
  - Slot name: age
  - Slot type: AMAZON.Number
  - Prompt: *What is your age?*



## Mark as Required

### Add IF Conditions (Conditional Logic)

Used to control flow based on slot values

### Example Condition:

If {age} < 18

### Steps:

1. Go to **Intent** → **Slots** -> under **Advance options**
2. Scroll to **Slot capture: success response** -> **Conditional branching**
3. Click **Add condition**

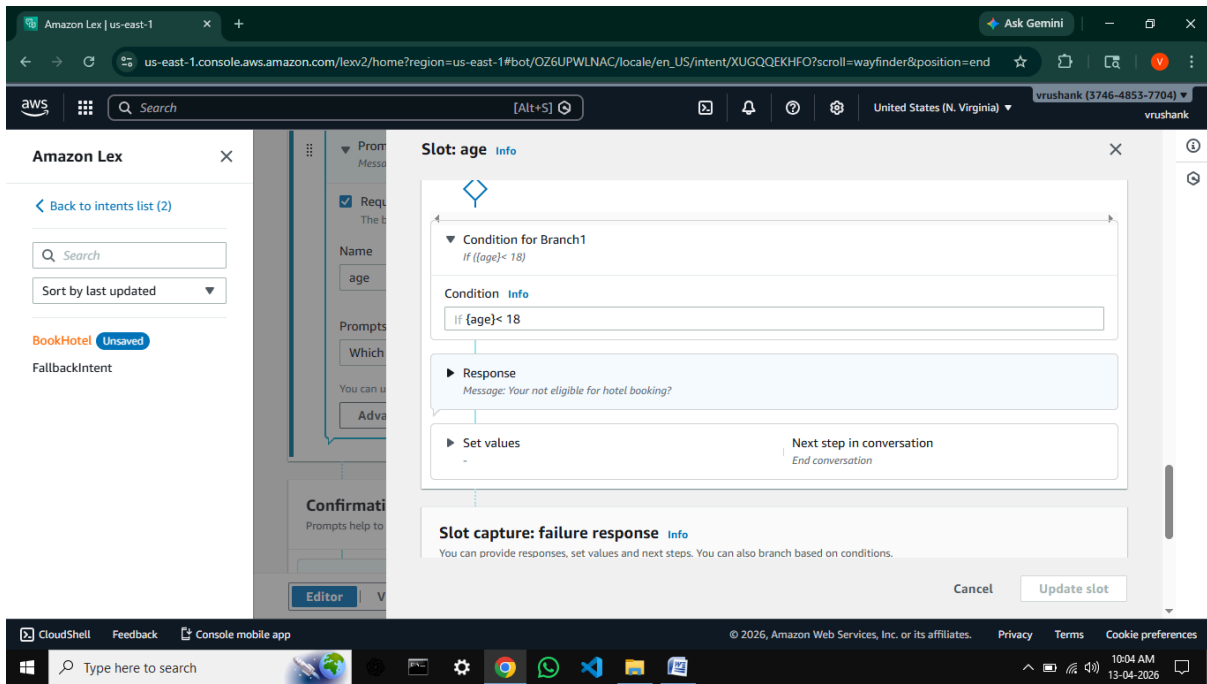
### Example:

#### Condition:

If {age} < 18

#### Response:

Your not eligible for hoot booking?

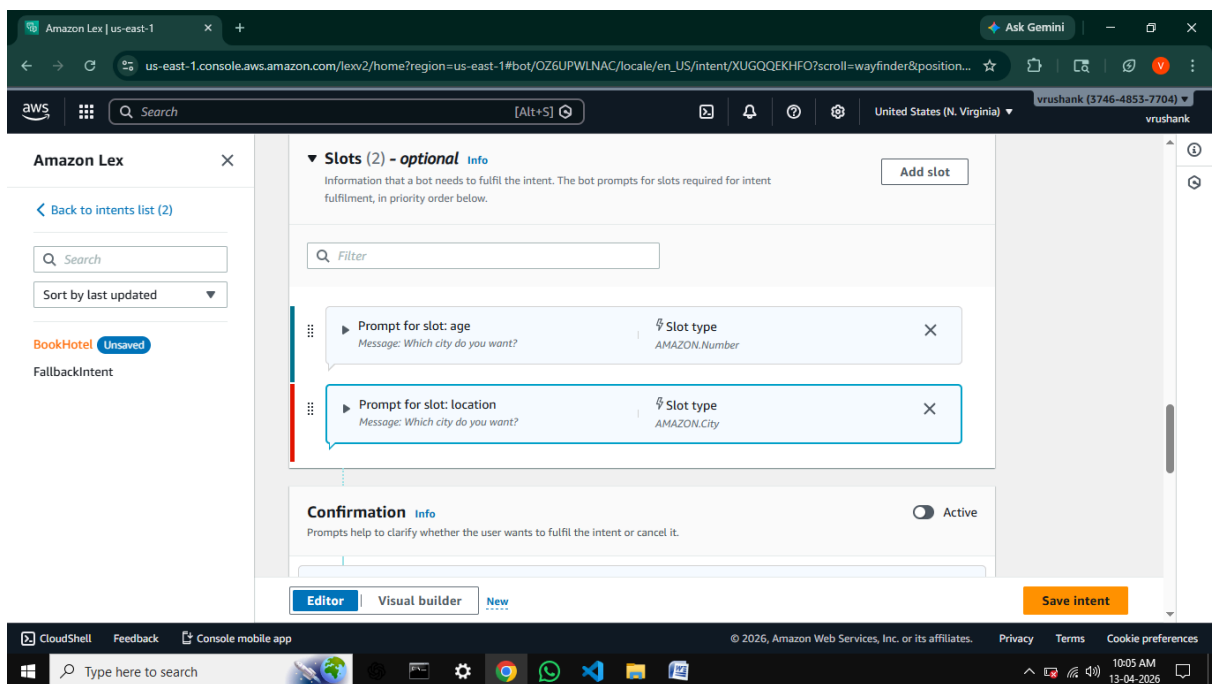


## Step 6: Create Slots (User Inputs)

Slots collect user information.

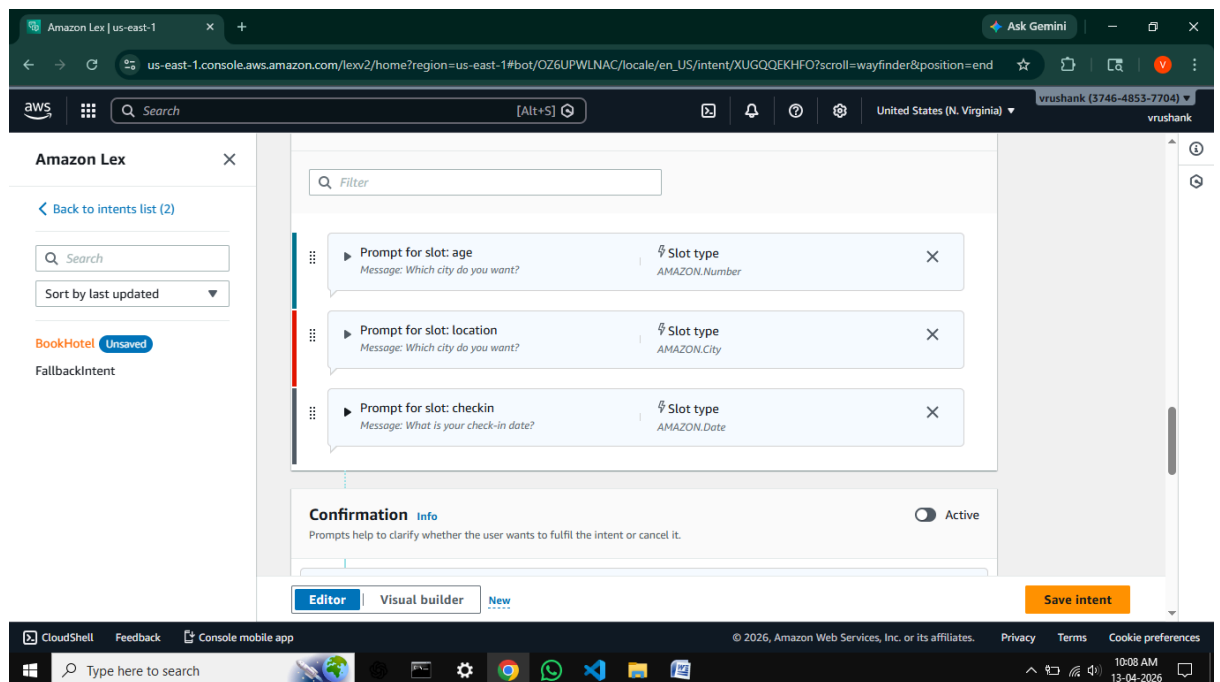
### Add Slot

3. Click **Add slot**
4. Fill:
  - Slot name: location
  - Slot type: AMAZON.City
  - Prompt: *Which city do you want?*
5. Mark as **Required**



Repeat for other slots.

1. Click **Add slot**
2. Fill details:
  - **Slot name:**checkin
  - **Slot type:**AMAZON.Date
  - **Prompt:**  
*"What is your check-in date?"*
3. Mark as **Required**  
(Optional but recommended)
  - **Add Retry prompt:**  
*"Please provide a valid date (e.g., 2026-03-25)"*
4. Click **Save**

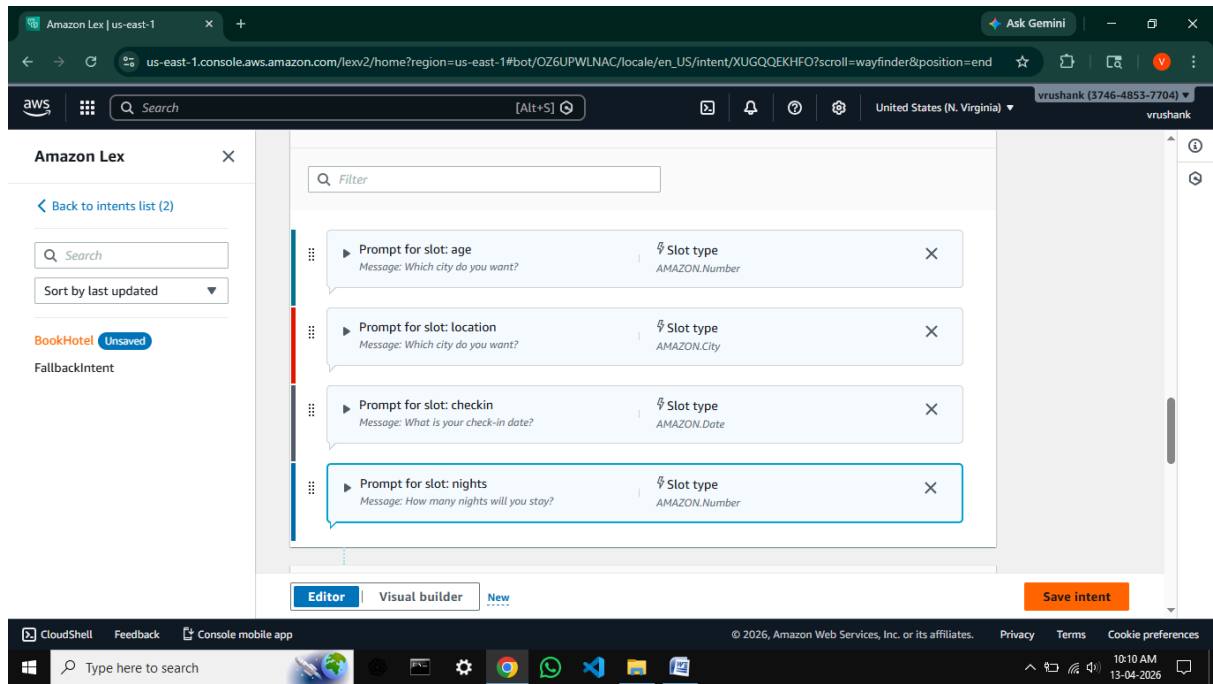


---

## Slot 7: Number of Nights (nights)

### Steps:

1. Click **Add slot** again
  2. Fill details:
    - **Slot name:** nights
    - **Slot type:**AMAZON.Number
    - **Prompt:**  
*"How many nights will you stay?"*
  3. Mark as **Required**
  4. Click **Save**
-



## Step 7: Create Custom Slot Type

### Save intent

#### Click on Back to intents

If you want custom values:

1. Go to **Slot Types**
2. Click **Add slot type** -> add blank slot type
3. Name: RoomType

Add values:

Single

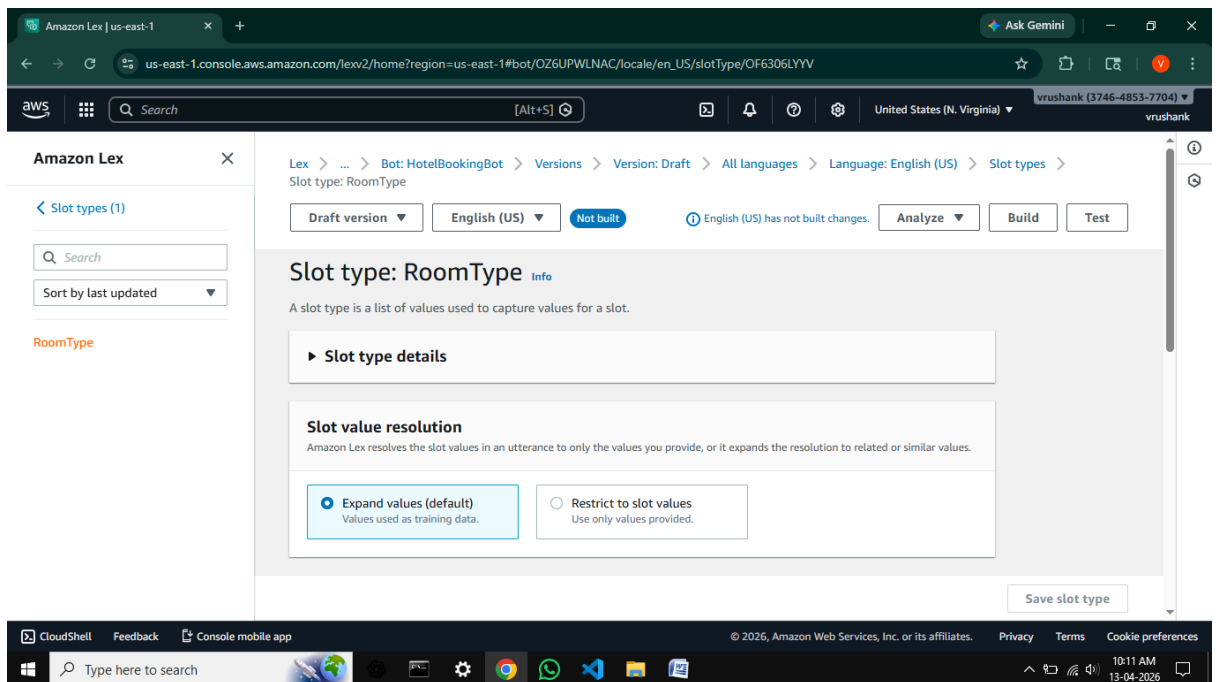
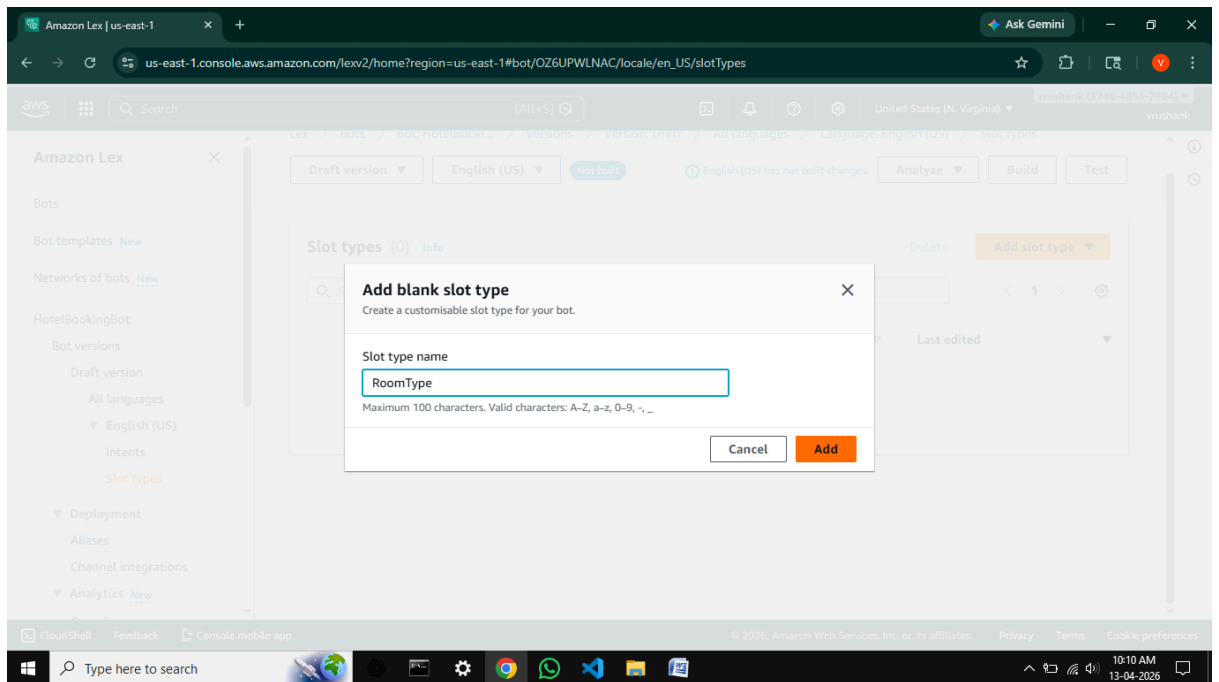
Double

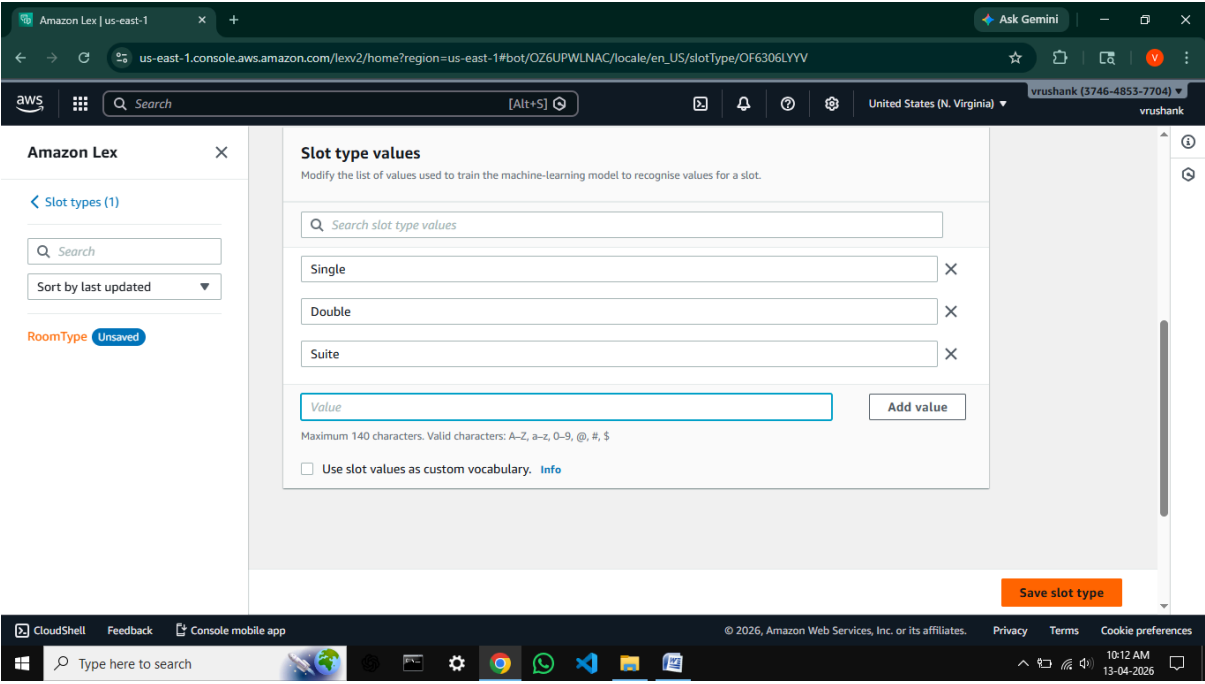
Suite

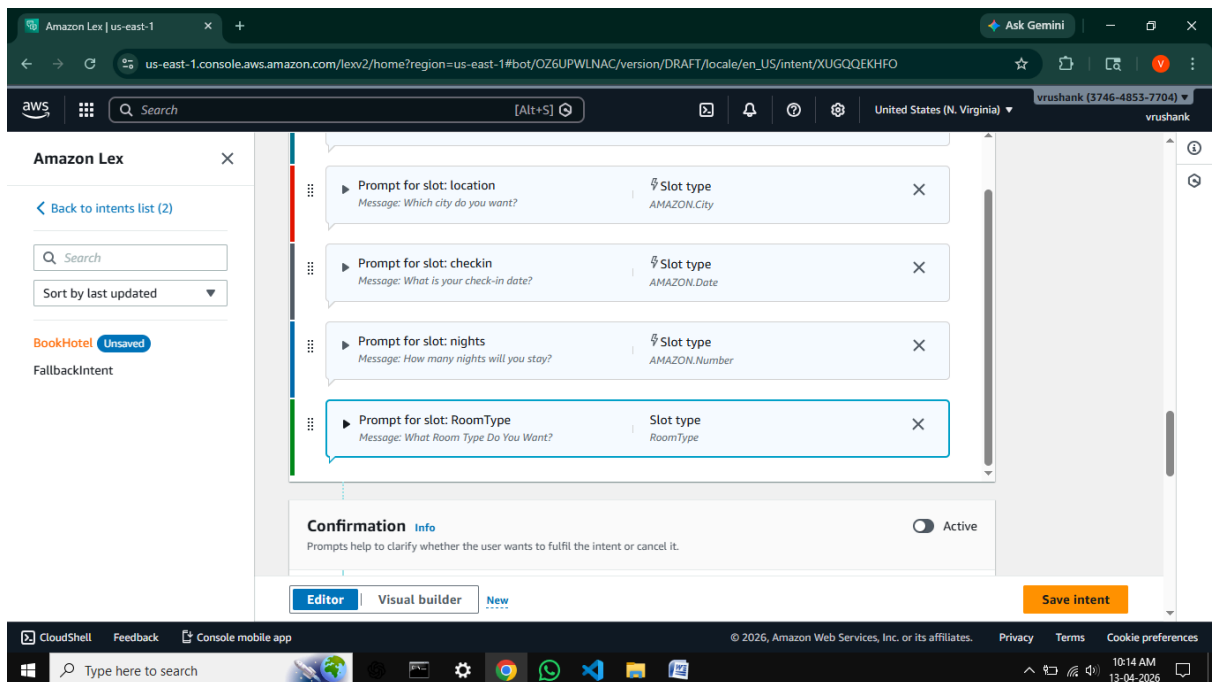
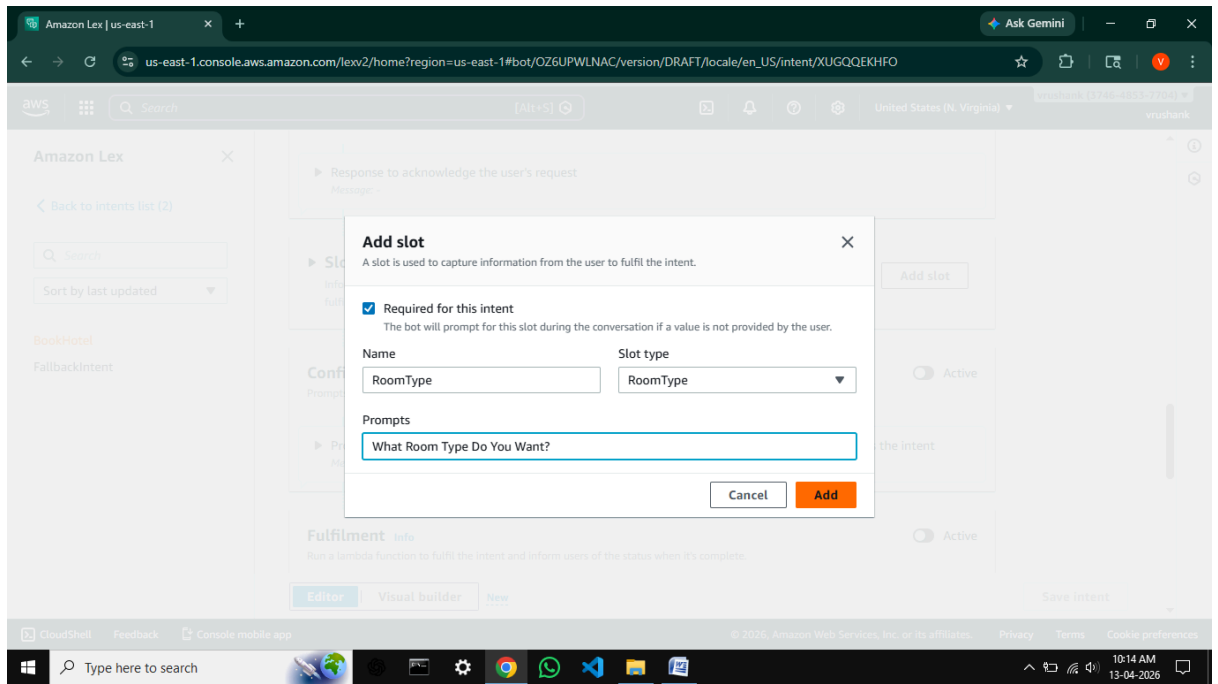
Save slot type

Now use this in slot: go back to intent

- Slot name: RoomType
- Slot type: RoomType







## Step 9: Add Response Cards (Card Groups)

Used for buttons (quick replies)

Steps:

1. Go to **slot prompts** -> **more promptoptions**
2. Click **Add** -> **add card groups**

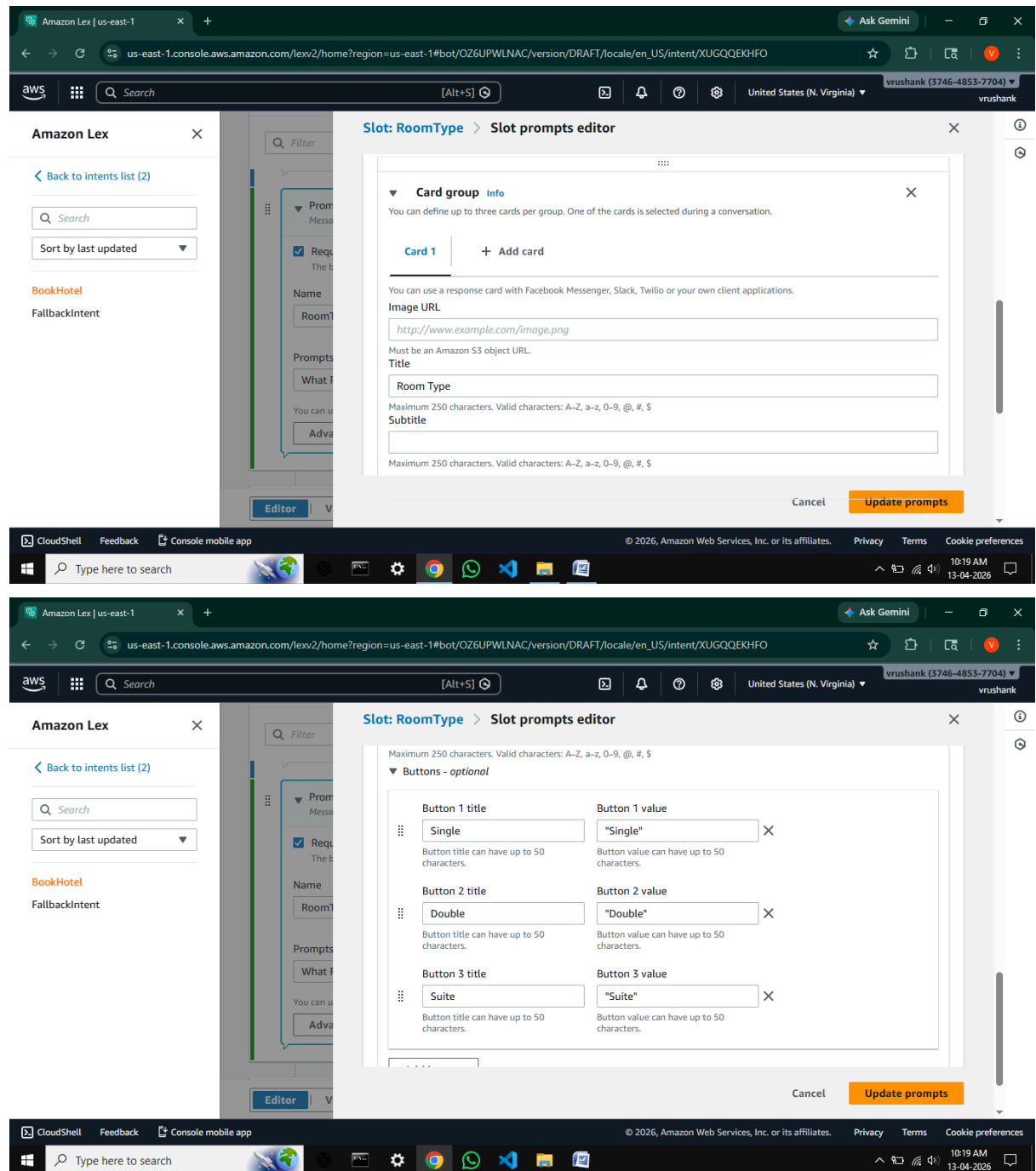
Example:

**Card Title:** Select Room Type

**click on add Buttons:**

- Single → "Single"
- Double → "Double"

- Suite → "Suite"



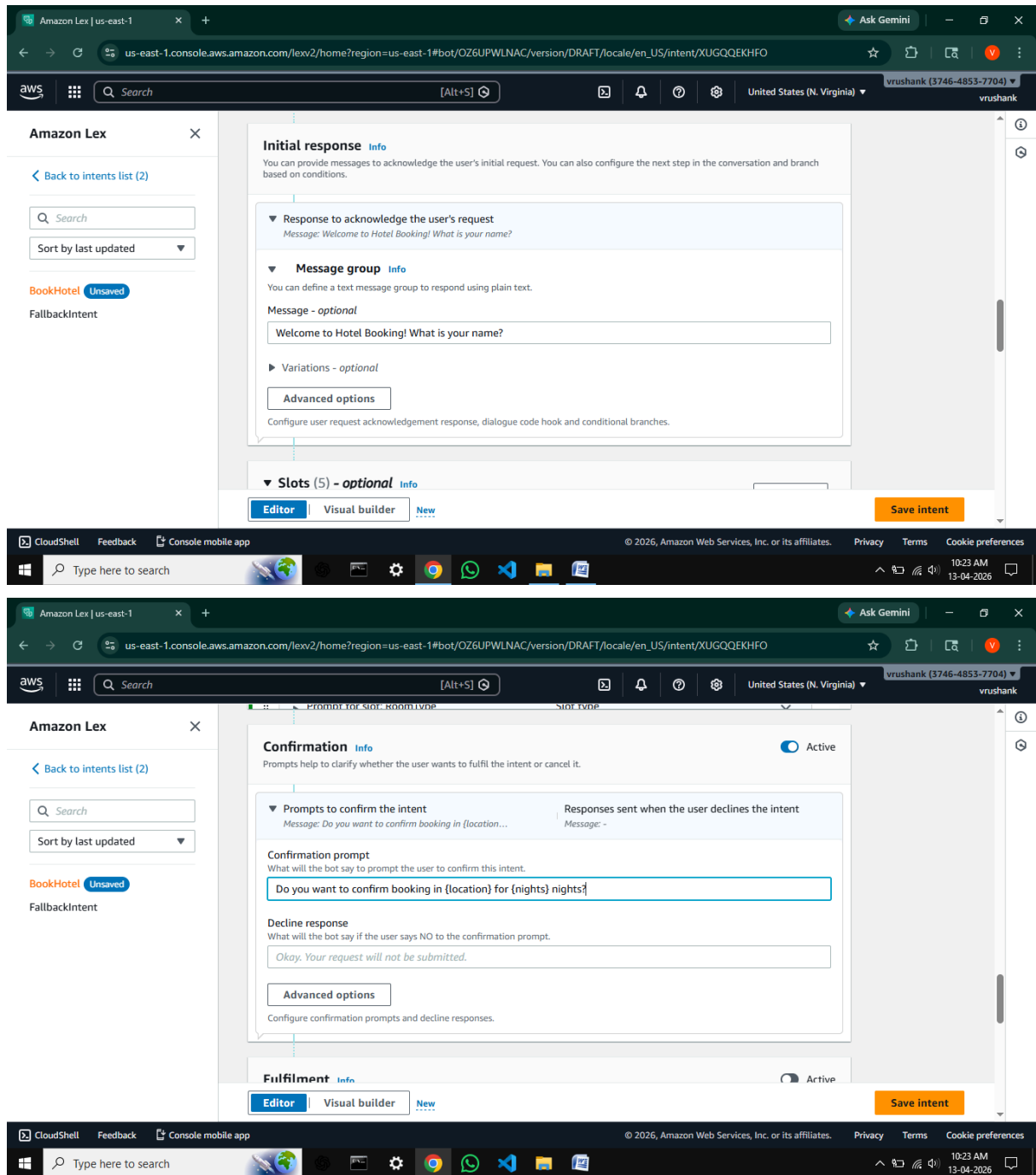
## Step 10: Configure Responses

### Initial Response:

Welcome to Hotel Booking! What is your name?

### Confirmation:

Do you want to confirm booking in {location} for {nights} nights?



## Step 11: Build & Test

1. Click **Build**
2. Use **Test chatbot** panel
3. Try:

Book a hotel

## Sample output

User: Book a hotel

Bot: What is your age?

User: 25

Bot: Which city do you want?

User: Hyderabad

Bot: Check-in date?

User: Tomorrow

Bot: Number of nights?

User: 2

Bot: Select room type

User: Double

Bot: Do you want to confirm booking in Hyderabad for 2 nights?

User: Yes

Bot: Booking confirmed.

Amazon Lex | us-east-1

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/OZ6UPWLNAC/version/DRAFT/locale/en\_US/intents

Amazon Lex

Bots

Bot templates [New](#)

Networks of bots [New](#)

HotelBookingBot

Bot versions

Draft version

All languages

English (US)

Intents

Slot types

Deployment

Aliases

Channel integrations

Analytics [New](#)

Building language English (US) in bot: HotelBookingBot. If your language contains external source slot types, the build might take longer to complete.

Lex > Bots > Bot: HotelBooki... > Versions > Version: DRAFT > All languages > Language: English (US) > Intents

Draft version English (US) Building Analyze Build Test

Intents (2) Info

An intent represents an action that the user wants to perform.

Search intents

	Name	Description	Last edited
<input checked="" type="radio"/>	BookHotel	-	Now
<input type="radio"/>	FallbackIntent	Default intent when no other intent matches	26 minutes ago

CloudShell Feedback Console mobile app

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Type here to search

10:24 AM 13-04-2026

Amazon Lex | us-east-1

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/OZ6UPWLNAC/version/DRAFT/locale/en\_US/intents

Amazon Lex

Bots

Bot templates [New](#)

Networks of bots [New](#)

HotelBookingBot

Bot versions

Draft version

All languages

English (US)

Intents

Slot types

Deployment

Aliases

Channel integrations

Analytics [New](#)

Successfully built language English (US) in bot: HotelBookingBot

Lex > Bots > Bot: HotelBooki... > Versions > Version: DRAFT > All languages > Language: English (US) > Intents

Draft version English (US) Successfully built

Intents (2) Info

An intent represents an action that the user wants to perform.

Search intents

	Name	Description
<input checked="" type="radio"/>	BookHotel	-
<input type="radio"/>	FallbackIntent	Default intent when no other intent matches

Test Draft version

Last build submitted: 1 minute ago

Inspect

Ready for complete testing

Type a message

CloudShell Feedback Console mobile app

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Type here to search

10:27 AM 13-04-2026

Amazon Lex | us-east-1

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/OZ6UPWLNAC/version/DRAFT/locale/en\_US/intents

Amazon Lex

Bots

Bot templates [New](#)

Networks of bots [New](#)

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Channel integrations

Analytics [New](#)

Successfully built language English (US) in bot: HotelBookingBot

Lex > Bots > Bot: HotelBooki... > Versions > Version: DRAFT > All languages > Language: English (US)

Draft version English (US) Successfully built

Intents (2) Info

An intent represents an action that the user wants to perform.

Search intents

	Name	Description
<input type="radio"/>	BookHotel	-
<input type="radio"/>	FallbackIntent	Default intent when no other intent matches

Test Draft version

Last build submitted: 1 minute ago

Inspect

Book a hotel

Welcome to Hotel Booking!  
What is your name?

What is your age?

25

Which city do you want?

Hyderabad

Ready for complete testing

Type a message

CloudShell Feedback Console mobile app

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Type here to search

10:33 AM 13-04-2026

Amazon Lex | us-east-1

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/OZ6UPWLNAC/version/DRAFT/locale/en\_US/intents

Amazon Lex

Bots

Bot templates [New](#)

Networks of bots [New](#)

HotelBookingBot

Bot versions

Draft version

All languages

English (US)

Intents

Slot types

Deployment

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Channel integrations

Analytics [New](#)

Successfully built language English (US) in bot: HotelBookingBot

Lex > Bots > Bot: HotelBooki... > Versions > Version: DRAFT > All languages > Language: English (US)

Draft version English (US) Successfully built

Intents (2) Info

An intent represents an action that the user wants to perform.

Search intents

	Name	Description
<input type="radio"/>	BookHotel	-
<input type="radio"/>	FallbackIntent	Default intent when no other intent matches

Test Draft version

Last build submitted: 1 minute ago

Inspect

25

Which city do you want?

Hyderabad

What is your check-in date?

Tomorrow

How many nights will you stay?

Ready for complete testing

Type a message

CloudShell Feedback Console mobile app

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Type here to search

10:33 AM 13-04-2026

Amazon Lex | us-east-1

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/OZ6UPWLNAC/version/DRAFT/locale/en\_US/intents

Search [Alt+S]

United States (N. Virginia) vrus Shank (3746-4853-7704) vrus Shank

**Amazon Lex**

- Bots
- Bot templates [New](#)
- Networks of bots [New](#)
- HotelBookingBot
  - Bot versions
    - Draft version
      - All languages
        - English (US)
          - Intents**
          - Slot types
    - Deployment
      - Aliases
      - Channel integrations
    - Analytics [New](#)

**Successfully built language English (US) in bot: HotelBookingBot**

Lex > Bots > Bot: HotelBooki... > Versions > Version: DRAFT > All languages > Language: English (US)

Draft version English (US) Successfully built

**Intents (2)** Info

An intent represents an action that the user wants to perform.

Search intents

	Name	Description
<input type="radio"/>	BookHotel	-
<input type="radio"/>	FallbackIntent	Default intent when no other intent matches

**Test Draft version**  
Last build submitted: 1 minute ago

Inspect

**Room Type**

Single Double Suite

"Double"

Ready for complete testing

Type a message

CloudShell Feedback Console mobile app

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Type here to search

10:34 AM 13-04-2026

Amazon Lex | us-east-1

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bot/OZ6UPWLNAC/version/DRAFT/locale/en\_US/intents

Search [Alt+S]

United States (N. Virginia) vrus Shank (3746-4853-7704) vrus Shank

**Amazon Lex**

- Bots
- Bot templates [New](#)
- Networks of bots [New](#)
- HotelBookingBot
  - Bot versions
    - Draft version
      - All languages
        - English (US)
          - Intents**
          - Slot types
    - Deployment
      - Aliases
      - Channel integrations
    - Analytics [New](#)

**Successfully built language English (US) in bot: HotelBookingBot**

Lex > Bots > Bot: HotelBooki... > Versions > Version: DRAFT > All languages > Language: English (US)

Draft version English (US) Successfully built

**Intents (2)** Info

An intent represents an action that the user wants to perform.

Search intents

	Name	Description
<input type="radio"/>	BookHotel	-
<input type="radio"/>	FallbackIntent	Default intent when no other intent matches

**Test Draft version**  
Last build submitted: 2 minutes ago

Inspect

"Double"

Do you want to confirm booking in Hyderabad for 2 nights?

Yes

Intent BookHotel is fulfilled

Ready for complete testing

Type a message

CloudShell Feedback Console mobile app

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Type here to search

10:34 AM 13-04-2026

IAM

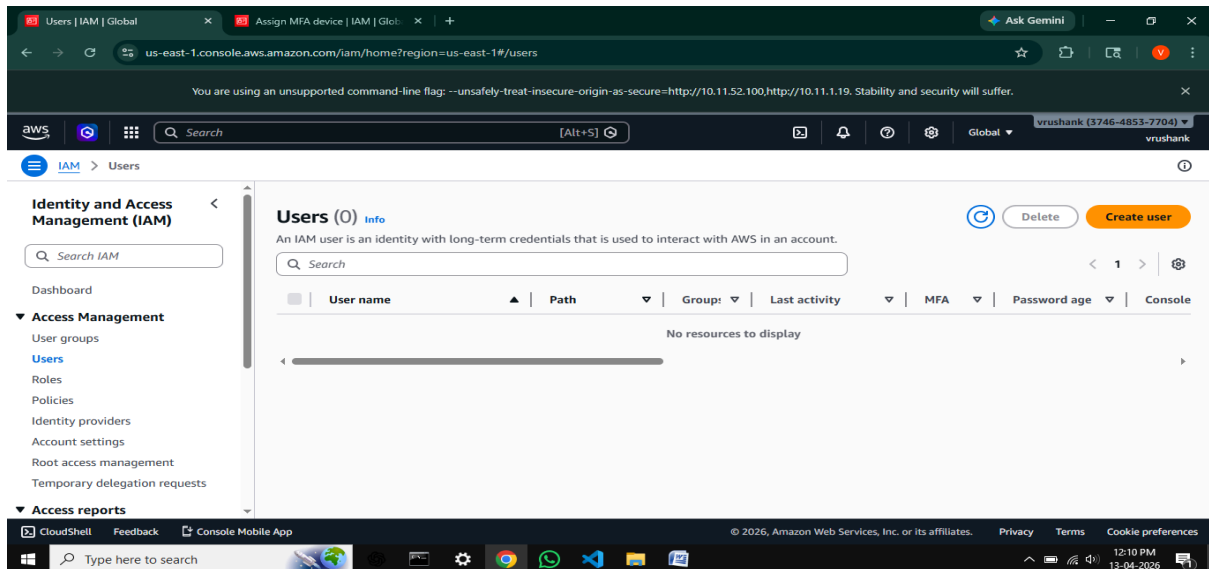
# WEEK-12

## IAM

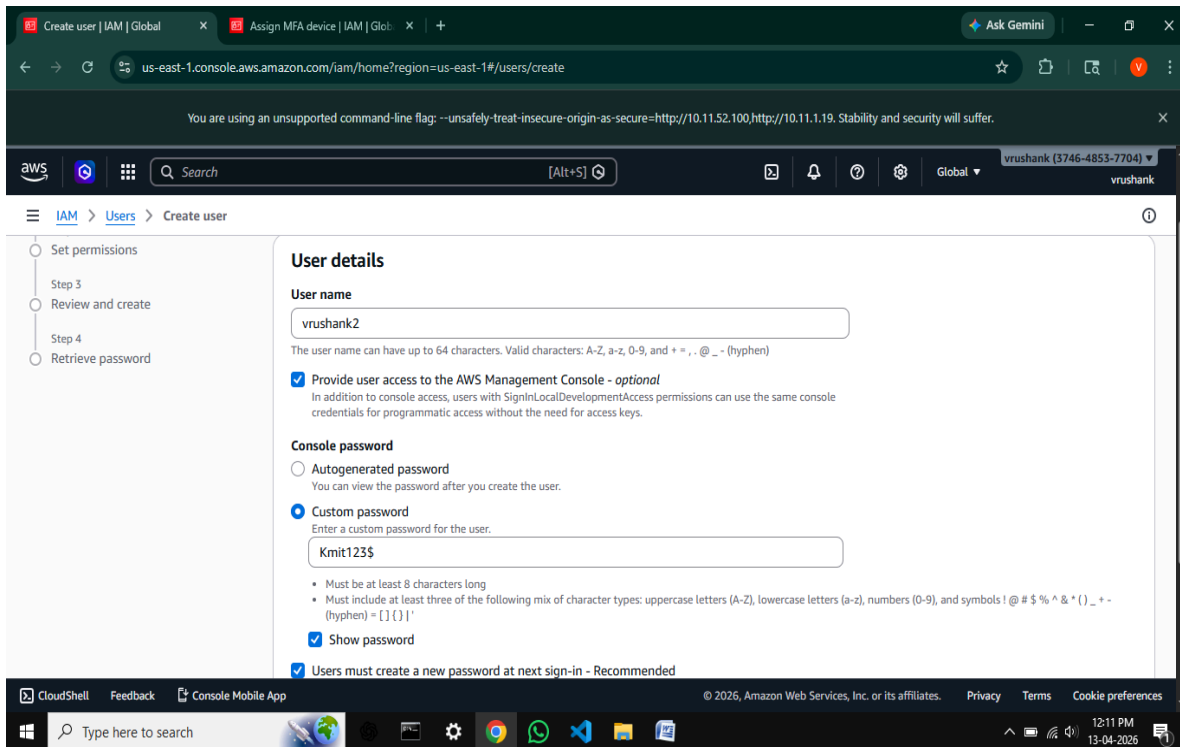
### 1. Administrative Setup (Root User)

Step 1: Log in to the AWS Management Console as the Root User.

Step 2: Search for IAM under the "Security, Identity, & Compliance" category. Step 3: Navigate to Users and click Create user.



Step 4: User Details: Enter a name (e.g., S3\_Specialist). Check the box for "Provide user access to the AWS Management Console." – custom password.



us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/create

You are using an unsupported command-line flag: --unsafely-treat-insecure-origin-as-secure=http://10.11.52.100,http://10.11.1.19. Stability and security will suffer.

aws Search [Alt+S]

vrushank (3746-4853-7704) vrushank

IAM > Users > Create user

Set permissions  
Step 3  
Review and create  
Step 4  
Retrieve password

### User details

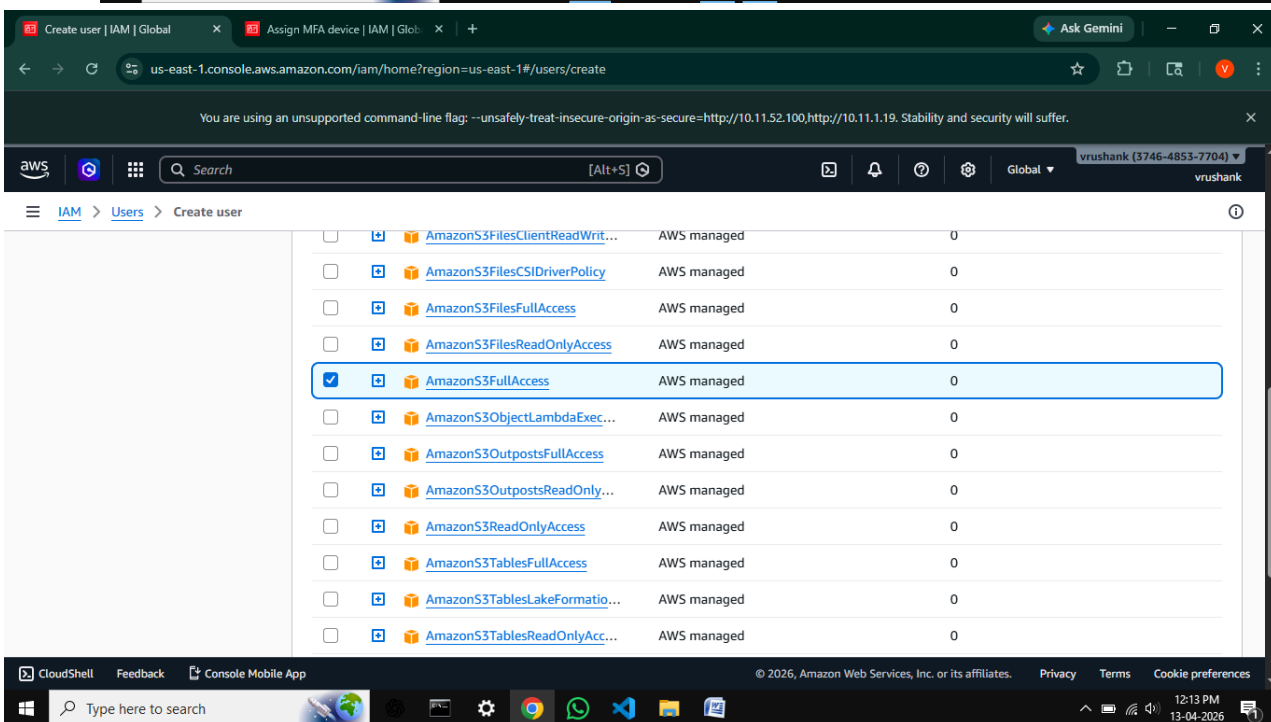
**User name**  
vrushank2  
The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ \_ - (hyphen)

☒ **Provide user access to the AWS Management Console - optional**  
In addition to console access, users with SignInLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.

**Console password**  
☐ Autogenerated password  
You can view the password after you create the user.  
☒ **Custom password**  
Enter a custom password for the user.  
Kmit123\$  
• Must be at least 8 characters long  
• Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & \* ( ) \_ + - (hyphen) = [ ] { } | '  
☒ Show password

☒ Users must create a new password at next sign-in - Recommended

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us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/create

You are using an unsupported command-line flag: --unsafely-treat-insecure-origin-as-secure=http://10.11.52.100,http://10.11.1.19. Stability and security will suffer.

aws Search [Alt+S]

vrushank (3746-4853-7704) vrushank

IAM > Users > Create user

<input type="checkbox"/>	AmazonS3FilesClientReadWrit...	AWS managed	0
<input type="checkbox"/>	AmazonS3FilesCSIDriverPolicy	AWS managed	0
<input type="checkbox"/>	AmazonS3FilesFullAccess	AWS managed	0
<input type="checkbox"/>	AmazonS3FilesReadOnlyAccess	AWS managed	0
<input checked="" type="checkbox"/>	AmazonS3FullAccess	AWS managed	0
<input type="checkbox"/>	AmazonS3ObjectLambdaExec...	AWS managed	0
<input type="checkbox"/>	AmazonS3OutpostsFullAccess	AWS managed	0
<input type="checkbox"/>	AmazonS3OutpostsReadOnly...	AWS managed	0
<input type="checkbox"/>	AmazonS3ReadOnlyAccess	AWS managed	0
<input type="checkbox"/>	AmazonS3TablesFullAccess	AWS managed	0
<input type="checkbox"/>	AmazonS3TablesLakeFormatio...	AWS managed	0
<input type="checkbox"/>	AmazonS3TablesReadOnlyAcc...	AWS managed	0

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Step 6: Review & Create: Complete the wizard. On the final page, click Download .csv file  
Now you see CC-LAB user created

**Step 7: Copy your 12-digit Account ID (found in the top-right dropdown or the .csv) and Sign Out.**

[illegible]

**Test 1 (Failure):** Navigate to **EC2**. Try to view instances. You will see "API Error" or "Access Denied." This proves the user is restricted.

The screenshot displays the AWS Management Console for the EC2 service in the eu-north-1 region. A blue notification bar at the top states: "You can change your default landing page for EC2. Permanently dismiss Change landing page".

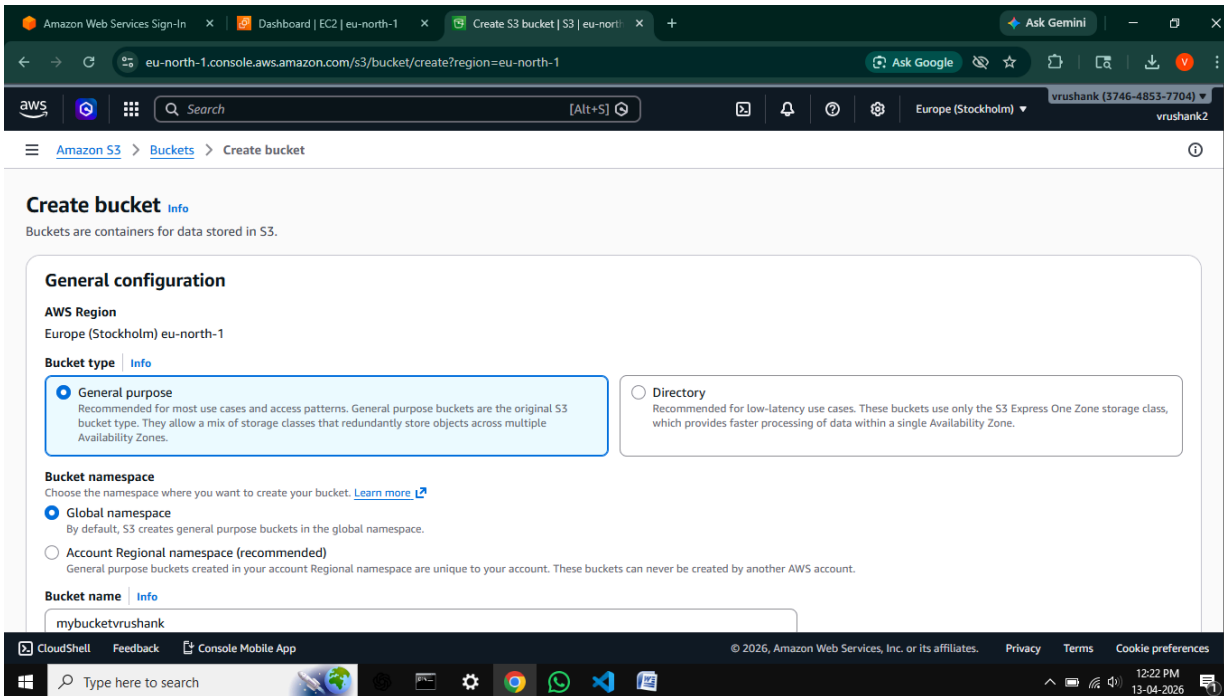
The main content area is titled "Resources" and lists various EC2 resources in the Europe (Stockholm) Region. The resources are organized into two columns:

Resource	Status
Instances (running)	0
Auto Scaling Groups	API Error
Capacity Reservations	API Error
Dedicated Hosts	API Error
Elastic IPs	API Error
Instances	API Error
Key pairs	API Error
Load balancers	API Error
Placement groups	API Error
Security groups	API Error
Snapshots	API Error
Volumes	API Error

On the right side, the "EC2 cost" section shows the date range as "Past 6 months" and the region as "Global". The "Total cost" and "Regions" are both marked as "Unable to load".

The bottom of the console shows the "Launch instance" button, "Service health", and "Account attributes". The footer includes the copyright notice "© 2026, Amazon Web Services, Inc. or its affiliates." and links for "Privacy", "Terms", and "Cookie preferences". The system clock indicates the time is 12:22 PM on 13-04-2026.

**Test 2 (Success):** Navigate to **S3**. Create a new bucket. It will work perfectly because of the attached policy.



Amazon Web Services Sign-InDashboard | EC2 | eu-north-1mybucketvrushank - S3 bucket +Ask Gemini

eu-north-1.console.aws.amazon.com/s3/buckets/mybucketvrushank?region=eu-north-1&tab=objectsAsk Google

awsSearch[Alt+S]Europe (Stockholm)vrushank (3746-4853-7704)vrushank2

Amazon S3Bucketsmybucketvrushank

Successfully created bucket "mybucketvrushank"

mybucketvrushankInfo

ObjectsMetadataPropertiesPermissionsMetricsManagementFile systems - newAccess Points

Objects (0)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

	Name	Type	Last modified	Size	Storage class
No objects					
You don't have any objects in this bucket.					
Upload					

CloudShellFeedbackConsole Mobile App

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Type here to search

12:24 PM13-04-2026

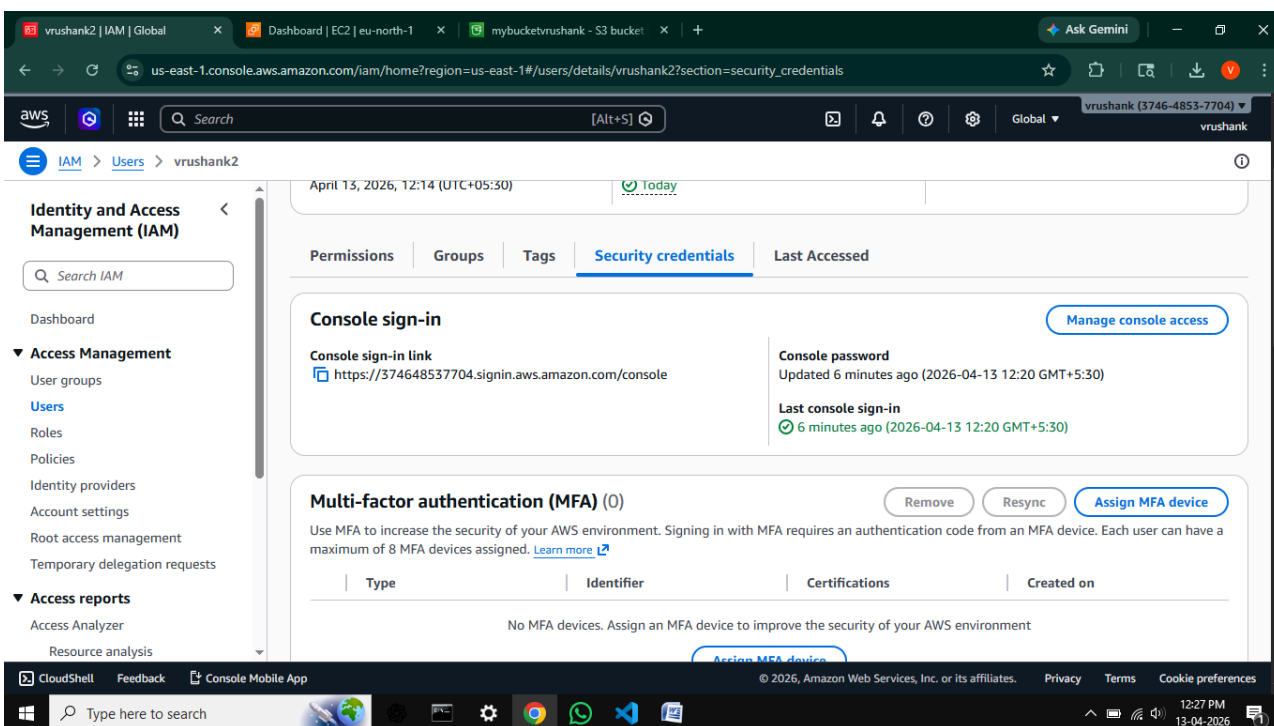
## Part B: CLI Access (Programmatic Access)

Objective: Access AWS resources from a standalone system (CMD/Terminal) using security keys.

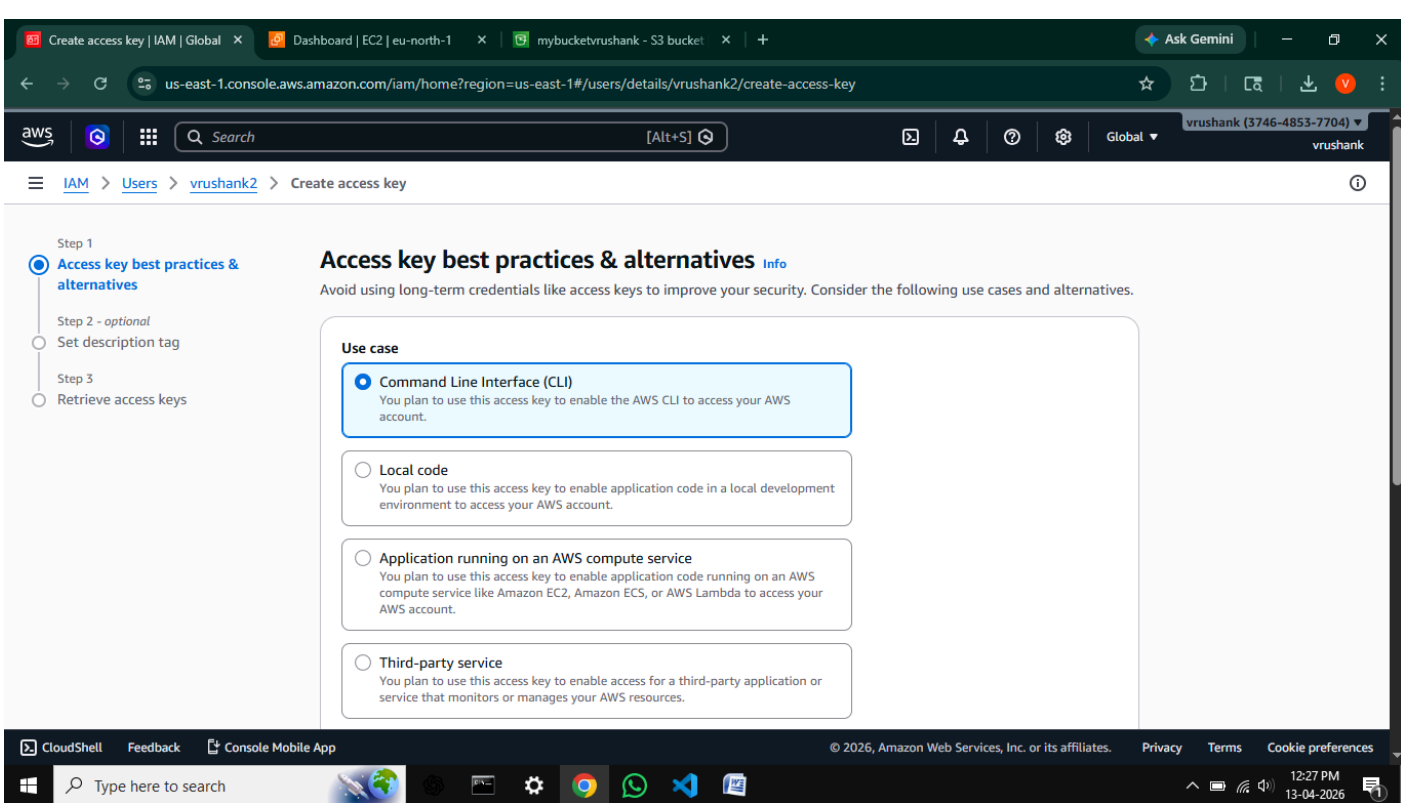
### 1. Credential Generation

Step 1: While logged in as an Admin/Root, go to IAM > Users and click on your user.

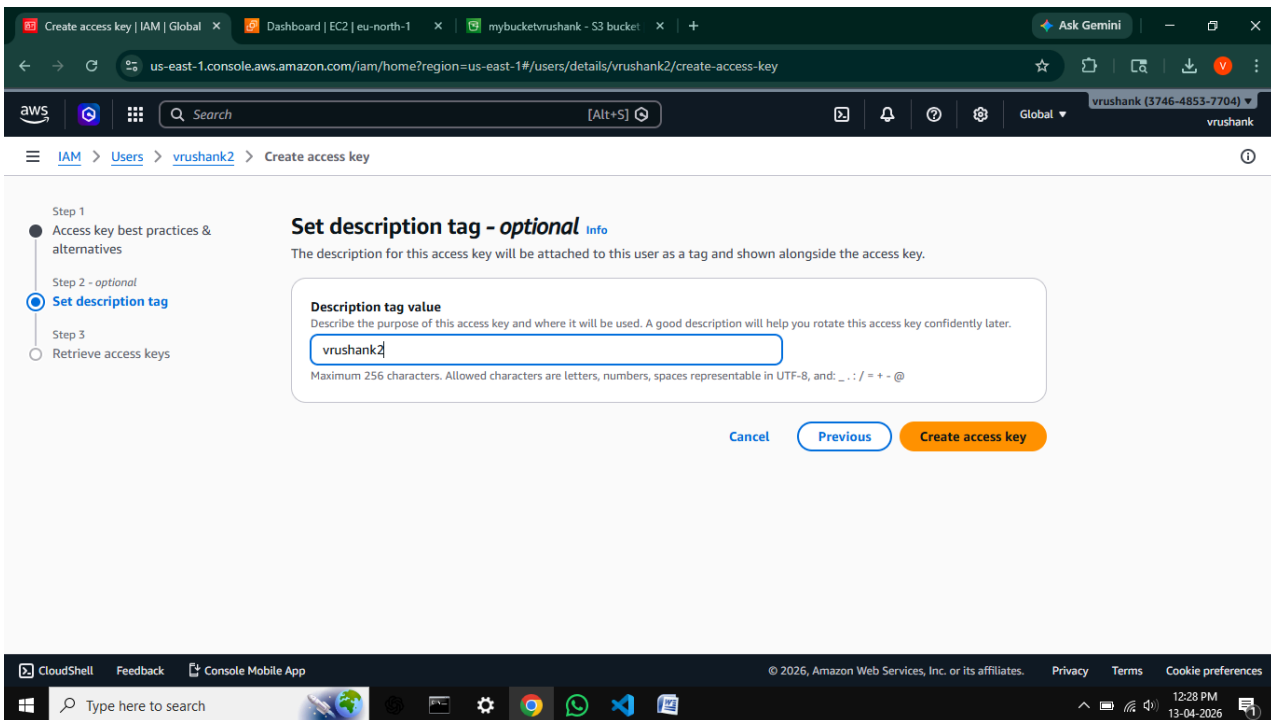
Step 2: Go to the Security credentials tab. Scroll to Access keys and click Create access key.



# Step 3: Select Command Line Interface (CLI). Acknowledge the warning and click Next.



Step 4: Download the .csv file. >  
Critical: This contains the Access Key ID and Secret Access Key. If lost, you must delete and recreate them.



Create access key | IAM | GlobalDashboard | EC2 | eu-north-1mybucketvrushank - S3 bucketAsk Geminius-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/vrushank2/create-access-keyvrushank (3746-4853-7704)vrushank

IAMUsersvrushank2Create access key

✔ This is the only time that the secret access key can be viewed or downloaded. You cannot recover it later. However, you can create a new access key any time.

Set description tag

Step 3

Retrieve access keys

Access key

Secret access key

AKIAVOOWLAJUUBGFQPB

\*\*\*\*\* Show

Access key best practices

- Never store your access key in plain text, in a code repository, or in code.
- Disable or delete access key when no longer needed.
- Enable least-privilege permissions.
- Rotate access keys regularly.

For more details about managing access keys, see the [best practices for managing AWS access keys](#).

Download .csv file

Done

CloudShellFeedbackConsole Mobile App

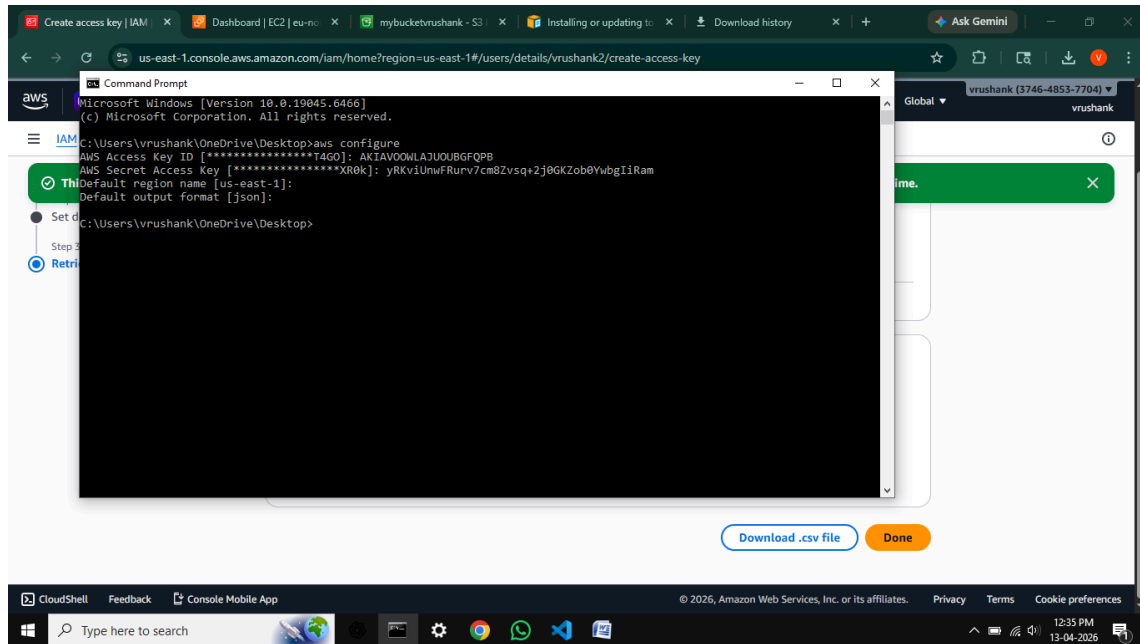
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Type here to search12:28 PM13-04-2026

## 2. Standalone System Configuration

Step 5: Install the AWS CLI Tool from google on your computer

Step 6: Open your Command Prompt (CMD) or Terminal. Step 7: Type `aws configure` and press Enter.



Step 8: Provide the following from your .csv: Access Key ID  
Secret Access Key

Default region: (e.g., ap-south-1)

Default output: json

Run `aws ec2 describe-nstances`.

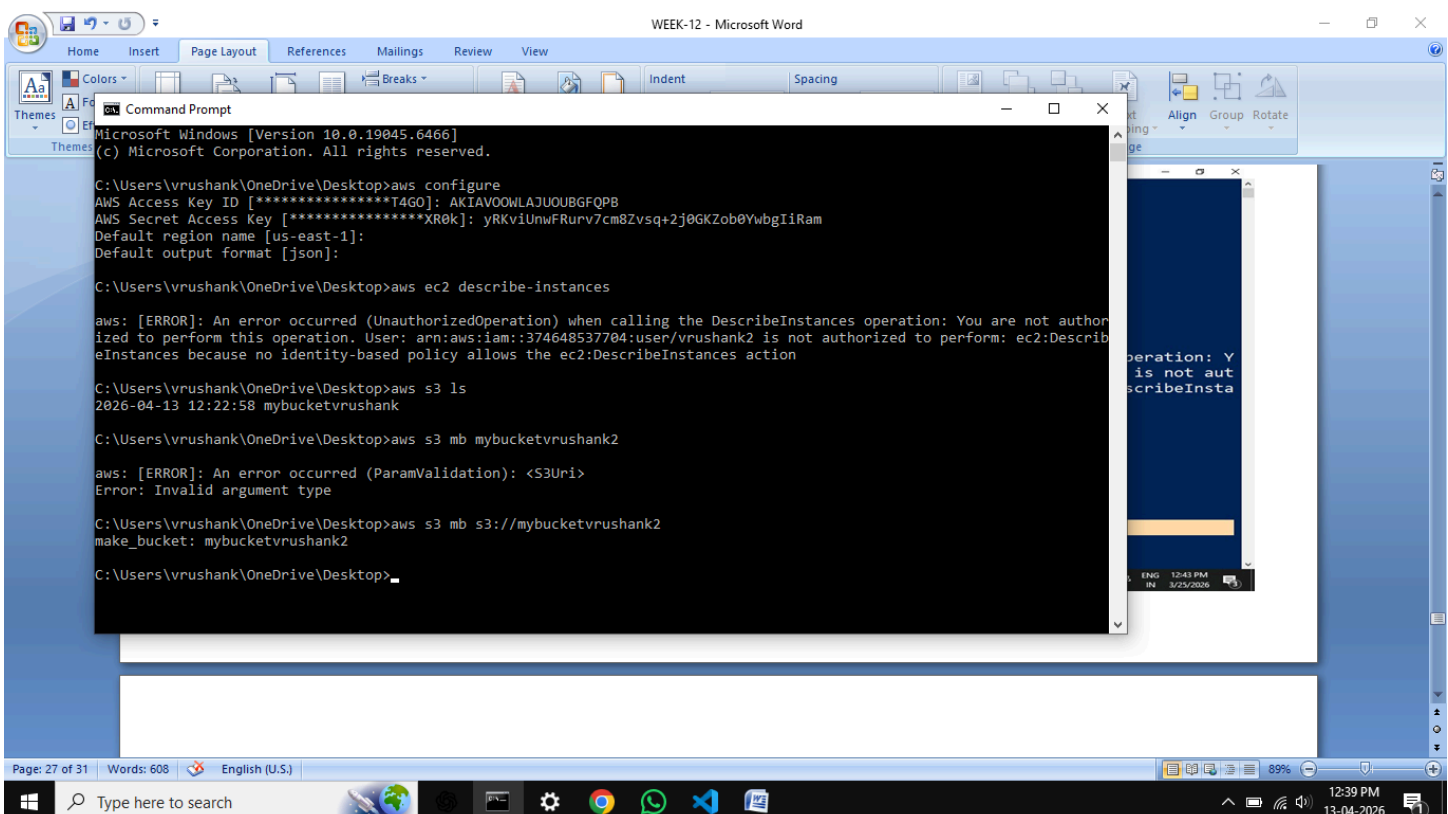
Result: You will get an (AccessDenied) error.

This confirms that the CLI is respecting the same IAM policy as the GUI.

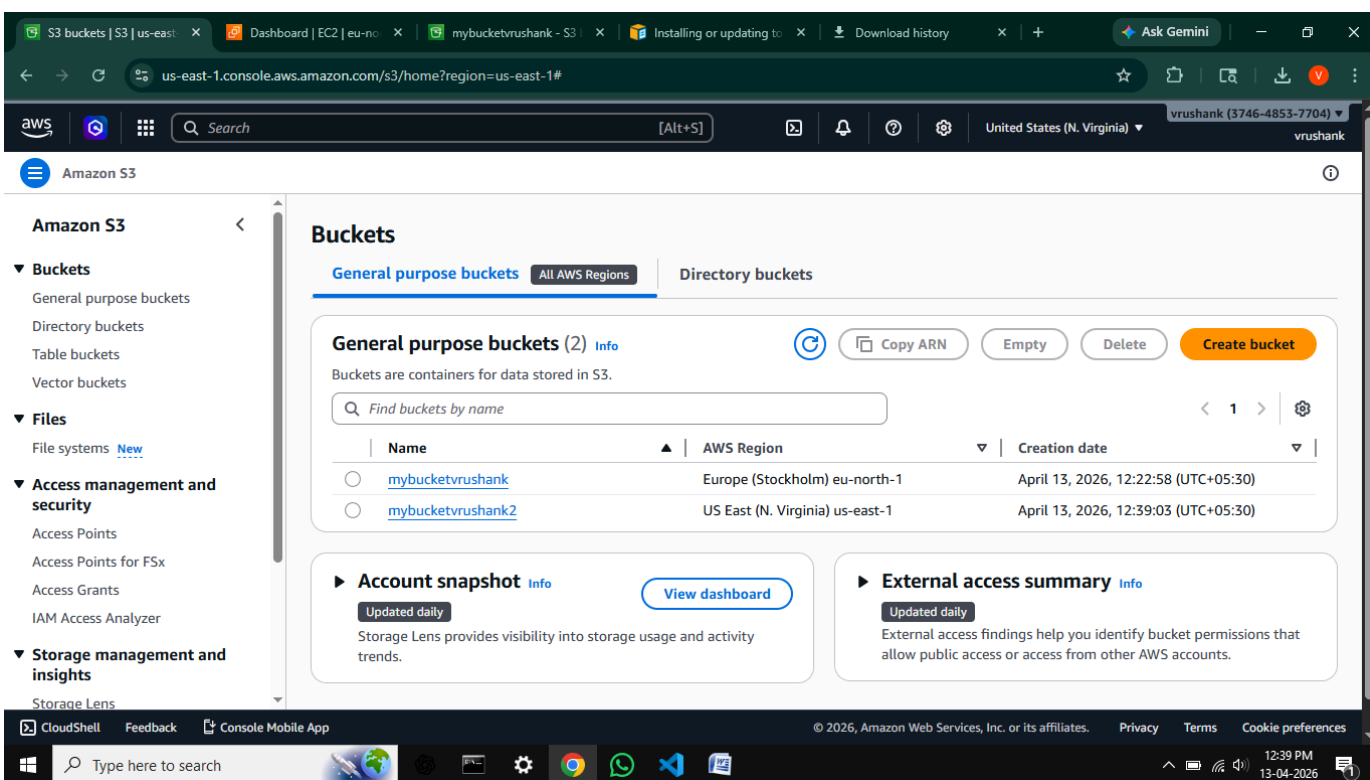
### 3. Verification via Commands

Step 9: Run `aws s3 ls`. This should list your buckets

Step 10: Run `aws s3 mb s3://your-unique-bucket-name`. (Success)



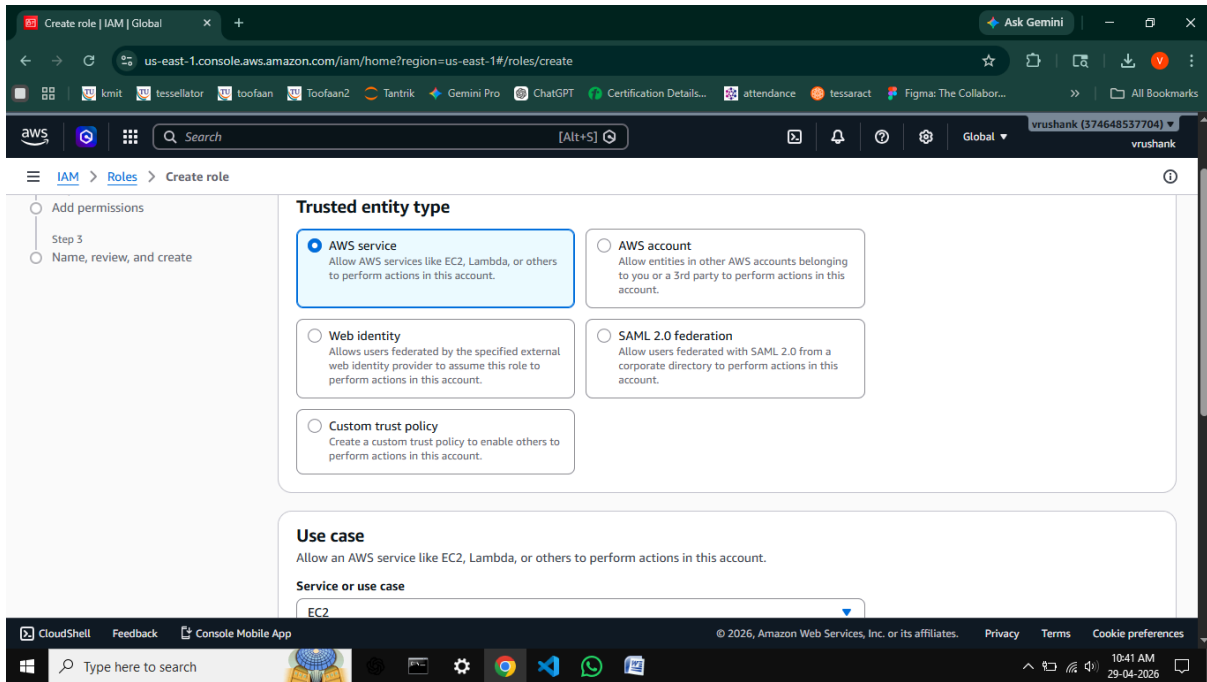
# Check in console you see the bucket created from CLI



IAM EC2

# Roles

## Procedure: Attaching a Role to an EC2 Instance



Under the "Service or use case" dropdown, select EC2. (This tells AWS: "I trust the EC2 service to use this role").

Add Permissions: Search for and check AmazonS3FullAccess.

Create role | IAM | Global

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/create?trustedEntityType=AWS\_SERVICE&selectedService=EC2&selectedUse...

vrushank (374648537704)

Step 1

Select trusted entity

Step 2

Add permissions

Step 3

Name, review, and create

### Add permissions Info

Permissions policies (1/1146) Info

Choose one or more policies to attach to your new role.

s3ful

Filter by Type

All types

1 match

<input checked="" type="checkbox"/>	Policy name Info	Type	Description
<input checked="" type="checkbox"/>	AmazonS3FullAccess	AWS managed	Provides full access to all buckets via the ...

► Set permissions boundary - optional

Cancel

Previous

Next

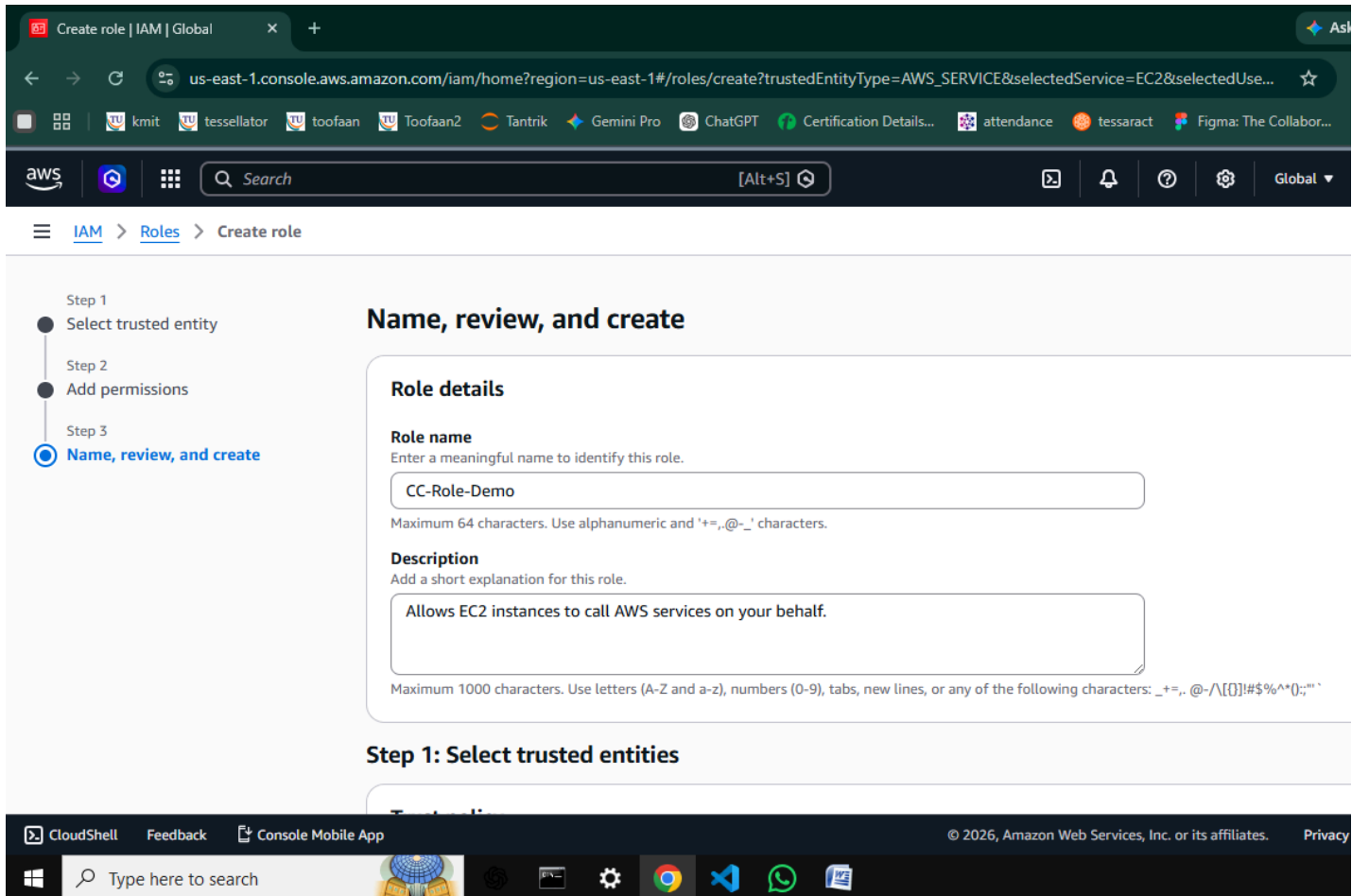
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Type here to search

10:45 AM29-04-2026

Name & Review: Name it (e.g., EC2-S3-ReadOnly-Role) and click Create role.



# Role has been created

The screenshot displays the AWS IAM console interface. At the top, a green banner confirms the creation of the role: **Role CC-Role-Demo created.**

The main content area is titled **Roles (5)** and includes a brief description: "An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. You trust..." Below this is a search bar and a table of roles.

<input type="checkbox"/>	Role name	Trusted entities
<input type="checkbox"/>	<a href="#">AWSServiceRoleForSupport</a>	AWS Service: support (Se...
<input type="checkbox"/>	<a href="#">AWSServiceRoleForTrustedAdvisor</a>	AWS Service: trustedadv...
<input type="checkbox"/>	<a href="#">CC-Role-Demo</a>	AWS Service: ec2

The left sidebar shows the navigation menu for Identity and Access Management (IAM), including sections for Access Management and Access reports.

The bottom of the image shows the Windows taskbar with various application icons and the system clock.

# Create the ec2 instance and and connect to it and check for aws ec2 describe-instances

The screenshot shows the AWS Management Console interface for launching an EC2 instance. The browser address bar indicates the URL: `us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:`. The console breadcrumb navigation shows `EC2 > Instances > Launch an instance`.

The main heading is **Launch an instance** with an [Info](#) link. Below the heading, a descriptive text states: "Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below."

The configuration steps are as follows:

- Name and tags** (with an [Info](#) link):
  - The **Name** field contains the text `DEF`.
  - A button labeled **Add additional tags** is located to the right of the name field.
- Application and OS Images (Amazon Machine Image)** (with an [Info](#) link):
  - A descriptive text states: "An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**."
  - A search bar contains the placeholder text: "Search our full catalog including 1000s of application and OS images".
  - Below the search bar are two tabs: **Recents** and **Quick Start** (which is currently selected).

On the right side of the console, a **Summary** panel is partially visible, showing the following configuration details:

- Number of instances:** 1
- Software Image (AMI):** -
- Virtual server type:** t3.micro
- Firewall (security group):** -
- Storage (volumes):** -

A **Cancel** button is located at the bottom of the summary panel.

The bottom of the screenshot shows the Windows taskbar with the search bar containing "Type here to search" and several application icons including CloudShell, Feedback, Console Mobile App, and various communication tools like Chrome, Teams, and WhatsApp.

 ec2-user@ip-172-31-38-194:~

Windows PowerShell

Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell <https://aka.ms/pscore6>

PS C:\WINDOWS\system32> ssh -i "id\_rsa" ec2-user@ec2-3-82-59-48.compute-1.amazonaws.com

Warning: Identity file id\_rsa not accessible: No such file or directory.

The authenticity of host 'ec2-3-82-59-48.compute-1.amazonaws.com (64:ff9b::352:3b30)' can't be established.

ED25519 key fingerprint is SHA256:09JrueplqlwwqwszOsxvdXWmnulR3vAnhRGyLzp0PnI.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

Warning: Permanently added 'ec2-3-82-59-48.compute-1.amazonaws.com' (ED25519) to the list of known hosts.

ec2-user@ec2-3-82-59-48.compute-1.amazonaws.com: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).

PS C:\WINDOWS\system32> cd "C:\Users\vrushank\OneDrive\Desktop\AWS"

PS C:\Users\vrushank\OneDrive\Desktop\AWS> ssh -i "week13pem.pem" ec2-user@ec2-3-92-238-28.compute-1.amazonaws.com

The authenticity of host 'ec2-3-92-238-28.compute-1.amazonaws.com (64:ff9b::35c:ee1c)' can't be established.

ED25519 key fingerprint is SHA256:3SkItAdLcy9qDOey0CakrfKHgJieCjq/tBCwtNwWpkc.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

Warning: Permanently added 'ec2-3-92-238-28.compute-1.amazonaws.com' (ED25519) to the list of known hosts.

```

 #_
 \ #####
 ~\ \#####\
 ~\ \###|
 ~\ \#/ ____
 ~\ V~' '->
 ~~~
    ~\  _
   _\  _\
  _\m/  '

```

Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

[ec2-user@ip-172-31-38-194 ~]\$ aws ec2 describe-instances

Unable to locate credentials. You can configure credentials by running "aws login".

[ec2-user@ip-172-31-38-194 ~]\$ aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws login".

[ec2-user@ip-172-31-38-194 ~]\$

Launch an instance | EC2 | us-east-1

Launch an instance | EC2 | us-east-1

Instances | EC2 | us-east-1

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

aws

Search

[Alt+S]

EC2

Instances

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Instances (1/2)

Info

Connect

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state
<input type="checkbox"/>	ABC	i-0d934a038af29e129	Running
<input checked="" type="checkbox"/>	DEF	i-0ef7f394a193f1eee	Running

i-0ef7f394a193f1eee (DEF)

Details

Status and alarms

Monitoring

Security

Network

Instance summary

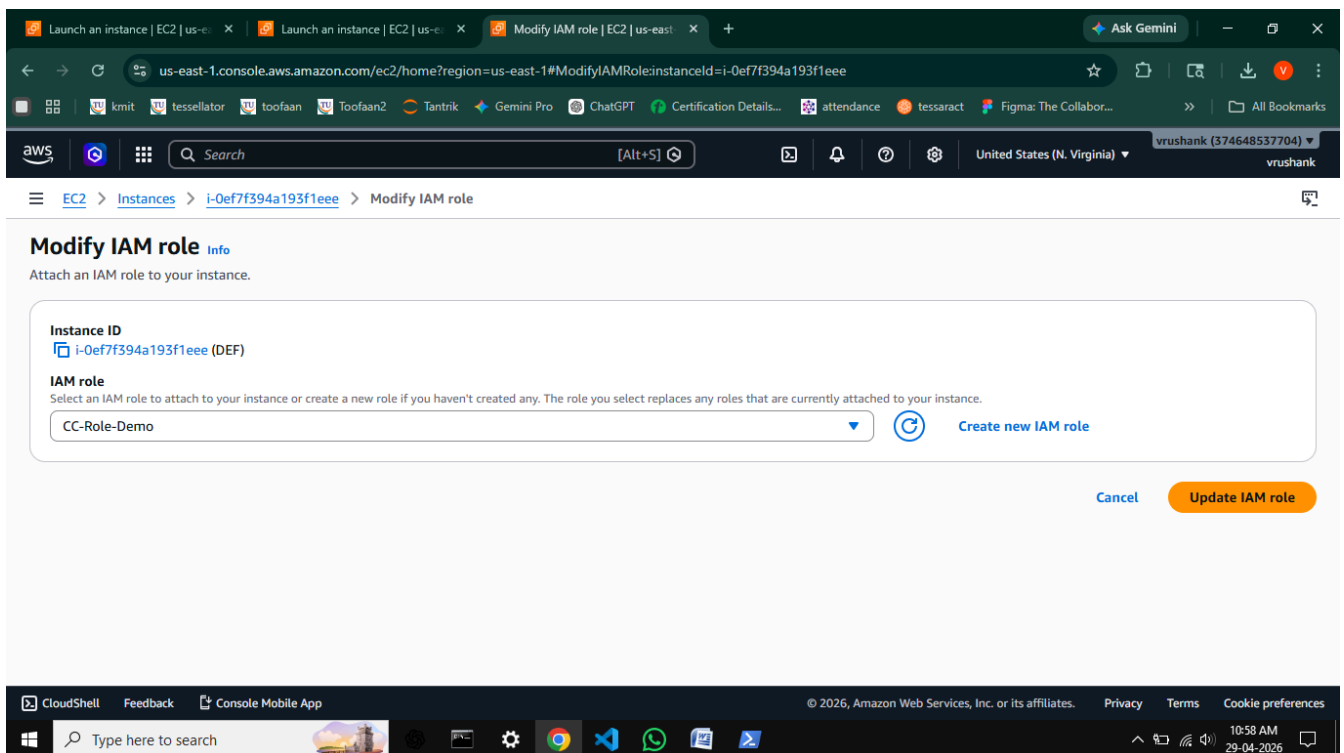
Instance ID

i-0ef7f394a193f1eee

Public IPv4 address

3.92.238.28 | open address

# Select the role you just created and click Update IAM role



### Verify via CLI:

SSH/Connect into your EC2 instance. Type `aws s3 ls`.

The Result: The command works immediately.

We can here role has only s3 full access only here-- aws ec2 describe-instances --

unauthorized

You did not have to provide an Access Key or Secret Key. AWS automatically provided

Temporary Security Credentials to the instance.

```
> ec2-user@ip-172-31-38-194:~
```

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/powershell

PS C:\WINDOWS\system32> ssh -i "id_rsa" ec2-user@ec2-3-82-59-48.compute-1.amazonaws.com
Warning: Identity file id_rsa not accessible: No such file or directory.
The authenticity of host 'ec2-3-82-59-48.compute-1.amazonaws.com (64:ff9b::352:3b30)' can't be established.
ED25519 key fingerprint is SHA256:09JrueplqlwwqwszOxsxdXWmnuIR3vAnhRGyLzp0PnI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-82-59-48.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
ec2-user@ec2-3-82-59-48.compute-1.amazonaws.com: Permission denied (publickey,gssapi-keyex,gssapi-withmic).
PS C:\WINDOWS\system32> cd "C:\Users\vrushank\OneDrive\Desktop\AWS"
PS C:\Users\vrushank\OneDrive\Desktop\AWS> ssh -i "week13pem.pem" ec2-user@ec2-3-92-238-28.compute-1.amazonaws.com
The authenticity of host 'ec2-3-92-238-28.compute-1.amazonaws.com (64:ff9b::35c:ee1c)' can't be established.
ED25519 key fingerprint is SHA256:3SkItAdLcy9qDOey0CakrfKHgJieCjq/tBCwtNWwpkc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-92-238-28.compute-1.amazonaws.com' (ED25519) to the list of known hosts.

      #_
     _#_   #####_    Amazon Linux 2023
    ~\  \_#####\_
   ~~  \_#####\_
   ~~   \###|
   ~~    \#/_____ https://aws.amazon.com/linux/amazon-linux-2023
   ~~     V~' '->
       ~~~
 ~_. . _/_/_/
 //_/_/_/
 _/m/'

[ec2-user@ip-172-31-38-194 ~]$ aws ec2 describe-instances

Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-172-31-38-194 ~]$ aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-172-31-38-194 ~]$ aws s3 ls
[ec2-user@ip-172-31-38-194 ~]$ aws s3 ls
2026-04-29 05:30:46 vrushankbucket123
[ec2-user@ip-172-31-38-194 ~]$
```